

K-POLYSTABILITY OF NILPOTENT ORBIT AND SLODOWY SLICE BASES

JIHAO LIU

ABSTRACT. We give affirmative answers to both parts of Xu–Zhuang Problem 23 in the normal/orbifold projective-base formulation, for every complex semisimple Lie algebra. For normalized nonzero nilpotent orbit closures, we prove K-polystability of the projective base of ordinary scalar dilation. For each positive-dimensional normalized irreducible component of a reduced Slodowy-slice intersection, we prove log K-polystability of the base of its effective Kazhdan action with the actual divisorial stabilizer boundary. Both conclusions include weak Kähler–Einstein currents and the equality case for all normal relatively ample test configurations. The proof constructs global homogeneous potentials through framed hyperkähler comparisons, transports them in the correct natural coordinates, and checks weighted descent and affine products. The main result of this paper was obtained using the Danus system.

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1. INTRODUCTION

Nilpotent orbit closures and Slodowy slices are affine symplectic varieties with contracting actions. Their projective quotients connect Lie theory with the existence of Kähler–Einstein metrics and the behavior of algebraic degenerations. Xu and Zhuang ask two questions: whether the natural projective bases of nilpotent orbit closures are K-polystable, and whether the natural projective bases of Slodowy slices are K-polystable [XZ26, Section 3.1, Problem 23]. We give affirmative answers to both parts, in the normal/orbifold projective-base formulation, for every complex semisimple Lie algebra. Theorem 1.1 answers the orbit question in part (1), and Theorem 1.2 answers the slice question in part (2).

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We now specify the normal/orbifold interpretation used in these answers. The projective-base discussion in [XZ26, Section 3.1, Paragraph 22] includes contact Fano orbifolds. For orbit closures, we take normalization and ordinary scalar dilation; the punctured action is free, so the divisorial boundary is zero. For slices, we normalize each irreducible component of the reduced orbit–slice intersection. On each positive-dimensional normalization, we remove the common finite kernel of the natural Kazhdan action and retain the boundary determined by its actual residual divisorial stabilizers. Orbifold K-polystability is expressed here as log K-polystability of this coarse base with its boundary. Zero-dimensional components have no projective base. These conventions make the normal/orbifold formulation precise; they are not additional clauses quoted from the original questions.

We now fix the conventions for the two main results. All varieties are over $\mathbb{C}$, and Norm, Spec and Proj denote normalization, affine spectrum and projective spectrum. Orbit closures and intersections have their reduced structures. For an $\mathfrak{sl}_2$ -triple (e, h, f) , put $\mathfrak{g}^f = \ker(\operatorname{ad} f)$; the zero triple is allowed when $e = 0$. The associated cocharacter defines the raw Kazhdan action $t^2 \operatorname{Ad}(t^{-h})$. A positive grading has degree-zero part $\mathbb{C}$. After removing its common finite kernel, the rational weight-one Seifert polarization H is the class of the character-one line on the quotient stack of the punctured cone. A prime divisor D on its coarse base contributes $(1 - 1/m_D)D$, where m_D is the generic stabilizer order for that effective action.

Here a log Fano pair (F, B) is a normal projective klt pair with effective rational boundary and ample rationally Cartier $-(K_F + B)$. We use normal test configurations with relatively ample rational polarization for this log-anticanonical class. An equivariantly product configuration may have a nontrivial one-parameter automorphism action on (F, B) . The weak Kähler–Einstein currents below have bounded potentials and full Monge–Ampère mass and satisfy the adapted-measure equation with probability normalization.

Theorem 1.1 (Answer to Xu–Zhuang Problem 23(1)). *Let $\mathfrak{g}$ be any complex semisimple Lie algebra, let $\mathcal{O} \subset \mathfrak{g}$ be a nonzero nilpotent adjoint orbit, and let Y be the normalization of $\overline{\mathcal{O}}$. Equip Y with the lifted scalar action, and write $\dim_{\mathbb{C}} Y = 2n$. The action on the punctured cone is free. Its projective base $F = \operatorname{Proj} \mathbb{C}[Y]$, with rational weight-one Seifert polarization H , is a klt Fano variety and satisfies*

$$-K_F \sim_{\mathbb{Q}} nH.$$

It admits a weak Kähler–Einstein current and is K-polystable: every normal relatively ample test configuration for $-K_F$ has nonnegative Donaldson–Futaki invariant, with equality only for an equivariantly product configuration.

Theorem 1.1 gives the affirmative answer to [XZ26, Problem 23(1)]. The scalar action is the lifted ordinary dilation. Its freeness holds on the entire punctured normalization, so no divisorial boundary occurs. The polarization H need only be a rational line bundle. In a semisimple algebra the relevant affine cone is the product of the simple-factor cones with the diagonal scalar action; Theorem 1.1 concerns its single projective base.

Theorem 1.2 (Answer to Xu–Zhuang Problem 23(2)). *Let $\mathfrak{g}$ be any complex semisimple Lie algebra, $\mathcal{O} \subset \mathfrak{g}$ a nilpotent adjoint orbit, and (e, h, f) an $\mathfrak{sl}_2$ -triple with $e \in \overline{\mathcal{O}}$. Let Z normalize an irreducible component of*

$$(\overline{\mathcal{O}} \cap (e + \mathfrak{g}^f))_{\operatorname{red}}.$$

If $\dim Z = 0$, Z is the point above e and has no projective base. If $\dim_{\mathbb{C}} Z = 2n > 0$, first give Z its lifted raw Kazhdan action $\kappa_t(x) = t^2 \operatorname{Ad}(t^{-h})x$, and then remove its common finite kernel μ_d . For the resulting effective total grading, put

$$F = \operatorname{Proj} \mathbb{C}[Z], \quad \ell = 2/d, \quad B = \sum_D \left(1 - \frac{1}{m_D}\right) D,$$

where m_D is the actual generic stabilizer order above D for the effective action. Then ℓ is a positive integer and, for the effective weight-one Seifert polarization H ,

$$-(K_F + B) \sim_{\mathbb{Q}} n\ell H.$$

The pair (F, B) is log Fano, admits a weak log Kähler–Einstein current, and is log K -polystable: every normal relatively ample log test configuration has nonnegative log Donaldson–Futaki invariant, with equality only for an equivariantly product configuration. In a decomposition into simple ideals, the affine factors are combined before taking this effective grading, projective base, and boundary.

Theorem 1.2 gives the affirmative answer to [XZ26, Problem 23(2)] in the normal/orbifold formulation. Its conclusion is log K -polystability of (F, B) with the actual boundary determined on the source component after effectivization. Removing the common kernel leaves the coarse projective base unchanged and fixes the effective weights and residual stabilizer orders. The boundary is not taken from an ambient matrix slice, and in a product it is not obtained by adding the factor boundaries. The point case is excluded from projective descent explicitly. Both Theorems 1.1 and 1.2 include the equality case for all normal relatively ample test configurations.

For a concrete instance, take the nonzero rank-one square-zero orbit in $\mathfrak{sl}_4$. Its Jordan row partition is $(2, 1, 1)$ and its complex dimension is six; this is the one-vertex quotient with dimensions $v_1 = 1$, $w_1 = 4$ examined below. Theorem 1.1 gives a five-dimensional K -polystable base with $-K_F \sim_{\mathbb{Q}} 3H$.

We now compare these answers with prior work. Xu and Zhuang, with an appendix by Guenancia, prove K -semistability of the normalized nilpotent-orbit bases for general semisimple Lie algebras, together with normalized-volume minimization by the scalar valuation [XZG26, Theorems 1.4 and 7.5]. They also record K -polystability for classical simple nilpotent orbit closures through finite-dimensional hyperkähler quotients and the boundary behavior of their potentials, and describe K -polystability of the general normalized orbit bases and natural Kazhdan-slice bases as expected [XZG26, Remark 7.6]. Theorem 1.1 includes these classical cases and strengthens the general semistability conclusion to polystability. Theorem 1.2 gives the componentwise log-polystability conclusion with the effective action and its residual boundary.

The constructions use established descriptions of nilpotent geometry. Kraft and Procesi prove normality of matrix orbit closures [KP79], and Beauville supplies the symplectic-singularity framework [Bea00]. Finiteness of nilpotent orbits, in the form recalled by Jablonski [Jab10, Theorem 5.2], combines with the Poisson transversality of Slodowy slices [CR20] and the description of transverse leaves [OR06] to give the smooth big opens used below. Kronheimer’s framed-trajectory construction gives hyperkähler structures and complex orbit-slice charts [Kro90]. Brylinski fixes the one-ended orbit normalization [Bry98], while Kobak and Swann construct potentials through finite-dimensional quotients and prove a conditional uniqueness theorem [KS01].

For the type-A comparison, Maciocia’s framed ADHM correspondence identifies the metric and half-norm potential [Mac91]. Nakajima describes the natural chosen-complex map through invariant instantons and trajectories [Nak94, Theorem 8.4]. The removable-connection input used in passing between trajectories and instantons is recorded in Platt’s appendix [Pla24, Theorem A.10 and Proposition A.11]. Maffei identifies the reduced image of a quiver resolution with the orbit-slice intersection [Maf05]; his dominant-data density argument [Maf02] is needed to reach the entire affine quotient. We also use Losev’s comparison and block-weight calculation [Los12], Ginzburg’s quotient and Poisson interfaces [Gin09], and the quiver smooth-locus criterion of Bellamy and Schedler [BS21]. The hypotheses of these results are kept separate: the reduced image, the whole affine quotient, and the regular simultaneous-zero hyperkähler quotient are not interchangeable without proof.

The analytic inputs likewise require control on the whole cone. Affine Kempf–Ness theory [Hos14] and Kähler reduction [GH10] let us descend the half-norm and its regularized logarithms, including nonfree strata. Scalar Hartogs extension [BO23, Lemma 2.6], combined with positive

grading, will extend a change of coordinates across the singular locus. On a smooth resolution, the noncharging theorem of Dinh and Sibony [DS10, Corollary A.32] and the mass formula of Botero, Burgos Gil, Holmes and de Jong [BGH⁺22, Lemma 2.24] provide the measure-theoretic inputs. The weak Kähler–Einstein framework of Berman, Boucksom, Eyssidieux, Guedj and Zeriahi [BBE⁺19] and Berman’s log-polystability theorem [Ber16, Theorem 4.8] then apply after the log-Fano and full-mass hypotheses have been proved. The geometric work is to construct global homogeneous potentials compatible with the required complex structures; a smooth-open metric or a chosen-complex comparison alone does not supply this input.

The proof separates the scalar-orbit construction from the slice construction. For orbit closures, the exact trace KKS normalization and the quaternionic real structure allow us to apply the Kobak–Swann uniqueness theorem to the matrix models. A faithful representation then pulls the globally defined matrix potential to the normalization of an arbitrary simple orbit closure. The trace constant scales the metric, symplectic form and potential consistently.

For slices, the essential compatibility concerns the actual ADHM–instanton–trajectory–Kronheimer composite. The equivariant monad determines the dominant lift and the correct endpoint modules. A right quaternionic rotation preserves that sector after a specified compensation at the retained positive frame; its change of negative frame is permitted because that frame is forgotten. Finite positive-normal-form uniqueness gives the inverse chart identity, and conjugating the chosen-complex comparison proves compatibility in every twistor direction. This uses neither differentiation of a normalizing gauge nor an unstated two-ended weighted gauge theorem.

The natural and algebraic quiver parametrizations are kept distinct. On the smooth big open their discrepancy is equivariant and biholomorphic. Hartogs extension and a positive-weight Taylor projection make it a graded algebraic automorphism of the whole cone. Pulling the physical potential back by the inverse automorphism gives precisely the potential compatible with the natural twistor structure. An all-oriented faithful chart comparison then permits covariant pullback to each general slice component, using its own top orbit to choose the ambient matrix slice. The construction supplies a suitable metric and does not require equality with an independently normalized slice metric.

Both constructions feed the same weighted descent argument. The canonical calculation records the actual cyclic stabilizer boundary. The cone determinant identity and globally bounded quotient weights give full mass, and finite adapted volume proves klt by the discrepancy integrability criterion. After the curvature and probability normalizations are checked, the weak Kähler–Einstein and log-polystability theorems apply. Positive sums of factor potentials preserve these properties on affine products. Removing the common kernel only after that product gives the two semisimple conclusions.

Section 2 fixes conventions. Sections 3 and 4 establish the affine geometry and quotient potentials. Section 5 develops framed charts and faithful naturality, and Section 6 treats scalar orbit closures. Sections 7 and 8 give the natural and physical type-A constructions; Section 9 compares their coordinates. Section 10 constructs the general slice potentials. Section 11 proves weighted descent, and Section 12 forms the affine products and proves the two main theorems.

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2. NOTATION AND CONVENTIONS

Set-up 2.1. All algebraic varieties are over $\mathbb{C}$. A normalization is the normalization of a reduced irreducible variety. For an $\mathfrak{sl}_2$ -triple (e, h, f) in a complex semisimple Lie algebra $\mathfrak{g}$, we write $\mathcal{S}_e = e + \mathfrak{g}^f$ and

$$\kappa_t(x) = t^2 \text{Ad}(t^{-h})x \quad (t \in \mathbb{C}^*).$$

Here t^h is the cocharacter obtained by integrating the triple. The zero triple is allowed when $e = 0$. Orbit closures and orbit–slice intersections carry their reduced structures. The expression “a component of a slice” means an irreducible component of $(\overline{\mathcal{O}} \cap \mathcal{S}_e)_{\text{red}}$ for a specified nilpotent orbit $\mathcal{O}$ and $e \in \overline{\mathcal{O}}$.

Definition 2.2. A positive grading on a normal affine domain A is a finitely generated grading $A = \bigoplus_{j \geq 0} A_j$ with $A_0 = \mathbb{C}$. Its vertex is the point defined by $\bigoplus_{j > 0} A_j$. If $A \neq \mathbb{C}$, the order of its ineffective kernel is

$$d = \gcd\{j > 0 : A_j \neq 0\}, \quad A_j^{\text{eff}} = A_{dj}.$$

The effective parameter is $s = t^d$. On a product of affine cones the grading is the total tensor grading. For an effective punctured cone $U \rightarrow F = \text{Proj } A$, the rational weight-one Seifert polarization H is characterized by the character-one line on $[U/\mathbb{G}_m]$. Its actual divisorial boundary is

$$B = \sum_D \left(1 - \frac{1}{m_D}\right) D,$$

where m_D is the generic stabilizer order above the prime divisor D for this effective action; terms with $m_D = 1$ are omitted.

Notation 2.3. We fix $d^c = \frac{i}{2}(\bar{\partial} - \partial)$, so $dd^c = i\partial\bar{\partial}$. If a Hermitian metric has local weight ψ and squared frame norm $e^{-\psi}$, its differential curvature is $C_\psi = dd^c\psi$ and its integral first-Chern current is

$$T_\psi = \frac{1}{2\pi} dd^c\psi.$$

Intersection numbers use integral first Chern classes. Thus a full-mass metric on an ample rational line bundle H on a variety of dimension N satisfies $\int T_\psi^N = H^N$. Hyperkähler forms use the differential convention $\omega_I = dd^c\Phi$ with $\Phi = \frac{1}{2}\|z\|^2$ in a flat quaternionic model. A hyperkähler triple is denoted by (I, J, K) , with $\omega_q(v, w) = g(qv, w)$ and $\Omega = \omega_J + i\omega_K$. A compact group named K_c is distinct from the third complex structure K .

Definition 2.4. Let $Z = \text{Spec } A$ be an irreducible normal positive affine cone of complex dimension $2n > 0$, with vertex z_0 and a possibly ineffective $\mathbb{G}_m$ -action. A *homogeneous hyperkähler potential package of weight a* , for an integer $a > 0$, consists of an invariant smooth open $M \subset Z \setminus \{z_0\}$ with $\text{codim}_Z(Z \setminus M) \geq 2$, a hyperkähler structure (g, I, J, K) on M whose I is algebraic, and a function $\Phi : Z \rightarrow [0, \infty)$ such that:

- (i) Φ is continuous and proper, vanishes exactly at z_0 , and is smooth and positive on M ; both Φ and $\log \Phi$, with value $-\infty$ at z_0 , are plurisubharmonic.
- (ii) On M , $dd^c\Phi = \omega_I$, and $\Omega = \omega_J + i\omega_K$ is a closed nondegenerate algebraic holomorphic two-form.
- (iii) The given action satisfies $\Phi(tz) = |t|^a \Phi(z)$ and $t^*\Omega = t^a \Omega$.

The logarithmic generator of the raw positive-real action is $E_{\text{raw}} = \frac{d}{du} \Big|_{u=0} \kappa_{e^u}$. After division by a kernel of order d , $E_{\text{raw}} = dE_{\text{eff}}$.

Definition 2.5. A log Fano pair (F, B) is normal and projective, B is an effective rational divisor, (F, B) is klt, and $-(K_F + B)$ is ample and rationally Cartier. We use normal log test configurations with relatively ample rational polarization for this log-anticanonical class, allowing positive integral multiples. A product configuration means a configuration equivariantly

isomorphic to $(F, B) \times \mathbb{A}^1$, with an action induced by a one-parameter subgroup of $\text{Aut}(F, B)$ and the standard action on $\mathbb{A}^1$. Log K-polystability means $\text{DF} \geq 0$ for every such normal configuration, with equality only for product configurations. When $B = 0$ we omit “log”.

3. NORMALIZED CONES AND ORBIT-PAIR GEOMETRY

We retain the conventions of Section 2. We first establish the contracting actions on the normalizations and their behavior under affine products. We then identify the smooth symplectic open on each component and prove that its complement has codimension at least two. An open subset with this property will be called *big*. We write G for the connected adjoint group of $\mathfrak{g}$ and fix a nondegenerate invariant symmetric bilinear form $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$ for the Lie–Poisson structure and its KKS forms.

We use nilpotent-orbit finiteness in [Jab10, Theorem 5.2], the Slodowy transversality statements in [CR20, Corollary 3.4 and Proposition 1.1(i)], and the Poisson-transversal leaf description in [OR06, Theorem 3(iv),(v)], in the following form.

Theorem 3.1. *A complex semisimple Lie algebra has finitely many nilpotent adjoint orbits. A Slodowy slice is a Poisson transversal to each adjoint orbit it meets. Its induced symplectic leaves are the connected components of the orbit intersections, with the induced symplectic forms; the intersections are smooth and transverse, and their dimensions are the orbit dimensions minus the dimension of the orbit through the slice vertex.*

In particular, each orbit intersection has finitely many connected components, all of the same even dimension. The dimension formula will exclude divisorial strata from the boundary of a dense leaf.

Lemma 3.2. *Let V_1, V_2 be irreducible affine complex varieties, let $R_i = \mathbb{C}[V_i]$ for $i = 1, 2$, and let $\nu_i : W_i \rightarrow V_i$ be their normalizations. Write $\bar{R}_i = \mathbb{C}[W_i]$ for the integral closure of R_i in its fraction field. Then $W_1 \times W_2$ is normal, and $\nu_1 \times \nu_2$ is the normalization of $V_1 \times V_2$. Equivalently, in the fraction field of $R_1 \otimes_{\mathbb{C}} R_2$,*

$$\overline{R_1 \otimes_{\mathbb{C}} R_2} = \bar{R}_1 \otimes_{\mathbb{C}} \bar{R}_2.$$

If the factor rings are graded and the normalization maps are equivariant, the diagonal action gives, for every integer m ,

$$(\bar{R}_1 \otimes_{\mathbb{C}} \bar{R}_2)_m = \bigoplus_{p+q=m} (\bar{R}_1)_p \otimes_{\mathbb{C}} (\bar{R}_2)_q.$$

This identifies the normalization before a projective quotient is taken.

Proof. We verify normality by Serre’s criterion, then identify the finite birational map and its grading.

Step 1. In this step, we prove normality of $W_1 \times W_2$ by Serre’s criterion. The product of two irreducible complex varieties is irreducible and reduced, so $W_1 \times W_2$ is integral. In local affine embeddings the Jacobian matrix for the product equations is block diagonal. Its singular locus is therefore

$$(\text{Sing } W_1 \times W_2) \cup (W_1 \times \text{Sing } W_2).$$

Normality of the factors makes each nonempty member of this union have codimension at least two. Thus the product satisfies (R_1) .

For (S_2) , consider any point w of the product, with image w_1 in W_1 , and set

$$R = \mathcal{O}_{W_1, w_1}, \quad T = \mathcal{O}_{W_1 \times W_2, w}, \quad P = T/\mathfrak{m}_R T.$$

The projection is flat. Its fiber is a base change of W_2 to the residue field of w_1 ; normal varieties over $\mathbb{C}$ are geometrically normal, so P is normal. Put $a = \dim R$ and $b = \dim P$. The local dimension formula gives $\dim T = a + b$. Regular sequences from R remain regular in T by

flatness. After quotienting by such a sequence, lifts of a regular sequence in the fiber remain regular; this is the flat local regular-sequence criterion. Consequently

$$\text{depth } T \geq \text{depth } R + \text{depth } P \geq \min\{2, a\} + \min\{2, b\} \geq \min\{2, a + b\}.$$

If $a, b > 0$, the two factors each supply a nonzerodivisor. If one dimension is zero and the sum is at least two, the other supplies a sequence of length two. When the sum is zero or one, the required depth is zero or one, respectively. This proves (S₂) at every point, and Serre's criterion proves normality.

Step 2. In this step, we identify the product normalization with the integral closure of $R_1 \otimes_{\mathbb{C}} R_2$. The product normalization map is a composition of two base changes of finite maps, hence is finite. Each ν_i is an isomorphism over a dense open subset of V_i , so the product map is birational. Thus $\overline{R}_1 \otimes_{\mathbb{C}} \overline{R}_2$ is an integral extension of $R_1 \otimes_{\mathbb{C}} R_2$ with the same fraction field, and is integrally closed by the preceding paragraph. Conversely, any element of that fraction field integral over $R_1 \otimes_{\mathbb{C}} R_2$ is also integral over $\overline{R}_1 \otimes_{\mathbb{C}} \overline{R}_2$, so belongs to it. This proves the asserted equality with the integral closure.

Step 3. In this step, we compute the grading induced by the diagonal action. For homogeneous $u \in (\overline{R}_1)_p$ and $v \in (\overline{R}_2)_q$, pullback by the diagonal action sends $u \otimes v$ to $t^{p+q}u \otimes v$. Decomposing the two factors into their weight spaces gives the displayed total grading. $\square$

Lemma 3.3. *Let $R = \bigoplus_{j \geq 0} R_j$ be a finitely generated complex domain with $R_0 = \mathbb{C}$, let $V = \text{Spec } R$, and let v_0 be the point defined by $R_{>0}$. Let $\nu : W \rightarrow V$ be the normalization. Then the grading action lifts uniquely to an algebraic $\mathbb{G}_m$ -action on W . Its coordinate ring is nonnegatively graded with degree-zero part $\mathbb{C}$. The lifted action extends to $\mathbb{A}^1 \times W$ and contracts W to a unique fixed point w_0 . Set-theoretically, $\nu^{-1}(v_0) = \{w_0\}$.*

The last assertion gives control of the full fiber over the vertex, even when the original variety is not normal.

Proof. We lift the action as a morphism, determine the integral grading, and then identify the vertex fiber.

Step 1. In this step, we lift the algebraic action uniquely to the normalization. The variety $\mathbb{G}_m \times W$ is normal by Lemma 3.2. The dominant map

$$\mathbb{G}_m \times W \longrightarrow V, \quad (t, w) \longmapsto t\nu(w)$$

therefore factors uniquely through the normalization W . The two maps expressing the action law agree after composition with ν , so the uniqueness of this factorization gives the action law; the identity element acts identically for the same reason. This produces the unique algebraic lift, and hence an integer weight decomposition on $A = \mathbb{C}[W]$.

Step 2. In this step, we prove positivity of the normalized grading and extend the contraction to $\mathbb{A}^1$. Let $u \in A$ be a nonzero homogeneous element of weight m . Since u is integral over R , it satisfies a monic equation of degree q . Taking the component of weight qm in that equation gives

$$u^q + a_1 u^{q-1} + \cdots + a_q = 0, \quad a_j \in R_{jm}.$$

If $m < 0$, all the a_j vanish, contrary to $u \neq 0$ in a domain. If $m = 0$, all the coefficients belong to $\mathbb{C}$, so $u \in \mathbb{C}$ because $\mathbb{C}$ is algebraically closed. Thus $A_j = 0$ for $j < 0$ and $A_0 = \mathbb{C}$. The coaction $u \mapsto t^j u$ for $u \in A_j$ now takes values in $A[t]$. It defines the extension to $\mathbb{A}^1$, whose value at $t = 0$ is the point w_0 defined by $A_{>0}$. At any fixed point every positive-weight function vanishes, so w_0 is the unique fixed point.

Step 3. In this step, we prove that the fiber over v_0 consists only of w_0 . The finite set $\nu^{-1}(v_0)$ is invariant. The connected group $\mathbb{G}_m$ fixes every point of this set. Each such point must therefore equal w_0 , proving the assertion about the fiber. $\square$

For an action with finite ineffective kernel, we use H_{raw} for the rational character-one Seifert class of the raw quotient and H_{eff} for that of the effective quotient.

Proposition 3.4. *Let $\mathfrak{g}$ be complex semisimple.*

- (i) *For a nonzero nilpotent orbit $\mathcal{O}$, its normalized closure Y has a unique vertex y_0 , an effective positive scalar grading, and a free scalar action on $Y \setminus \{y_0\}$. Its divisorial quotient boundary is zero.*
- (ii) *For every component X of an orbit-slice intersection and its normalization Z , the raw Kazhdan action preserves X , lifts to Z , and contracts to the unique point above e . If $\dim Z = 0$, Z is this point and has no projective base. Otherwise its kernel is finite, effectivization leaves $\text{Proj } \mathbb{C}[Z]$ unchanged as a scheme, and $H_{\text{eff}} = dH_{\text{raw}}$.*
- (iii) *For a decomposition of $\mathfrak{g}$ into simple ideals, orbit closures, reduced orbit-slice intersections, their irreducible components, and their normalizations factor as affine products. The actions are diagonal and the gradings are total. The diagonal raw kernel order on a positive-dimensional slice component is the gcd of the kernel orders of its positive-dimensional factors.*

The scalar action is free at every point of the punctured normalization. For slices, the raw and effective gradings give the same coarse base but different character-one polarizations.

Proof. We apply the normalization lemma to the scalar and Kazhdan actions, compute effectivization on the graded ring, and then establish the factorization over the simple ideals.

Step 1. In this step, we prove scalar contraction and freeness on the punctured normalization. Every nilpotent orbit is stable under nonzero scalar multiplication. Indeed, if x is nilpotent, choose a triple (x, h_x, f_x) . For $t \in \mathbb{C}^*$ choose s with $s^2 = t$; then

$$\text{Ad}(s^{h_x})x = s^2x = tx.$$

The assertion for the zero orbit holds directly. Hence the reduced closure of $\mathcal{O}$ has its ordinary nonnegative polynomial grading, with degree-zero part $\mathbb{C}$. Lemma 3.3 gives the unique lifted action and the singleton fiber over zero. Since $\mathcal{O} \neq \{0\}$, some linear function on $\mathfrak{g}$ restricts nontrivially to its closure. The normalized ring therefore contains a nonzero element of degree one, which proves effectivity.

Let $y \neq y_0$ and write x for its image in $\overline{\mathcal{O}}$. The vertex-fiber assertion gives $x \neq 0$. If t fixes y , equivariance implies $tx = x$, and hence $t = 1$. The punctured action is free. Every generic stabilizer above a divisor of the quotient consequently has order one, so its actual divisorial boundary is zero. This proves (i).

Step 2. In this step, we lift the Kazhdan contraction to each normalized component and identify the zero-dimensional case. In the adjoint $\mathfrak{sl}_2$ -module, vectors killed by f are lowest-weight vectors. Write

$$\mathfrak{g}^f = \bigoplus_{m \geq 0} (\mathfrak{g}^f)_{-m}, \quad x = e + \sum_{m \geq 0} x_{-m}.$$

The action has the explicit form

$$\kappa_t(x) = e + \sum_{m \geq 0} t^{m+2} x_{-m}.$$

All transverse weights are positive. This preserves $\mathcal{S}_e$ and extends to $t = 0$ with value e . It also preserves $\overline{\mathcal{O}}$, since its two factors are scalar multiplication and an inner automorphism. The connected group $\mathbb{G}_m$ preserves each irreducible component of the reduced intersection. Thus X is invariant, contains e , and has a nonnegative coordinate grading with degree-zero part $\mathbb{C}$. Lemma 3.3 gives its unique lift and the contraction of Z to the unique point above e .

If $\dim Z = 0$, the reduced irreducible complex variety X is the point e , and Z is also a point. Its positive-degree ideal is zero, so its Proj is empty. Its raw kernel is all of $\mathbb{G}_m$; no finite kernel order is assigned to this case.

Step 3. In this step, we compute the effective grading, its polarization, and its residual stabilizers. Suppose now that $\dim Z > 0$, and write $A = \mathbb{C}[Z] = \bigoplus_{j \geq 0} A_j$. An element t acts identically if and only if $t^j = 1$ for every occurring positive degree. Thus the kernel is μ_d , where

$$d = \gcd\{j > 0 : A_j \neq 0\}.$$

The grading $A_k^{\text{eff}} = A_{dk}$ has gcd one, and the raw action factors through the effective action by $s = t^d$. For a homogeneous u of raw degree $dk > 0$, a fraction v/u^a has raw degree zero exactly when it has effective degree zero. The chart rings $(A_u)_0$ and their gluings are unchanged. These charts identify $\text{Proj } A$ and $\text{Proj } A^{\text{eff}}$ as schemes. The effective character $s \mapsto s$ pulls back to the raw character $t \mapsto t^d$. Consequently

$$\mathcal{O}_{\text{eff}}(1) \mapsto \mathcal{O}_{\text{raw}}(d), \quad H_{\text{eff}} = dH_{\text{raw}}.$$

This proves (ii).

At a punctured point some positive-degree function is nonzero, so its stabilizer is finite. Its effective stabilizer is its raw stabilizer modulo μ_d . In particular, the order m_D used in Definition 2.2 is the residual order at a general point above D . The common ineffective kernel has already been removed when the coefficient $1 - 1/m_D$ is formed.

Step 4. In this step, we prove affine factorization over the simple ideals and compute the diagonal kernel. Write $\mathfrak{g} = \bigoplus_{i=1}^r \mathfrak{g}_i$. The adjoint action factors through the product of the adjoint groups of the simple ideals, so

$$\mathcal{O} = \prod_i \mathcal{O}_i, \quad \overline{\mathcal{O}} = \prod_i \overline{\mathcal{O}_i}.$$

The second equality follows because the product orbit is dense in the product of the closures. The triple splits into triples (e_i, h_i, f_i) , allowing zero triples, and

$$\mathfrak{g}^f = \bigoplus_i \mathfrak{g}_i^{f_i}, \quad \mathcal{S}_e = \prod_i \mathcal{S}_{e_i}.$$

Put $V_i = (\overline{\mathcal{O}_i} \cap \mathcal{S}_{e_i})_{\text{red}}$. Finite-type reduced complex algebras are geometrically reduced, so their tensor product is reduced. Equality of the underlying intersections therefore gives

$$(\overline{\mathcal{O}} \cap \mathcal{S}_e)_{\text{red}} = \prod_i V_i.$$

To identify its components, let $R_i = \mathbb{C}[V_i]$. For a choice of a minimal prime $\mathfrak{p}_i \subset R_i$ in each factor, the ideal

$$P_{(\mathfrak{p}_i)} = \sum_i R_1 \otimes \cdots \otimes \mathfrak{p}_i \otimes \cdots \otimes R_r$$

is prime: its quotient is the tensor product of the geometrically integral domains $R_i/\mathfrak{p}_i$. Conversely, if P is a minimal prime of $\bigotimes_i R_i$, choose a minimal prime $\mathfrak{p}_i$ inside its contraction to each factor. Then $P_{(\mathfrak{p}_i)} \subset P$, so minimality gives equality. Contraction recovers each $\mathfrak{p}_i$, since tensoring the quotient domains over $\mathbb{C}$ is faithfully flat. These ideals are themselves minimal: a smaller minimal prime would give smaller factor minimal primes on contraction. Thus the irreducible components are uniquely the products $X = \prod_i X_i$ of factor components.

Lemma 3.2, applied inductively, gives

$$\text{Norm}(X) = \prod_i \text{Norm}(X_i), \quad \text{Norm}(\overline{\mathcal{O}}) = \prod_i \text{Norm}(\overline{\mathcal{O}_i}).$$

Dimensions add under these products. Both actions are diagonal by their formulas, and uniqueness of their lifts preserves this property on the normalizations. If $A_i = \mathbb{C}[\text{Norm}(X_i)]$, the raw grading is

$$\left(\bigotimes_i A_i \right)_m = \bigoplus_{m_1 + \cdots + m_r = m} \bigotimes_i (A_i)_{m_i}.$$

If $\dim X > 0$, let d_i be the gcd of the positive occurring degrees on each positive-dimensional factor. Every total degree is divisible by $\gcd_i d_i$. Conversely, each factor weight occurs in the tensor product by tensoring a homogeneous element with units in all other factors. The gcd of all total weights must therefore divide every d_i , proving that it equals $\gcd_i d_i$. Point factors contribute only degree zero. If all factors are points, the product is a point and its raw kernel is all of $\mathbb{G}_m$, as in (ii). This proves (iii) and completes the proof. $\square$

We next use the Poisson structure to identify the big open on a slice component. The orbit that contains this open is selected by the component itself; the proof does not require it to be the orbit $\mathcal{O}$ originally used to specify the intersection.

Proposition 3.5. *Let X be a positive-dimensional component of $(\overline{\mathcal{O}} \cap \mathcal{S}_e)_{\text{red}}$ and $\nu : Z \rightarrow X$ its normalization. There is a unique orbit $\mathcal{O}_{\text{top}}$ for which $L = X \cap \mathcal{O}_{\text{top}}$ is open dense. The set L is connected, smooth and symplectic, and $X \setminus L$ has codimension at least two. The map ν identifies $M = \nu^{-1}(L)$ with L , and $Z \setminus M$ has codimension at least two. Its transverse algebraic symplectic form has raw Kazhdan weight two and extends uniquely as a reflexive two-form on Z . Likewise, the nonzero orbit is a scalar-invariant smooth big open in its normalized closure, and its KKS form has scalar weight one and extends reflexively.*

Thus normalization leaves the dense connected symplectic leaf unchanged. The codimension assertion is a consequence of the finite orbit stratification and the even dimensions of its symplectic leaves.

Proof. We show that each component is a union of finitely many symplectic leaves, identify the unique dense leaf, and pass to normalization. We then compute the weights and extend the forms reflexively.

Step 1. In this step, we prove that each irreducible component is a union of finitely many symplectic leaves. Let $V = (\overline{\mathcal{O}} \cap \mathcal{S}_e)_{\text{red}}$. The transversal structure of Theorem 3.1 is algebraic. Indeed, for the algebraic Lie–Poisson bivector π , its conormal splitting along the slice is

$$T\mathfrak{g}|_{\mathcal{S}_e} = T\mathcal{S}_e \oplus \pi^\#(N^*\mathcal{S}_e).$$

The map from $N^*\mathcal{S}_e$ to the normal bundle induced by $\pi^\#$ is an algebraic bundle isomorphism. Inverting its matrix locally and projecting π to $T\mathcal{S}_e$ therefore gives a regular algebraic Poisson tensor. Its leaves are the connected components of the orbit intersections, and their symplectic forms are restrictions of the algebraic KKS forms.

The subset V is a union of these leaves. If a regular function on $\mathcal{S}_e$ vanishes on V , its derivative along every Hamiltonian vector field also vanishes on V , because the field is tangent to the leaf through each point. The reduced ideal of V is consequently a Poisson ideal. Thus $R = \mathbb{C}[V]$ is a reduced Poisson algebra.

Let $\mathfrak{p}$ be the minimal prime defining X , let $b \in \mathfrak{p}$, and let $D_a = \{a, -\}$ be any Hamiltonian derivation of R . The reduced local ring $R_{\mathfrak{p}}$ is a field, so $b/1 = 0$. There is $s \notin \mathfrak{p}$ with $sb = 0$. Applying the derivation gives

$$D_a(s)b + sD_a(b) = 0.$$

Modulo $\mathfrak{p}$ this says $sD_a(b) = 0$, and primality gives $D_a(b) \in \mathfrak{p}$. Every Hamiltonian derivation therefore preserves X . Local Hamiltonian flows preserve it as well, and their orbits are the connected symplectic leaves. Hence a leaf meeting X is wholly contained in X .

There are finitely many nilpotent orbits by Theorem 3.1. Each orbit intersection is a smooth algebraic variety of finite type and has finitely many connected components. Each such component is algebraic and locally closed; smoothness makes a connected component irreducible. It follows that X is a finite union of locally closed connected symplectic leaves. For a leaf $C \subset \mathcal{O}' \cap \mathcal{S}_e$ their dimensions satisfy

$$\dim C = \dim \mathcal{O}' - \dim(G \cdot e).$$

In particular, they are even and are equal for all components of a fixed orbit intersection.

Step 2. In this step, we identify the unique dense connected leaf and prove the codimension bound for its boundary. The closures of the finitely many leaves cover the irreducible variety X . One of these closures must be X . Denote the corresponding leaf by L_0 and its orbit by $\mathcal{O}_{\text{top}}$. A dense locally closed subset is open in its closure, so L_0 is open dense in X . Two distinct leaves cannot both be open dense, since nonempty open subsets of an irreducible variety intersect.

If another connected component of $\mathcal{O}_{\text{top}} \cap \mathcal{S}_e$ met X , the saturation just proved would put all of it in X . The dimension formula gives it the same dimension as L_0 , hence as X . Its closure would be X , making it a second open dense leaf. This is impossible. Thus

$$X \cap \mathcal{O}_{\text{top}} = L_0 = L.$$

This also proves uniqueness of $\mathcal{O}_{\text{top}}$. The set L is smooth and symplectic, and it is connected both as a leaf and as an open subset of the irreducible variety X .

Every other leaf contained in X has dimension strictly less than $\dim X$; equality would again make it open dense. Since both its dimension and $\dim X = \dim L$ are even, its dimension is at most $\dim X - 2$. The complement $X \setminus L$ is the finite union of these leaves. Its dimension is therefore at most $\dim X - 2$, which proves the required codimension bound.

Step 3. In this step, we pass the dense leaf to the normalization and compute its symplectic Kazhdan weight. The open subset $L \subset X$ is smooth and hence normal. Normalization commutes with restriction to an open subset, so $\nu : M = \nu^{-1}(L) \rightarrow L$ is an isomorphism. Any irreducible component of $Z \setminus M$ maps finitely onto a subvariety of $X \setminus L$, and thus has dimension at most $\dim X - 2$. Since $\dim Z = \dim X$, the complement of M also has codimension at least two. In particular, M is smooth and connected.

Write Ω_L for the transverse symplectic form on L and $\Omega_M = \nu^* \Omega_L$. These are closed nondegenerate algebraic two-forms. For any nilpotent adjoint orbit $\mathcal{O}'$ the KKS formula is

$$(\Omega_{\mathcal{O}'})_x([u, x], [v, x]) = \langle x, [u, v] \rangle.$$

For $c \in \mathbb{C}^*$ it gives

$$(\Omega_{\mathcal{O}'})_{cx}([u, cx], [v, cx]) = c (\Omega_{\mathcal{O}'})_x([u, x], [v, x]).$$

Inner automorphisms preserve this form. The raw Kazhdan action is an inner automorphism followed by scalar multiplication by t^2 , so it multiplies the KKS form by t^2 . It preserves the slice and every nilpotent orbit. Proposition 3.4 also gives invariance of X and its normalization. Hence L and M are invariant, and restriction to the transverse leaf gives

$$\kappa_t^* \Omega_L = t^2 \Omega_L, \quad \kappa_t^* \Omega_M = t^2 \Omega_M.$$

Step 4. In this step, we prove the scalar-orbit assertions and extend both symplectic forms reflexively. For a nonzero nilpotent orbit $\mathcal{O}$, the orbit is smooth and locally closed, hence open in $\overline{\mathcal{O}}$. Normalization is therefore an isomorphism over it. Its boundary is a finite union of strictly smaller nilpotent orbits. Their KKS forms make their dimensions even, so each has dimension at most $\dim \mathcal{O} - 2$. Finiteness of normalization preserves the resulting codimension bound upstairs. Thus $\mathcal{O}$ is a smooth big open in its normalized closure Y . The scalar invariance proved in Proposition 3.4 holds on this open, and the displayed KKS calculation gives scalar weight one.

It remains to extend the forms. For the inclusion $j : M \hookrightarrow Z$, put $\Omega_Z^{[2]} = (\Omega_Z^2)^{**}$. A reflexive sheaf on a normal variety is determined by its stalks at codimension-one points, viewed inside its generic fiber. The open M contains every such point, and on M this sheaf is the locally free sheaf Ω_M^2 . Consequently

$$\Omega_Z^{[2]} = j_* \Omega_M^2.$$

The form Ω_M therefore determines a unique reflexive two-form on Z . Pullback by κ_t and multiplication by t^2 give sections that agree on M , so uniqueness preserves the weight on Z . Applying the same argument to $\mathcal{O} \hookrightarrow Y$ gives the unique reflexive extension of its KKS form, with scalar weight one. This completes both assertions. $\square$

4. GLOBAL POTENTIALS ON FINITE-DIMENSIONAL QUOTIENTS

We retain the curvature and hyperkähler conventions of Notation 2.3. We first construct the half-norm potential on an affine quotient, including its singular strata. We then apply this construction to the doubled image flag of a nilpotent matrix. The trace calculation fixes the symplectic normalization, while the distinction between arrow scaling and matrix scaling fixes the weight of the potential.

Theorem 4.1. *Let a connected compact group K_c act unitarily on a finite-dimensional Hermitian space E , with complexification G . For the standard half-squared-norm moment map, inclusion of its zero level induces the affine Kempf–Ness homeomorphism with $\text{Spec } \mathbb{C}[E]^G$ in the trivial-character case. Kähler reduction descends invariant local potentials to continuous plurisubharmonic potentials on the analytic quotient. This applies on the full quotient for an invariant strictly plurisubharmonic potential when its analytic semistable set is all of E .*

We use the affine Kempf–Ness theorem in [Hos14, Theorem 4.2], with trivial character, and the potential construction of [GH10, Theorem 1.2 and Proposition 1.1]. The semistable-set hypothesis will be checked both for the half-norm and for its regularized logarithm.

Proposition 4.2. *Let E, K_c, G be as above, let $C \subset E$ be a closed G -invariant complex cone, and let $Q = \text{Spec } \mathbb{C}[C]^G$. The function $\frac{1}{2}\|z\|^2$ on $C \cap \mu_{\mathbb{R}}^{-1}(0)$ descends to a continuous proper nonnegative plurisubharmonic function ρ_Q on Q , of scalar modulus weight two, vanishing exactly at the quotient origin. If Q is positive-dimensional and normal, $\log \rho_Q$, extended by $-\infty$ at the origin, is plurisubharmonic. These assertions require no freeness assumption.*

In particular, the assertions concern the function on the entire affine quotient, including points with positive-dimensional stabilizers.

Proof. We descend the potential on the full ambient quotient, restrict it to the closed cone quotient, and obtain the logarithm by a decreasing limit of strictly plurisubharmonic potentials.

Step 1. In this step, we descend the half-norm to a continuous proper plurisubharmonic function on the full ambient quotient. Put

$$\phi(z) = \frac{1}{2}\|z\|^2, \quad A = \text{Spec } \mathbb{C}[E]^G, \quad L = \mu_{\mathbb{R}}^{-1}(0).$$

We normalize the moment map associated to ϕ to vanish at the origin. In unitary coordinates,

$$\text{dd}^c \phi = \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j.$$

For the trivial character every point is semistable: the invariant constant 1 is nonzero everywhere. By Theorem 4.1, every complex orbit closure therefore meets L , and the unique closed complex orbit in a quotient fibre meets L in one K_c -orbit. Thus $L/K_c \simeq A$, and the K_c -invariant function $\phi|_L$ descends to a continuous function ρ_A .

The analytic semistable set for this moment map is

$$\{z \in E : \overline{Gz} \cap L \neq \emptyset\} = E.$$

The potential ϕ is invariant, strictly plurisubharmonic and exhaustive on every closed quotient fibre. The potential-reduction part of Theorem 4.1 applies on all of E and makes ρ_A plurisubharmonic across every orbit-type stratum. For $R \geq 0$ its sublevel is the image of the compact set

$$L \cap \{\|z\|^2 \leq 2R\}.$$

Since A is Hausdorff, a closed inverse image of a compact subset of $[0, \infty)$ is closed in one of these compact sublevels. This proves properness.

Step 2. In this step, we restrict the descended potential to Q and verify its homogeneity and zero set. Let I_C be the ideal of C . Exactness of invariants for the reductive group G , applied degree by degree, gives the surjection

$$\mathbb{C}[E]^G \longrightarrow (\mathbb{C}[E]/I_C)^G = \mathbb{C}[C]^G.$$

Hence Q is a closed subvariety of A . If an orbit lies in C , its closure and its closed orbit lie in C as well. The preceding Kempf–Ness identification consequently restricts to

$$(C \cap L)/K_c \simeq Q.$$

The descended function is $\rho_Q = \rho_A|_Q$. It is continuous and plurisubharmonic, and restriction to the closed subset Q preserves properness. The real moment map is quadratic, so $\mu_{\mathbb{R}}(\lambda z) = |\lambda|^2 \mu_{\mathbb{R}}(z)$. Scalar multiplication therefore acts on this zero-level quotient and gives

$$\rho_Q(\lambda q) = |\lambda|^2 \rho_Q(q).$$

Moreover, $\phi(z) = 0$ if and only if $z = 0$. Thus ρ_Q vanishes precisely at the quotient origin, which we denote by q_0 .

Step 3. In this step, we prove global plurisubharmonicity of $\log \rho_Q$ by descending regularized logarithms. For $\epsilon > 0$ set $\phi_\epsilon = \log(\phi + \epsilon)$ on E . Let $\mathcal{L}_\phi$ denote the Hermitian Levi form of ϕ . Since

$$|\partial\phi|_{\mathcal{L}_\phi}^2 = \phi,$$

Cauchy–Schwarz gives, for each complex tangent vector v ,

$$\begin{aligned} \mathcal{L}_{\phi_\epsilon}(v, \bar{v}) &= \frac{\mathcal{L}_\phi(v, \bar{v})}{\phi + \epsilon} - \frac{|\partial\phi(v)|^2}{(\phi + \epsilon)^2} \\ &\geq \frac{\epsilon}{(\phi + \epsilon)^2} \mathcal{L}_\phi(v, \bar{v}). \end{aligned}$$

Thus ϕ_ϵ is smooth, invariant and strictly plurisubharmonic. The formula $d^c \phi_\epsilon = (\phi + \epsilon)^{-1} d^c \phi$ gives its associated moment map, with the same zero normalization:

$$\mu_\epsilon = (\phi + \epsilon)^{-1} \mu_{\mathbb{R}}, \quad \mu_\epsilon^{-1}(0) = L.$$

Its analytic semistable set is still all of E , since this set is defined by the orbit-closure condition of meeting the zero level. Also ϕ_ϵ tends to infinity with $\|z\|$, so it is exhaustive on each closed quotient fibre. Theorem 4.1 therefore descends it to the plurisubharmonic function $\log(\rho_A + \epsilon)$ on the same analytic quotient A . Restriction gives $\log(\rho_Q + \epsilon)$ on Q .

As ϵ decreases to zero, these functions decrease to $\log \rho_Q$ off q_0 and to $-\infty$ at q_0 . The limit is upper semicontinuous. On an analytic disc the decreasing subharmonic functions have a subharmonic limit, allowing the identically $-\infty$ limit for a disc mapped to q_0 . On a nonconstant disc the limit is finite at every point mapped away from q_0 . The positive-dimensional normal cone has no component supported only at q_0 : its scalar contraction makes it connected, and the irreducible components of a normal space are disjoint. Thus the limit is not identically $-\infty$ on a component. The disc criterion proves the asserted global plurisubharmonicity. $\square$

We now specify the matrix quotient and its flat normalization.

Definition 4.3. Let $a \in \mathfrak{sl}_N(\mathbb{C})$ be nonzero nilpotent of nilpotency index k . Set $V_i = \text{Im}(a^{k-i})$ for $0 \leq i \leq k$, so $V_0 = 0$ and $V_k = \mathbb{C}^N$, and give these spaces the Hermitian metrics induced from $\mathbb{C}^N$. The *doubled image-flag space* is

$$W_A = \bigoplus_{i=0}^{k-1} (\text{Hom}(V_i, V_{i+1}) \oplus \text{Hom}(V_{i+1}, V_i)).$$

Write its coordinates as $(\alpha_i, \beta_i)_{i=0}^{k-1}$, and put

$$K_A = \prod_{i=1}^{k-1} \mathrm{U}(V_i), \quad G_A = \prod_{i=1}^{k-1} \mathrm{GL}(V_i).$$

With g_0 and g_k the identity maps, change of basis acts by

$$\alpha_i \mapsto g_{i+1} \alpha_i g_i^{-1}, \quad \beta_i \mapsto g_i \beta_i g_{i+1}^{-1}.$$

Let W_A° be the open locus where all α_i are injective and all β_i are surjective. The terminal product is

$$p_A(\alpha, \beta) = \alpha_{k-1} \beta_{k-1} \in \mathrm{End}(V_k).$$

The Hermitian norm and the flat hyperkähler structure on this space are fixed by

$$\begin{aligned} \|w\|_{\mathrm{HS}}^2 &= \sum_{i=0}^{k-1} \mathrm{Tr}(\alpha_i^* \alpha_i + \beta_i \beta_i^*), & \phi_A(w) &= \frac{1}{2} \|w\|_{\mathrm{HS}}^2, \\ I(\alpha_i, \beta_i) &= (i\alpha_i, i\beta_i), & J(\alpha_i, \beta_i) &= (-\beta_i^*, \alpha_i^*), & K &= IJ, \\ \Omega_{W_A} &= \omega_J + i\omega_K = \sum_{i=0}^{k-1} \mathrm{Tr}(d\alpha_i \wedge d\beta_i). \end{aligned}$$

Here the trace of the wedge is evaluated by composition in $\mathrm{End}(V_{i+1})$. The simultaneous complex and real zero equations are, for $1 \leq i < k$,

$$(4.1) \quad \begin{aligned} &\alpha_{i-1} \beta_{i-1} - \beta_i \alpha_i = 0, \\ &\alpha_{i-1} \alpha_{i-1}^* - \beta_{i-1}^* \beta_{i-1} + \beta_i \beta_i^* - \alpha_i^* \alpha_i = 0. \end{aligned}$$

We denote the hyperkähler moment map by μ_A , its complex component by $\mu_{A, \mathbb{C}}$, and the common zero set by L_A . The second equation specifies the real zero set independently of the identification of the dual of $\mathfrak{u}(V_i)$ with Hermitian matrices.

Theorem 4.4. *Complex matrix conjugacy-class closures are normal. The normalization of a nilpotent orbit closure in a complex simple Lie algebra has symplectic singularities. For a nonzero type-A nilpotent orbit, the doubled image-flag zero hyperkähler quotient is homeomorphic to its full closure, and its injective-forward, surjective-backward open quotient is diffeomorphic to the orbit with its natural complex symplectic structure. Under a free proper hyperkähler reduction compatible with the quaternionic scalar action, the flat hyperkähler potential descends.*

The normality assertion is [KP79, Section 0.1, p. 227], and the symplectic-singularity assertion is [Bea00, Section 2.6]. The quotient and potential statements are [KS01, Section 2.1, equations (2.2)–(2.5), and Theorem 2.1]. We use the full-quotient identification as a homeomorphism. The algebraic identification and the constants in our curvature convention are checked below.

Theorem 4.5. *Bedford–Taylor products of locally bounded plurisubharmonic functions on a complex manifold do not charge pluripolar subsets. On a smooth projective variety, a semipositive Hermitian metric with locally bounded weights on a rational line bundle has top Monge–Ampère mass equal to the top self-intersection of that line bundle.*

The noncharging assertion is [DS10, Appendix A, Corollary A.32]; the equality of total mass and intersection number is [BGH⁺22, Lemma 2.24]. The first assertion applies to the smooth resolution used below.

Theorem 4.6. *Let $a \in \mathfrak{sl}_N(\mathbb{C})$ be nonzero nilpotent, and let $Y_A = \overline{\mathrm{SL}_N \cdot a}$. The doubled image-flag quotient gives a homogeneous hyperkähler potential package of weight one on Y_A , whose smooth big open is exactly $\mathrm{SL}_N \cdot a$. The potential is the descended half-squared Hilbert–Schmidt norm. Its holomorphic symplectic form is exactly*

$$\Omega_{A,x}([\xi, x], [\eta, x]) = \mathrm{Tr}(x[\xi, \eta]).$$

The associated top Monge–Ampère measure gives no mass to the boundary of the orbit.

The weight here refers to ordinary scalar multiplication of matrices. Simultaneous scalar multiplication of the arrows has weight two for the half-norm and induces the square of that scalar on matrices.

Proof. We identify the affine quotient with the normal matrix closure, determine its smooth locus, and then descend the flat metric and potential with their exact normalizations. Finally, we compute the boundary mass on a smooth resolution.

Step 1. In this step, we identify the affine quotient algebraically with the normal matrix orbit closure. Write $\mathcal{O}_A = \mathrm{SL}_N \cdot a$ and use Definition 4.3. The GL_N - and SL_N -orbits of a coincide. Indeed, for a conjugator $b \in \mathrm{GL}_N$ choose $c \in \mathbb{C}^*$ with $c^N \det b = 1$; then cb gives the same conjugation and lies in SL_N . Thus Theorem 4.4 makes Y_A normal.

Set

$$C_A = (\mu_{A,\mathbb{C}}^{-1}(0))_{\mathrm{red}}, \quad Q_A = \mathrm{Spec} \mathbb{C}[C_A]^{G_A}.$$

The first equation of (4.1) defines a closed G_A -invariant complex cone. The affine Kempf–Ness identification gives $Q_A \simeq L_A/K_A$. The invariant polynomial p_A therefore induces a morphism $\bar{p}_A : Q_A \rightarrow Y_A$ whose underlying analytic map is the homeomorphism of Theorem 4.4.

We check that this morphism is an algebraic isomorphism; a homeomorphism alone would not transport plurisubharmonicity. Let $R = \mathbb{C}[Q_A]$, graded by arrow degree, and view $S = \mathbb{C}[Y_A]$ as its subalgebra via $\bar{p}_A^*$. Give matrix coordinates degree two, so this inclusion respects the gradings. The ring R is reduced and $R_0 = \mathbb{C}$. Its open subset over $\mathcal{O}_A$ is irreducible and dense, by the open-orbit identification and the homeomorphism, so R is a domain. Let $f_1, \dots, f_m$ be the pullbacks of matrix coordinates. The fibre over zero consists only of the quotient origin. Hence

$$\sqrt{(f_1, \dots, f_m)R} = R_{>0},$$

and $R/(f_1, \dots, f_m)R$ is finite-dimensional. Choose homogeneous representatives of a vector-space basis of this quotient. They generate R as an S -module: subtracting their constant linear combination from a homogeneous element leaves $\sum_j f_j r_j$ with each r_j of smaller degree, and induction on degree applies. Thus $\bar{p}_A$ is finite. Its generic fibre has one point; in characteristic zero this makes the extension of function fields have degree one. Since S is integrally closed, the integral extension $S \subset R$ inside that common function field is equality. This proves the required isomorphism.

Step 2. In this step, we prove that the smooth locus of Y_A is exactly $\mathcal{O}_A$. By Theorem 4.4, Y_A has symplectic singularities. On $(Y_A)_{\mathrm{reg}}$ the inverse of the symplectic form extending the orbit form agrees with the Lie–Poisson bivector on the dense open $\mathcal{O}_A$. The two holomorphic bivectors consequently agree everywhere on $(Y_A)_{\mathrm{reg}}$. At a point y , the rank of the Lie–Poisson bivector is the dimension of its adjoint orbit. Every orbit in $Y_A \setminus \mathcal{O}_A$ has dimension smaller than $\dim Y_A$, so none can be a smooth point of Y_A . Conversely, $\mathcal{O}_A$ is a smooth open homogeneous variety. It follows that

$$(Y_A)_{\mathrm{reg}} = \mathcal{O}_A.$$

Normality then gives $\mathrm{codim}_{Y_A}(Y_A \setminus \mathcal{O}_A) \geq 2$.

Step 3. In this step, we verify free reduction on the orbit and compute the exact trace KKS form. The action of K_A is proper because K_A is compact. To check freeness on W_A° , suppose $(g_1, \dots, g_{k-1})$ fixes a point and set $g_k = 1$. The equality $\alpha_{k-1} g_{k-1}^{-1} = \alpha_{k-1}$ and injectivity of α_{k-1} give $g_{k-1} = 1$. The equation $g_{i+1} \alpha_i g_i^{-1} = \alpha_i$ gives the same conclusion successively for every internal vertex. The free zero-level quotient

$$M_A = (W_A^\circ \cap L_A)/K_A$$

therefore carries the reduced positive hyperkähler metric and is identified with $\mathcal{O}_A$ by the terminal product.

For the last pair put $A = \alpha_{k-1}$ and $B = \beta_{k-1}$. The residual $\mathrm{GL}(V_k)$ -action is $(A, B) \mapsto (gA, Bg^{-1})$. For $\xi \in \mathfrak{gl}(V_k)$ its fundamental vector is $(\xi A, -B\xi)$. For an arbitrary variation (C, D) , the flat holomorphic symplectic form gives

$$\begin{aligned}\Omega_{W_A}((\xi A, -B\xi), (C, D)) &= \mathrm{Tr}(\xi AD + CB\xi) \\ &= d\mathrm{Tr}(AB\xi)(C, D).\end{aligned}$$

All other arrow components of this fundamental vector vanish. Since $d(AB)(\eta A, -B\eta) = [\eta, AB]$, we obtain

$$\Omega_{W_A}(\xi^\#, \eta^\#) = \mathrm{Tr}([\eta, AB]\xi) = \mathrm{Tr}(AB[\xi, \eta]).$$

These calculations take place on the complex moment level. The residual $\mathrm{GL}(V_k)$ -action preserves that level and commutes with G_A , and the functions $\mathrm{Tr}(p_A\xi)$ are G_A -invariant. Thus the identities descend to the holomorphic symplectic quotient identified in Theorem 4.4. The fundamental vectors span the tangent space of the orbit, so the reduced form is exactly the stated trace KKS form, with coefficient +1. This form is closed, nondegenerate and algebraic on $\mathcal{O}_A$.

Step 4. In this step, we descend the half-norm as a hyperkähler potential and compute its gradient. In unitary linear coordinates on W_A ,

$$\mathrm{dd}^c \phi_A = \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j = \omega_{I, W_A}, \quad \nabla \phi_A = \mathcal{R}, \quad |\mathcal{R}|^2 = 2\phi_A,$$

where $\mathcal{R}_w = w$ is the real radial vector. The corresponding identities hold for J and K as well. The group K_A commutes with the quaternionic scalar action and preserves the half-norm. That action rotates the three quadratic moment components and preserves their common zero set, as required in the potential-descent assertion of Theorem 4.4.

More explicitly, homogeneity of the moment components makes $\mathcal{R}$ tangent to L_A . Invariance of ϕ_A gives $g(\mathcal{R}, \xi^\#) = d\phi_A(\xi^\#) = 0$, so $\mathcal{R}$ is horizontal on the free locus. The horizontal space is preserved by I, J, K . Let ι denote the inclusion of $W_A^\circ \cap L_A$ in W_A , and let $\pi : W_A^\circ \cap L_A \rightarrow M_A$ be the quotient map. The descended function Φ_A thus satisfies

$$\pi^* d_q^c \Phi_A = \iota^* d_q^c \phi_A \quad (q = I, J, K).$$

Taking exterior derivatives proves $dd_q^c \Phi_A = \omega_{q, A}$. In particular,

$$\mathrm{dd}^c \Phi_A = \omega_{I, A}, \quad |\nabla \Phi_A|^2 = 2\Phi_A \quad \text{on } \mathcal{O}_A.$$

Here the normalization comes from the displayed flat calculation, without a change in the Hilbert–Schmidt norm. The residual $\mathrm{SU}(N)$ -action preserves the metric and potential. The quaternionic map J on the arrows induces $x \mapsto -x^*$ on the orbit.

Step 5. In this step, we extend the potential to the full closure and compute its weight under matrix scaling. Apply Proposition 4.2 to $C_A \subset W_A$ and transport its descended function through the algebraic isomorphism $\bar{p}_A$. This gives the same half-norm function Φ_A on the full closure. It is continuous, proper and nonnegative, its zero set is $\{0\}$, and both Φ_A and its extended logarithm are plurisubharmonic. This argument applies to all zero-level strata, without a freeness assumption there.

Simultaneous multiplication of all arrows by $\lambda \in \mathbb{C}^*$ multiplies the complex moment equations by λ^2 and the real ones by $|\lambda|^2$. It preserves L_A and gives

$$p_A(\lambda w) = \lambda^2 p_A(w), \quad \phi_A(\lambda w) = |\lambda|^2 \phi_A(w), \quad \lambda^* \Omega_{W_A} = \lambda^2 \Omega_{W_A}.$$

For an ordinary matrix scalar t , choose λ with $\lambda^2 = t$. The quotient identification makes the induced action independent of this choice, and hence

$$\Phi_A(tx) = |t| \Phi_A(x), \quad t^* \Omega_A = t \Omega_A.$$

Equivalently, the real radial vector on the arrows projects to twice the logarithmic generator of positive matrix scaling. Thus, if that matrix generator is E_A , then $\nabla \Phi_A = 2E_A$. The

matrix action is effective and free off zero, since $tx = x \neq 0$ forces $t = 1$. These conclusions verify all parts of the weight-one potential package in Definition 2.4 and the exact symplectic normalization.

Step 6. In this step, we prove that the top Monge–Ampère measure gives zero mass to the orbit boundary. Write $\dim_{\mathbb{C}} Y_A = 2n$ and choose a proper resolution $\tau : \tilde{Y}_A \rightarrow Y_A$ which is an isomorphism over $\mathcal{O}_A$, and set $u = \Phi_A \circ \tau$. The function u is continuous and plurisubharmonic, hence locally bounded. Its Bedford–Taylor top product is a locally finite positive measure on the smooth space $\tilde{Y}_A$. The set $\tau^{-1}(Y_A \setminus \mathcal{O}_A)$ is a proper analytic subset and is locally pluripolar. Theorem 4.5 therefore gives

$$(\mathrm{dd}^c u)^{2n}(\tau^{-1}(Y_A \setminus \mathcal{O}_A)) = 0.$$

We identify its push-forward with the product on Y_A rather than applying the smooth-space theorem directly to the singular closure. Regard the products as currents on a local ambient embedding. Proper birationality gives $\tau_*[\tilde{Y}_A] = [Y_A]$. If the push-forward identity holds for the j th Bedford–Taylor product, its recursive definition gives

$$\begin{aligned} \tau_*((\mathrm{dd}^c u)^{j+1}) &= \tau_* \mathrm{dd}^c(u(\mathrm{dd}^c u)^j) \\ &= \mathrm{dd}^c(\Phi_A \tau_*((\mathrm{dd}^c u)^j)) = (\mathrm{dd}^c \Phi_A)^{j+1}. \end{aligned}$$

Here proper push-forward commutes with differentiation and with multiplication by a continuous function pulled back from the target. Induction, starting with the integration currents, proves

$$\tau_*\left(\frac{(\mathrm{dd}^c u)^{2n}}{(2n)!}\right) = \frac{(\mathrm{dd}^c \Phi_A)^{2n}}{(2n)!}.$$

Properness makes this push-forward locally finite. Its value on the orbit boundary is the value upstairs on $\tau^{-1}(Y_A \setminus \mathcal{O}_A)$, which is zero. This proves the last assertion. $\square$

Lemma 4.7. *Let (M, g, I, J, K) be positive hyperkähler of real dimension $4n$, oriented by I , and put $\Omega = \omega_J + i\omega_K$. Then*

$$\frac{\omega_I^{2n}}{(2n)!} = \frac{\Omega^n \wedge \bar{\Omega}^n}{4^n(n!)^2}.$$

This identity fixes the volume coefficient for the metric and its own holomorphic symplectic form.

Proof. The calculation is pointwise. Choose an orthonormal quaternionic basis of the tangent space and corresponding I -complex covectors $dz_1, dw_1, \dots, dz_n, dw_n$. They give

$$\omega_I = \frac{i}{2} \sum_{j=1}^n (dz_j \wedge d\bar{z}_j + dw_j \wedge d\bar{w}_j), \quad \Omega = \sum_{j=1}^n dz_j \wedge dw_j.$$

Put

$$\nu_j = \left(\frac{i}{2} dz_j \wedge d\bar{z}_j\right) \wedge \left(\frac{i}{2} dw_j \wedge d\bar{w}_j\right).$$

With the I -complex orientation,

$$\frac{\omega_I^{2n}}{(2n)!} = \bigwedge_{j=1}^n \nu_j, \quad \Omega^n = n! \bigwedge_{j=1}^n (dz_j \wedge dw_j).$$

The holomorphic and antiholomorphic two-form blocks commute. For each quaternionic pair, moving $d\bar{z}_j$ past dw_j gives

$$dz_j \wedge dw_j \wedge d\bar{z}_j \wedge d\bar{w}_j = -dz_j \wedge d\bar{z}_j \wedge dw_j \wedge d\bar{w}_j = 4\nu_j.$$

Consequently

$$\Omega^n \wedge \bar{\Omega}^n = 4^n(n!)^2 \bigwedge_{j=1}^n \nu_j = 4^n(n!)^2 \frac{\omega_I^{2n}}{(2n)!},$$

which proves the formula. $\square$

On the matrix orbit the preceding calculation gives, in the fixed curvature convention and with $\dim_{\mathbb{C}} Y_A = 2n$,

$$\frac{(\mathrm{dd}^c \Phi_A)^{2n}}{(2n)!} = \frac{\Omega_A^n \wedge \bar{\Omega}_A^n}{4^n (n!)^2}.$$

Extend the measure on the right by zero over $Y_A \setminus \mathcal{O}_A$. Theorem 4.6 shows that the measure on the left also gives that boundary zero mass. Thus this equality holds on all of Y_A , and the extended symplectic-volume measure is locally finite.

5. FRAMED TRAJECTORIES AND NATURAL CHARTS

We use the hyperkähler conventions fixed above. For a compact group K_c , we write $\mathfrak{k}$ for its Lie algebra. We next prepare the comparison maps used to pull type-A potentials back to general orbit closures and slices. We identify the trajectory space with framed invariant connections, keeping track of the endpoint estimates and the two framings. We then compute the effect of dilation in the orbit–slice charts and prove that these charts commute with faithful unitary representations.

5.1. Temporal normalization and the negative frame.

Definition 5.1. Let K_c be a compact connected semisimple matrix group and $p_{\pm} : \mathrm{SU}(2) \rightarrow K_c$ homomorphisms. Choose an oriented basis (e_1, e_2, e_3) with $[e_b, e_c] = -2e_a$ cyclically, and set $\tau_a^{\pm} = dp_{\pm}(e_a)$. The *positive-framed trajectory space* $\mathcal{M}(p_-, p_+)$ consists of smooth triples $T : \mathbb{R} \rightarrow \mathfrak{k}^3$ satisfying

$$T'_a = -2T_a - [T_b, T_c], \quad T_a - \tau_a^+ = O(e^{-2t}) \ (t \rightarrow +\infty), \quad T_a - \mathrm{Ad}_g \tau_a^- = O(e^{2t}) \ (t \rightarrow -\infty)$$

for some $g \in K_c$, with the same bounds for every temporal derivative. The positive frame is fixed, the negative frame g is forgotten, and these terminal temporal representatives undergo no further quotient. Equivalently, in covariant quadruples the gauges have positive limit 1, negative limit in the simultaneous centralizer of τ^- , and all-derivative rate-two convergence at both ends.

The negative endpoint determines a conjugate of τ^- , but does not specify a conjugating element. We make this distinction explicit in the covariant description. Every asymptotic estimate below includes derivative order zero.

Definition 5.2. For the endpoint data of Definition 5.1, put

$$Z_{\pm} = \{z \in K_c : \mathrm{Ad}_z \tau_a^{\pm} = \tau_a^{\pm} \text{ for } a = 1, 2, 3\}.$$

The *covariant trajectory space* $\mathcal{C}$ consists of smooth quadruples $Q = (Q_0, Q_1, Q_2, Q_3) : \mathbb{R} \rightarrow \mathfrak{k}^4$ satisfying

$$Q'_a + [Q_0, Q_a] = -2Q_a - [Q_b, Q_c]$$

cyclically, and, for every integer $m \geq 0$,

$$\begin{aligned} \partial_t^m Q_0, \partial_t^m (Q_a - \tau_a^-) &= O(e^{2t}) \quad (t \rightarrow -\infty), \\ \partial_t^m Q_0, \partial_t^m (Q_a - \tau_a^+) &= O(e^{-2t}) \quad (t \rightarrow +\infty). \end{aligned}$$

The *positive-based gauge group* $\mathcal{G}_+$ consists of smooth maps $v : \mathbb{R} \rightarrow K_c$ with limits $v_+ = 1$ and $v_- \in Z_-$, satisfying

$$\partial_t^m (v - 1) = O(e^{-2t}), \quad \partial_t^m (v - v_-) = O(e^{2t})$$

at the respective ends for every $m \geq 0$. Its subgroup $\mathcal{G}_{\mathrm{db}}$ has $v_- = 1$ as well. Their action is

$$(v * Q)_0 = \mathrm{Ad}_v Q_0 - v'v^{-1}, \quad (v * Q)_a = \mathrm{Ad}_v Q_a.$$

The *space with a negative frame* is

$$\mathcal{P} = \{(T, q) : T \in \mathcal{M}(p_-, p_+), \ q \in K_c, \ \lim_{t \rightarrow -\infty} T_a(t) = \mathrm{Ad}_q \tau_a^- \text{ for } a = 1, 2, 3\}.$$

Lemma 5.3. *For the endpoint data of Definition 5.1, terminal temporal normalization gives canonical bijections*

$$\mathcal{C}/\mathcal{G}_{\text{db}} \simeq \mathcal{P}, \quad \mathcal{C}/\mathcal{G}_+ \simeq \mathcal{M}(p_-, p_+).$$

The fibers of $\mathcal{P} \rightarrow \mathcal{M}(p_-, p_+)$ are right Z_- -torsors with action $(T, q) \cdot z = (T, qz)$.

This describes precisely which information terminal normalization retains: the temporal triple and, before the negative frame is forgotten, one element of a centralizer torsor.

Proof. We construct the terminal gauge with its derivative bounds, and then identify its behavior under the two gauge groups.

Step 1. In this step, we construct the unique terminal gauge with all required endpoint derivative estimates. For $Q \in \mathcal{C}$, solve

$$(5.1) \quad u' = uQ_0, \quad \lim_{t \rightarrow +\infty} u(t) = 1.$$

Both tails of Q_0 are integrable. To construct u , solve the equation with $u_R(R) = 1$ at a finite terminal point. Its integral equation is

$$u_R(t) = 1 - \int_t^R u_R(s)Q_0(s) ds.$$

Solutions remain in K_c . Gronwall's inequality and the vanishing integral of the tail of Q_0 show that they converge on every compact interval as $R \rightarrow \infty$. The limit solves (5.1), and

$$u(t) - 1 = - \int_t^\infty u(s)Q_0(s) ds = O(e^{-2t}).$$

Compactness of K_c bounds u . Integrability at the other end gives a limit $q \in K_c$ and

$$u(t) - q = \int_{-\infty}^t u(s)Q_0(s) ds = O(e^{2t}).$$

For $m \geq 1$, differentiating the equation gives

$$\partial_t^m u = \sum_{j=0}^{m-1} \binom{m-1}{j} (\partial_t^j u) (\partial_t^{m-1-j} Q_0).$$

Induction proves the same decay for every derivative of $u - 1$ and $u - q$ at the corresponding ends. Two terminal solutions have constant ratio and the same positive limit, so the solution is unique.

Set $T_a = \text{Ad}_u Q_a$. The zeroth component of $u * Q$ vanishes, and direct differentiation gives

$$\frac{d}{dt} \text{Ad}_u Q_a = \text{Ad}_u (Q'_a + [Q_0, Q_a]) = -2T_a - [T_b, T_c].$$

The product rule and the bounds just proved give every derivative bound in Definition 5.1, with negative limit $\text{Ad}_q \tau^-$. Thus normalization produces $(T, q) \in \mathcal{P}$.

Step 2. In this step, we identify the two gauge quotients and the right Z_- -torsor of negative frames. Replacing Q by $v * Q$ replaces its terminal gauge by uv^{-1} . For $v \in \mathcal{G}_{\text{db}}$, this leaves both T and q unchanged. Conversely, if Q and $\tilde{Q}$ have the same pair, their terminal gauges satisfy

$$\tilde{Q} = (\tilde{u}^{-1}u) * Q.$$

The comparison gauge has limit one at both ends and has every required derivative bound, so it belongs to $\mathcal{G}_{\text{db}}$. Given $(T, q) \in \mathcal{P}$, connectedness of K_c gives a smooth path u equal to q on a negative tail and to 1 on a positive tail. Then $Q = u^{-1} * (0, T)$ lies in $\mathcal{C}$ and normalizes to (T, q) . Different interpolations give the same double-based class.

For $v \in \mathcal{G}_+$, normalization leaves T unchanged and replaces q by qv^{-1} . If two quadruples normalize to the same T , their negative limits obey

$$\text{Ad}_q \tau^- = \text{Ad}_{\tilde{q}} \tau^-.$$

Consequently $\tilde{q}^{-1}q \in Z_-$, and their comparison gauge belongs to $\mathcal{G}_+$. This proves the second bijection. Finally, all choices of q for a fixed T differ uniquely by right multiplication by an element of Z_- . An interpolation with negative limit z^{-1} and positive limit 1 realizes the action $(T, q) \mapsto (T, qz)$ on double-based classes. $\square$

The subset of pairs $(T, 1)$ identifies with temporal triples whose negative endpoint is exactly τ^- . An arbitrary pair with that same endpoint may still have any $q \in Z_-$. In particular, $\mathcal{P} = \mathcal{M}(p_-, p_+) \times K_c$ when $p_- = 0$.

5.2. Equivariant compactification and complex charts. We use the following removable-singularity result in its framed form.

Theorem 5.4. *For a compact structure group and smooth background metric, an $L^2_{1,\text{loc}}$ anti-self-dual connection on a trivial bundle over a punctured four-ball with finite curvature energy has smooth removable-extension data. A transition gauge between two such extension data extends continuously across the puncture.*

This is [Pla24, Appendix A, Theorem A.10 and Proposition A.11]. The continuity conclusion concerns the transition between extensions. The next lemma supplies the smoothness needed to extend the symmetry.

Lemma 5.5. *Let K_c be a compact matrix group. The following hold.*

- (i) *If a_1, a_2 are smooth connections on the trivial K_c -bundle over a four-ball and v is a continuous gauge, smooth off the center, with $a_2 = v * a_1$ off the center, then v is smooth and the identity holds on the whole ball.*
- (ii) *Let a compact Lie group L_c act smoothly on a four-ball, fixing its center. Suppose a smooth connection on the trivial K_c -bundle has a prescribed jointly smooth, connection-preserving lifted L_c -action off the center. If each lifted map extends continuously to the central fiber, then the action extends uniquely to a jointly smooth, connection-preserving action on the whole bundle.*

The second assertion includes smooth dependence on the acting group, which is needed to identify the fixed-fiber isotropy homomorphisms.

Proof. We prove the two extension statements in order.

Step 1. In this step, we prove smoothness of the continuous transition gauge. For (i), the gauge identity is

$$dv = va_1 - a_2v.$$

Its right-hand side is continuous across the center. Integrating along coordinate segments, including segments through the center, shows that v is C^1 and that the equation holds there. Repeated differentiation then proves C^m regularity for every m .

Step 2. In this step, we extend the lifted action smoothly, including its dependence on the group parameter. For (ii), fix $\ell \in L_c$ and write its punctured lift as S_ℓ . Its gauge factor compares the smooth connection with its pullback by the smooth base action. Part (i) makes S_ℓ smooth across the central fiber. The group law holds there by continuity from the punctured ball. To prove joint smoothness, let $\varpi(\ell)$ denote the action on the central fiber. Fix a noncentral point x and a path γ_x from the center to x . Connection preservation gives, for a point p in the central fiber,

$$\varpi(\ell)p = \text{PT}_{\ell\gamma_x}^{-1}(S_\ell(\text{PT}_{\gamma_x}p)).$$

The right-hand side is smooth in (ℓ, p) , since the prescribed action at x is jointly smooth and parallel transport depends smoothly on parameters. Hence ϖ is smooth. For radial paths γ_y depending smoothly on y , connection preservation now gives

$$S_\ell(\text{PT}_{\gamma_y}p) = \text{PT}_{\ell\gamma_y}(\varpi(\ell)p).$$

This proves joint smoothness also at $y = 0$. Agreement on the dense punctured bundle proves uniqueness. $\square$

We specify the complex endpoint notation before stating the chart theorem. For the chosen oriented decomposition, set

$$H_{\pm} = i\tau_1^{\pm}, \quad X_{\pm} = \frac{-\tau_2^{\pm} + i\tau_3^{\pm}}{2}, \quad Y_{\pm} = \frac{\tau_2^{\pm} + i\tau_3^{\pm}}{2}.$$

The bracket convention gives $[H_{\pm}, X_{\pm}] = 2X_{\pm}$, $[H_{\pm}, Y_{\pm}] = -2Y_{\pm}$, and $[X_{\pm}, Y_{\pm}] = H_{\pm}$. Thus, for the connected complexification of K_c , $\mathcal{O}(p_{\pm})$ is the adjoint orbit of $Y_{\pm}$, and

$$\mathcal{S}(p_{+}) = Y_{+} + (\mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C})^{X_{+}}.$$

For a covariant quadruple, T_0 denotes its temporal component; for a terminal temporal triple we set $T_0 = 0$.

Theorem 5.6. *For compact connected semisimple K_c and endpoints $p_{\pm}$, Kronheimer's positive-framed solution space has a natural hyperkähler structure. For each oriented decomposition $\mathbb{R}^3 = \mathbb{R} \oplus \mathbb{C}$, its associated complex chart is a biholomorphism onto $\mathcal{O}(p_{-}) \cap \mathcal{S}(p_{+})$, possibly empty or disconnected. In complex variables $\alpha = (T_0 + iT_1)/2$, $\beta = (T_2 + iT_3)/2$, the unique positive-normalizing complex gauge has positive limit one and gives*

$$2\alpha = H_{+}, \quad \beta(t) = Y_{+} + e^{-2t} \text{Ad}(e^{-tH_{+}})\varepsilon, \quad [X_{+}, \varepsilon] = 0.$$

The chart value is $Y_{+} + \varepsilon$; its membership in the negative orbit uses the two-ended classification. The complex structures from the oriented decompositions form the natural twistor sphere.

Here the chart and hyperkähler structure are those of [Kro90, Theorem 1 and Remark (ii)]. The complex equation and its gauge action are

$$\beta' + 2\beta + 2[\alpha, \beta] = 0, \quad \alpha \mapsto \text{Ad}_v \alpha - \frac{1}{2}v'v^{-1}, \quad \beta \mapsto \text{Ad}_v \beta.$$

The positive normal form is [Kro90, Section 5, Lemma 11]. The negative normal form of [Kro90, Section 5, Lemma 10] and the classification in [Kro90, Section 4, Proposition 7] put its value in $\mathcal{O}(p_{-})$. The real trajectories represent these complex classes by [Kro90, Section 4, Proposition 8]. These have separate roles: the positive-normalizing gauge itself is not required to satisfy a negative-end framing condition.

Proposition 5.7. *The space $\mathcal{M}(p_{-}, p_{+})$ is canonically identified with the $\text{SU}(2)$ -invariant anti-self-dual connections on S^4 with endpoint isotropies p_{-}, p_{+} , retaining the positive frame and forgetting the negative frame. Radial compactification, terminal temporal normalization and their inverses preserve the all-derivative rate-two endpoint conditions. The resulting charts and smooth hyperkähler structures are those of Theorem 5.6. Retaining a negative frame instead gives a centralizer torsor over this space.*

The connection space in this statement consists of smooth equivariant bundles with invariant connections, up to equivariant connection-preserving isomorphism preserving the positive frame. The negative isotropy is specified up to conjugacy. A negative frame is additional data identifying it with the fixed homomorphism p_{-} .

Proof. We first compactify the cylinder connection and extend its symmetry. Radial parallel transport then identifies the isotropies and gives the endpoint bounds. Finally we check the frames and the inverse construction.

Step 1. In this step, we identify the trajectory equation with the radial ASD equation. Identify $\mathbb{R}^4 \setminus \{0\}$ with $\mathbb{R} \times \text{SU}(2)$ by $(t, u) \mapsto e^t u$. Let θ_a be the left-invariant coframe dual to e_a . The bracket normalization gives

$$d\theta_a = 2\theta_b \wedge \theta_c$$

cyclically. Give the cylinder its product metric and orientation $dt \wedge \theta_1 \wedge \theta_2 \wedge \theta_3$. The connection $\nabla_T = d + \sum_a T_a(t)\theta_a$ is invariant under left multiplication on $SU(2)$, and

$$(5.2) \quad F_{\nabla_T} = \sum_a T'_a dt \wedge \theta_a + \sum_{\text{cyclic } (a,b,c)} (2T_a + [T_b, T_c])\theta_b \wedge \theta_c.$$

Since $*(dt \wedge \theta_a) = \theta_b \wedge \theta_c$, the equation $F_{\nabla_T} + *F_{\nabla_T} = 0$ is exactly the trajectory equation. Anti-self-duality and curvature energy are conformally invariant in dimension four, so this also describes the connection on the punctured conformal compactification S^4 .

Step 2. In this step, we prove finite energy and extend the connection and its symmetry smoothly across both poles. The endpoint forms $\sum_a \tau_a^\pm \theta_a$ are flat. At each end write T as its limiting triple plus an error. Formula (5.2) expresses the curvature coefficients as linear combinations of that error and its derivative, and quadratic terms in the error. Their bounds are $O(e^{2t})$ at the negative end and $O(e^{-2t})$ at the positive end. The two energy tails are therefore bounded by constants times

$$\int_{-\infty}^{-R} e^{4t} dt, \quad \int_R^{\infty} e^{-4t} dt.$$

On a punctured ball the bundle is trivial and the connection is smooth, hence $L_{1,\text{loc}}^2$. The background conformal metric is smooth on the ball. Theorem 5.4 applies at both punctures and gives smooth extensions.

For each $\ell \in SU(2)$, pull back an extension by the base action of ℓ . The prescribed punctured lift identifies this pullback with the original punctured connection. The transition conclusion of Theorem 5.4 extends that lift continuously to the fixed fiber. Lemma 5.5 then gives a jointly smooth $SU(2)$ -action preserving the extension. The same lemma makes transitions between any two chosen extensions smooth. They agree with the prescribed identification on the cylinder, so the two endpoint extensions glue canonically up to equivariant connection-preserving isomorphism.

Step 3. In this step, we identify the endpoint isotropies and prove every rate-two derivative estimate by radial transport. Denote the two fixed points of the base by o_- and o_+ , and use radial coordinates $r_- = e^t$ and $r_+ = e^{-t}$ near them. Choose a point in either fixed fiber and let $\varpi_\pm : SU(2) \rightarrow K_c$ be its actual isotropy homomorphism. Radial parallel transport gives a smooth radial section $s_\pm^{\text{rad}}$. In this gauge the local connection one-form $\mathcal{A} = \sum_i \mathcal{A}_i dx^i$ satisfies

$$\mathcal{A}_i(x) = \int_0^1 \sigma x^j (F_{\mathcal{A}})_{ji}(\sigma x) d\sigma.$$

Here the coordinate index j is summed. For completeness, radial gauge means $x^j \mathcal{A}_j = 0$, so $x^j (F_{\mathcal{A}})_{ji} = x^j \partial_j \mathcal{A}_i + \mathcal{A}_i$; integration along the ray gives the displayed identity. Thus $\mathcal{A}_i = x^j b_{ji}(x)$ for smooth functions b_{ji} . Evaluation on an angular vector at radius $r_\pm$ contributes a second factor of $r_\pm$.

Equivariance of radial transport makes the corrected section

$$s_\pm^{\text{inv}}(t, u) = s_\pm^{\text{rad}}(t, u) \varpi_\pm(u)$$

invariant and temporal. Its angular form is

$$\text{Ad}_{\varpi_\pm(u)^{-1}} \mathcal{A} + \varpi_\pm(u)^{-1} d\varpi_\pm(u) = \sum_a (d\varpi_\pm(e_a) + R_a^\pm(r_\pm)) \theta_a.$$

Invariance makes the coefficients independent of u . Along any fixed angular ray, the error is $r_\pm^2$ times a function smooth up to $r_\pm = 0$. Repeated application of $r_\pm \partial_{r_\pm}$ preserves this property. Since

$$\partial_t = r_- \partial_{r_-} = -r_+ \partial_{r_+}$$

in the respective endpoint charts, every temporal derivative of the error is $O(r_\pm^2)$.

Compare this corrected section with the original invariant temporal section s^T . Their transition is constant in t , since both temporal components vanish, and is constant in u , since

both sections are invariant and the left action on the sphere is transitive. Write $s_{\pm}^{\text{inv}} = s^T c_{\pm}$. Set $T_{\pm} = \lim_{t \rightarrow \pm\infty} T(t)$, viewed as linear maps on $\mathfrak{su}(2)$. The limiting angular forms give

$$T_{\pm} = \text{Ad}_{c_{\pm}} d\varpi_{\pm}.$$

At the positive end $T_+ = dp_+$, and at the negative end $T_- = \text{Ad}_q dp_-$ for some $q \in K_c$. Equality of these differential homomorphisms identifies $\varpi_{\pm}$ with $p_{\pm}$ up to conjugacy, since $\text{SU}(2)$ is connected. Replacing the positive radial frame by its translate by c_+^{-1} makes its isotropy exactly p_+ and its corrected section exactly s^T . This determines the retained positive frame, including its value within the possible Z_+ -family.

At the negative end choose a frame with isotropy exactly p_- . Its corrected section is $s^T q$, and the preceding estimates give, for every $m \geq 0$,

$$(5.3) \quad \partial_t^m (T_a - \text{Ad}_q \tau_a^-) = O(e^{2t}), \quad \partial_t^m (T_a - \tau_a^+) = O(e^{-2t})$$

at the respective ends. Transitions between different removable extensions preserve parallel transport and the identification on the cylinder. They therefore preserve the normalized positive frame and all these endpoint comparisons.

Step 4. In this step, we construct the inverse correspondence and verify its framing and chart identifications. Start with a smooth invariant connection and a positive frame of isotropy p_+ . Radially transport that frame from o_+ , correct the section by $p_+(u)$, and continue it by temporal parallel transport over the cylinder. The section is invariant, so the connection has the form $d + \sum_a T_a(t) \theta_a$. Formula (5.2) gives its trajectory equation. Near the negative pole, compare with a radial section corrected by $p_-(u)$. Its transition to the global temporal section is a constant q . The calculation leading to (5.3) proves every required bound at both ends. Thus the inverse takes values in $\mathcal{M}(p_-, p_+)$.

The two constructions are inverse on the punctured cylinder. Smooth uniqueness of transitions makes them inverse on the compactified framed isomorphism classes as well. A retained negative frame chooses q . Changing it on the right by $z \in Z_-$ changes the corrected section from $s^T q$ to $s^T qz$. Thus it gives exactly the torsor of Lemma 5.3. A change of positive frame acts on temporal triples by constant conjugation through Z_+ ; we retain this frame throughout.

Kronheimer's equation (1), boundary condition (2), and framing description on printed pages 474–475 identify this same invariant instanton space [Kro90, Section 2]. The inverse just constructed shows that its smooth representatives have the explicit rate-two bounds. Hence its natural charts and smooth hyperkähler structure are precisely those in Theorem 5.6. These structures belong to $\mathcal{M}(p_-, p_+)$; the additional negative frame is only the Z_- -torsor just described. $\square$

5.3. Translation and the Kazhdan parameter. In the endpoint notation above, the ordered triple (e, h, f) for the slice is $(Y_+, -H_+, X_+)$. Thus its compact endpoint has complex triple $(X_+, H_+, Y_+) = (f, -h, e)$ when the nilpotent entries are listed on either side of the semisimple entry.

Proposition 5.8. *In the signed convention where the positive compact endpoint has complex triple $(f, -h, e)$, translation $T(t) \mapsto T(t + \lambda)$ becomes $\kappa_{e^{-\lambda}}$ under the positive-normalized orbit-slice chart. Inverse spatial dilation $x \mapsto s^{-1}x$, for $s > 0$, becomes $T(t) \mapsto T(t - \log s)$ and hence κ_s . When $p_+ = 0$, the distinguished orbit chart is exactly*

$$\chi(T) = \frac{1}{2} \lim_{t \rightarrow +\infty} e^{2t} (T_2(t) + iT_3(t)).$$

The parameter s here is the raw Kazhdan parameter, before any removal of a finite kernel.

Proof. We compute the translated normal form, compare it with spatial pullback, and then extract the coordinate at a zero positive endpoint.

Step 1. In this step, we compute the Kazhdan parameter induced by temporal translation. Let $v_+(t)$ be the unique positive-normalizing complex gauge. For the translated trajectory its translated gauge $v_+(t + \lambda)$ still tends to one at the positive end. Applying it gives

$$2\alpha_\lambda = H_+, \quad \beta_\lambda(t) = Y_+ + e^{-2t} \text{Ad}(e^{-\lambda H_+})\varepsilon_\lambda, \quad \varepsilon_\lambda = e^{-2\lambda} \text{Ad}(e^{-\lambda H_+})\varepsilon.$$

The subspace centralizing X_+ is preserved by $\text{Ad}(e^{-\lambda H_+})$, so this is already the positive normal form. Put $s = e^{-\lambda}$. Since $h = -H_+$ and $[H_+, Y_+] = -2Y_+$, we obtain

$$\begin{aligned} \kappa_s(Y_+ + \varepsilon) &= s^2 \text{Ad}(s^{H_+})(Y_+ + \varepsilon) \\ &= Y_+ + s^2 \text{Ad}(s^{H_+})\varepsilon = Y_+ + \varepsilon_\lambda. \end{aligned}$$

This proves the first assertion with its sign and its exponent.

Step 2. In this step, we identify inverse spatial dilation with temporal translation and the raw Kazhdan action. For $\delta_{s^{-1}}(x) = s^{-1}x$, cylinder coordinates give

$$\delta_{s^{-1}}(e^t u) = e^{t - \log s} u, \quad \delta_{s^{-1}}^* \theta_a = \theta_a.$$

Thus $\delta_{s^{-1}}^* \nabla_T = \nabla_{T(\cdot - \log s)}$. Both poles are fixed, so the retained frame pulls back to the same framed fiber. The uniqueness in Proposition 5.7 identifies the compactifications as well. Translation by λ multiplies the constants in the positive and negative rate estimates by $e^{-2\lambda}$ and $e^{2\lambda}$, respectively, for every derivative. It leaves the negative comparison q unchanged and commutes with both constant frame actions. Taking $\lambda = -\log s$ gives κ_s .

Step 3. In this step, we derive the orbit-chart formula with coefficient $1/2$ when $p_+ = 0$. Suppose $p_+ = 0$. The smooth radial expansion from Proposition 5.7, with $r_+ = e^{-t}$, has the form

$$T_a = r_+^2 U_a + r_+^3 V_a + O(r_+^4).$$

Since $\partial_t = -r_+ \partial_{r_+}$ and the bracket term is $O(r_+^4)$, its cubic coefficient in the trajectory equation gives $-3V_a = -2V_a$. Hence $V_a = 0$ and

$$T_a = e^{-2t} U_a + O(e^{-4t}).$$

In this case the positive normal form is $2\alpha^{\text{nf}} = 0$, $\beta^{\text{nf}}(t) = e^{-2t}\varepsilon$. The normalizing gauge tends to one, so

$$\varepsilon = \lim_{t \rightarrow \infty} e^{2t} \text{Ad}_{v_+(t)} \beta(t) = \frac{1}{2}(U_2 + iU_3).$$

Theorem 5.6 identifies this value with $\chi(T)$, proving the displayed formula. $\square$

For any oriented orthonormal basis (a, b, c) of $\mathbb{R}^3$, the same zero-positive-endpoint formula reads $\chi_a(T) = \frac{1}{2} \lim_{t \rightarrow \infty} e^{2t}(T_b + iT_c)$, where $T_b = \sum_j b_j T_j$ and $T_c = \sum_j c_j T_j$. It is the chart for the corresponding member of the twistor sphere.

5.4. Faithful representations and gauge lifting. We first isolate the real gauge argument. The terminal subgroup condition is what forces the entire gauge to stay in the subgroup.

Lemma 5.9. *Let L_c be a closed Lie subgroup of $\text{U}(V)$ with Lie algebra $\mathfrak{l}$, and let $a, b : [t_0, \infty) \rightarrow \mathfrak{l}$ be continuous. Suppose a C^1 map $v : [t_0, \infty) \rightarrow \text{U}(V)$ satisfies*

$$b = vav^{-1} - v'v^{-1}, \quad \lim_{t \rightarrow \infty} v(t) \in L_c.$$

Then $v(t) \in L_c$ for every $t \geq t_0$.

Proof. Let $U_a(s, t)$ be parallel transport, defined by $\partial_s U_a(s, t) = -a(s)U_a(s, t)$ and $U_a(t, t) = 1$. Solving this equation within L_c and using uniqueness in $\text{U}(V)$ shows that $U_a(s, t)$ and $U_b(s, t)$ belong to L_c . Differentiating gives the transport identity

$$U_b(s, t) = v(s)U_a(s, t)v(t)^{-1}, \quad v(t) = U_b(s, t)^{-1}v(s)U_a(s, t).$$

Choose a bi-invariant metric on $U(V)$. Left and right multiplication by elements of L_c preserve distance to L_c , so

$$\text{dist}(v(t), L_c) = \text{dist}(v(s), L_c)$$

for every $s \geq t$. The right-hand side tends to zero as $s \rightarrow \infty$. Closedness of L_c proves the assertion. $\square$

Theorem 5.10. *Let K_c be compact connected with semisimple Lie algebra and let $r : K_c \rightarrow U(V)$ be faithful. Pointwise application of dr induces an injective triholomorphic immersion*

$$\mathcal{M}_{K_c}(p_-, p_+) \longrightarrow \mathcal{M}_{\text{SU}(V)}(rp_-, rp_+).$$

In every oriented complex chart this map satisfies $\chi_V \circ F = dr_{\mathbb{C}} \circ \chi_{K_c}$. Every allowed ambient positive-framed gauge relating two image trajectories is the image of a unique allowed source gauge, including its endpoint limits, centralizer condition, and all derivative decay estimates.

Faithfulness here is a hypothesis on the compact group representation. It supplies a smooth inverse on the image subgroup, which is used to lift the gauge together with its endpoint estimates.

Proof. We establish the source gauge lift, compare positive normal forms, and use the chart identity in every oriented complex structure.

Step 1. In this step, we prove that the representation lands in $\text{SU}(V)$ and defines the trajectory map. Since $\mathfrak{k} = [\mathfrak{k}, \mathfrak{k}]$ and traces of commutators vanish,

$$\text{Tr}(dr(\xi)) = 0 \quad (\xi \in \mathfrak{k}).$$

The character $\det \circ r$ has zero differential. Connectedness of K_c and its value at the identity imply $\det \circ r = 1$. Thus $r(K_c) \subset \text{SU}(V)$. Application of dr preserves the equations, the endpoint triples, and every derivative estimate. It also intertwines the real gauge actions. This defines F . Equivalently, it extends the structure group of the connections in Proposition 5.7; parallel transport and the terminal ODE commute with this extension.

Step 2. In this step, we lift every allowed ambient gauge uniquely with its endpoint conditions and derivative estimates. Take $Q, \tilde{Q} \in \mathcal{C}$ and an allowed ambient gauge v with $v * dr(Q) = dr(\tilde{Q})$. Its temporal components satisfy

$$dr(\tilde{Q}_0) = v dr(Q_0) v^{-1} - v' v^{-1}, \quad v_+ = 1.$$

The image $L_c = r(K_c)$ is closed because K_c is compact. Lemma 5.9, applied on every half-line, gives $v(t) \in L_c$ for all $t \in \mathbb{R}$. Faithfulness makes $r : K_c \rightarrow L_c$ a diffeomorphism. Define the unique smooth lift $u = r^{-1} \circ v$. Its limits are $u_+ = 1$ and $u_- = r^{-1}(v_-)$.

We check the rates for this actual lift. In endpoint coordinate neighborhoods write v as $z(t)$, its limiting value as $z_{\pm}$, and $u = \eta(z(t))$ for the smooth inverse map in these coordinates. Shrink the neighborhoods so that each derivative of any fixed order of η is bounded. The mean value estimate gives the required order-zero bound on $u - u_{\pm}$. For $m \geq 1$, the chain rule writes $\partial_t^m u$ as a finite sum of terms of the form

$$(D^k \eta)_{z(t)}(\partial_t^{j_1} z(t), \dots, \partial_t^{j_k} z(t)), \quad j_{\nu} \geq 1, \quad j_1 + \dots + j_k = m.$$

The ambient gauge estimates pass to z by the smooth coordinate map. Each term above has bounded coefficient and at least one factor with the prescribed exponential decay. Consequently, for every $m \geq 0$,

$$\partial_t^m (u - 1) = O(e^{-2t}) \quad (t \rightarrow +\infty), \quad \partial_t^m (u - u_-) = O(e^{2t}) \quad (t \rightarrow -\infty).$$

The negative ambient limit centralizes every $dr(\tau_a^-)$, so

$$dr(\text{Ad}_{u_-} \tau_a^- - \tau_a^-) = \text{Ad}_{v_-} dr(\tau_a^-) - dr(\tau_a^-) = 0.$$

Injectivity of dr gives $u_- \in Z_-$. Finally, differentiating $r(u) = v$ and using the gauge action gives $dr(u * Q) = v * dr(Q) = dr(\tilde{Q})$. The same injectivity yields $u * Q = \tilde{Q}$. Thus $u \in \mathcal{G}_+$ and is

the required source gauge, uniquely by faithfulness. This also proves injectivity on gauge classes. If an ambient gauge was initially described in $U(V)$, taking traces in its temporal equation gives $\text{Tr}(v'v^{-1}) = 0$; its positive limit one then forces $\det v = 1$ by Jacobi's formula.

Step 3. In this step, we compare positive normal forms and prove the chart identity. Fix an oriented decomposition and a source trajectory. Let w_+ be its positive-normalizing complex gauge from Theorem 5.6. The complexification of r sends w_+ to a complex gauge with positive limit one. Applying $dr_{\mathbb{C}}$ to the normalized pair gives

$$\begin{aligned} 2\alpha_V^{\text{nf}} &= dr_{\mathbb{C}}(H_+), \\ \beta_V^{\text{nf}}(t) &= dr_{\mathbb{C}}(Y_+) + e^{-2t} \text{Ad}(e^{-t dr_{\mathbb{C}}(H_+)}) dr_{\mathbb{C}}(\varepsilon). \end{aligned}$$

Moreover, $[dr_{\mathbb{C}}(X_+), dr_{\mathbb{C}}(\varepsilon)] = 0$. This is the target positive normal form, with slice value $dr_{\mathbb{C}}(Y_+ + \varepsilon)$. Its uniqueness gives

$$(5.4) \quad \chi_V \circ F = dr_{\mathbb{C}} \circ \chi_{K_c}.$$

Membership in the negative orbit and the identification with real trajectories are supplied by the two-ended classification in Theorem 5.6. The normal-form comparison uses no negative-end condition on w_+ .

Step 4. In this step, we deduce triholomorphicity and injectivity of the differential from the identities in all oriented charts. The argument acts on the Lie algebra factor and commutes with a change of oriented basis in $\mathbb{R}^3$. Equation (5.4) therefore holds for each of the complex charts in the natural twistor sphere. Since $dr_{\mathbb{C}}$ is complex linear and the charts are biholomorphisms, F is holomorphic for each such complex structure. In particular, its differential intertwines each of I, J, K . Differentiating the same identity in any one chart gives

$$d\chi_V \circ dF = dr_{\mathbb{C}} \circ d\chi_{K_c}.$$

The chart differentials are isomorphisms and $dr_{\mathbb{C}}$ is injective, so dF is injective. Together with the gauge-class injectivity proved above, this establishes the asserted injective triholomorphic immersion for every semisimple source. $\square$

6. POTENTIALS ON NILPOTENT ORBIT CLOSURES

We retain the curvature convention of Section 2 and the positive-framed trajectory spaces of Section 5. We construct the potential on a normalized nilpotent orbit closure by pulling back the matrix half-norm. The comparison on the open orbit fixes the metric and its symplectic normalization; finiteness of the map controls the potential on the entire closure. Invariant forms on compact real Lie algebras are extended complex bilinearly whenever their arguments lie in the complexification.

Theorem 6.1. *For a compact simple Lie algebra with invariant form minus its Killing form, the one-ended Kronheimer orbit chart has its corresponding KKS holomorphic symplectic form and a homogeneous hyperkähler potential; the induced quaternionic scalar action realizes the compact real structure by quaternion j . On a nilpotent orbit with its fixed natural complex structure and KKS form, there is at most one compatible compact-invariant hyperkähler structure admitting a hyperkähler potential whose quaternion j induces that compact real structure.*

The normalization and quaternionic action are those of [Bry98, Theorem 5.1(ii)–(iv), Corollaries 5.2–5.4, and equations (64), (104)]. The uniqueness assertion is [KS01, Proposition 2.2]. Its real-structure hypothesis will be checked for both matrix models below.

Proposition 6.2. *Let $\mathfrak{g}$ be simple, let b be minus the Killing form of a compact real form, and let $r : G \hookrightarrow \text{GL}(V)$ be a faithful algebraic representation of its connected adjoint group, unitary on a compact form. Set $\rho = dr$, and let $c_\rho > 0$ satisfy*

$$-\text{Tr}(\rho(u)\rho(v)) = c_\rho b(u, v).$$

On each nonzero nilpotent orbit, its embedding into the represented matrix orbit is triholomorphic and pulls back the trace-normalized matrix Kronheimer metric to c_ρ times the source Kronheimer metric. The trace-normalized matrix metric is exactly the doubled image-flag quotient metric, including its half-norm potential. Its pulled-back holomorphic symplectic form is

$$\Omega_x([\xi, x], [\eta, x]) = -c_\rho b(x, [\xi, \eta]).$$

Thus the same positive constant scales the metric and all three Kähler forms. The trace convention fixes that constant before the matrix potential is pulled back.

Proof. We first fix the one-ended KKS normalization, then obtain the metric comparison from faithful naturality. We finally identify the matrix trajectory metric and its homogeneous potential with the quotient model.

Step 1. In this step, we determine the trace scaling constant and fix the KKS sign. Let K_c be the compact real form of G , with Lie algebra $\mathfrak{k}$. Cyclicity of trace makes

$$b_\rho(u, v) = -\text{Tr}(\rho(u)\rho(v))$$

an invariant symmetric form on $\mathfrak{k}$. For $u \neq 0$, faithfulness and skew-Hermitianity give

$$b_\rho(u, u) = \text{Tr}(\rho(u)^*\rho(u)) > 0.$$

Invariant symmetric forms on the compact simple algebra $\mathfrak{k}$ are scalar multiples of b , so $b_\rho = c_\rho b$ with $c_\rho > 0$. Since $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, traces of commutators also give $\rho(\mathfrak{g}) \subset \mathfrak{sl}(V)$. The determinant character of $r|_{K_c}$ has zero differential and K_c is connected; hence $r(K_c) \subset \text{SU}(V)$.

Choose the nonzero negative endpoint corresponding to the given orbit and take positive endpoint zero. Proposition 5.8 fixes the orbit chart as

$$\chi(T) = \zeta_2(T) + i\zeta_3(T), \quad \zeta_a(T) = \frac{1}{2} \lim_{t \rightarrow +\infty} e^{2t} T_a(t).$$

Let $\Omega_{\mathfrak{g}}$ be the source holomorphic symplectic form and write $B_{\mathfrak{g}}$ for the complex Killing form, so $B_{\mathfrak{g}} = -b$. In the convention of Theorem 6.1, the infinitesimal vector associated with ξ is $-\xi$ and its Hamiltonian is $B_{\mathfrak{g}}(\xi, x)$, with $\iota_{-\xi}\Omega_{\mathfrak{g}} = -dB_{\mathfrak{g}}(\xi, x)$. Consequently,

$$\begin{aligned} (\Omega_{\mathfrak{g}})_x(-[\xi, x], -[\eta, x]) &= -dB_{\mathfrak{g}}(\xi, x)(-[\eta, x]) \\ &= B_{\mathfrak{g}}(\xi, [\eta, x]) = B_{\mathfrak{g}}(x, [\xi, \eta]). \end{aligned}$$

The two signs in the tangent arguments cancel. Thus, in the tangent convention used here,

$$(6.1) \quad (\Omega_{\mathfrak{g}})_x([\xi, x], [\eta, x]) = -b(x, [\xi, \eta]).$$

The factor $1/2$ in χ is part of this normalization.

On $\mathfrak{su}(V)$ use the positive form $q(U, W) = -\text{Tr}(UW)$. Passing from minus the matrix Killing form to q multiplies the path-space metric and its three symplectic forms by the same positive scalar. It leaves the trajectory equation, the quotient, and the complex structures unchanged. The reduced tensors therefore scale by that scalar as well. Formula (6.1) then becomes the exact matrix formula

$$(\Omega_V)_A([U, A], [W, A]) = \text{Tr}(A[U, W]).$$

Step 2. In this step, we prove that the faithful orbit embedding pulls back the matrix metric to $c_\rho g_{K_r}$. Pointwise application of ρ preserves the trajectory equation and all endpoint estimates. By Theorem 5.10, it gives an injective triholomorphic immersion F of the genuine positive-framed spaces. The endpoint formula gives

$$\chi_V(F(T)) = \frac{1}{2} \lim_{t \rightarrow +\infty} e^{2t} \rho(T_2(t) + iT_3(t)) = \rho(\chi(T)).$$

Thus F is exactly the algebraic orbit embedding in these charts. Its symplectic pullback is computed on orbit tangent vectors by

$$\begin{aligned}
 (\rho^* \Omega_V)_x([\xi, x], [\eta, x]) &= \text{Tr}(\rho(x)[\rho(\xi), \rho(\eta)]) \\
 (6.2) \qquad \qquad \qquad &= \text{Tr}(\rho(x)\rho([\xi, \eta])) \\
 &= -c_\rho b(x, [\xi, \eta]) = c_\rho(\Omega_{\mathfrak{g}})_x([\xi, x], [\eta, x]).
 \end{aligned}$$

Let g_{Kr} denote the source metric and $g_{V, \text{Kr}}$ the trace-normalized target metric. Taking real and imaginary parts of (6.2) scales ω_J and ω_K by c_ρ . Since F intertwines J , for real tangent vectors v, w we obtain

$$\begin{aligned}
 g_{V, \text{Kr}}(dFv, dFw) &= \omega_{J, V}(dFv, J_V dFw) \\
 &= \omega_{J, V}(dFv, dF(Jw)) = c_\rho \omega_J(v, Jw) = c_\rho g_{\text{Kr}}(v, w).
 \end{aligned}$$

Triholomorphicity now gives the same scaling for ω_I .

Step 3. In this step, we identify the matrix trajectory and quotient metrics and their normalized homogeneous potentials. Let $\mathcal{O}_V$ be the matrix orbit and write g_Q and Φ_V for its doubled image-flag metric and half-norm. Both g_Q and $g_{V, \text{Kr}}$ have the natural complex structure. By Theorem 4.6 and the preceding normalization, both have holomorphic symplectic form Ω_V , with coefficient one in the trace formula. Both are $\text{SU}(V)$ -invariant: the terminal unitary action preserves the flat quotient data, and the compact adjoint action preserves the trajectory construction.

We check the potential and real structure on the quotient. Write its arrows as $\alpha_i : V_i \rightarrow V_{i+1}$ and $\beta_i : V_{i+1} \rightarrow V_i$, for $0 \leq i < k$, with $V_0 = 0$ and $V_k = V$. On the open quotient the forward arrows are injective and the backward arrows are surjective. An internal unitary stabilizer is trivial: starting with the identity at V_k , the equation $\alpha_{k-1} g_{k-1}^{-1} = \alpha_{k-1}$ gives $g_{k-1} = 1$, and descending induction gives every $g_i = 1$. The internal group is compact, so its action is proper. Quaternionic multiplication commutes with this group and rotates the moment-map triple, preserving its zero level and its smooth quotient stratum. The potential descent in Theorem 4.4 therefore makes

$$\Phi_V = \frac{1}{2} \sum_i (\|\alpha_i\|_{\text{HS}}^2 + \|\beta_i\|_{\text{HS}}^2)$$

a hyperkähler potential there, in our curvature convention. The action of quaternion j is

$$j(\alpha_i, \beta_i) = (-\beta_i^*, \alpha_i^*).$$

It preserves the indicated injectivity and surjectivity conditions and sends the terminal product to

$$A = \alpha_{k-1} \beta_{k-1} \mapsto -\beta_{k-1}^* \alpha_{k-1}^* = -A^*.$$

This is precisely the compact real structure $\sigma(A) = -A^*$, as in [KS01, Section 2.1].

On the trajectory side, Theorem 6.1 provides the homogeneous hyperkähler potential and its quaternionic action; a positive rescaling of both metric and potential leaves that action unchanged. Conjugation of the imaginary quaternionic moment map by j fixes ζ_2 and reverses ζ_3 . Hence its action in the orbit chart is

$$\zeta_2 + i\zeta_3 \mapsto \zeta_2 - i\zeta_3 = -(\zeta_2 + i\zeta_3)^*,$$

because ζ_2, ζ_3 are skew-Hermitian. Thus this action also realizes σ . The fixed complex structure, exact KKS form, compact invariance, and potential with the required real structure now agree. The uniqueness assertion in Theorem 6.1 gives $g_Q = g_{V, \text{Kr}}$ and equality of their three Kähler forms.

Their homogeneous potentials have the same normalization as well. Let E_V be the generator of $A \mapsto e^u A$ on $\mathcal{O}_V$. The positive real quaternion s acts on the orbit by $A \mapsto s^2 A$ in the trajectory model, by [Bry98, Corollaries 5.2–5.4]. Its generator is the gradient of the normalized homogeneous potential, namely $2E_V$. In the quotient model the gradient of the flat half-norm

is the radial arrow vector. This vector is tangent to the zero moment level by quadratic homogeneity and orthogonal to the internal group orbits by norm invariance. Its quotient is therefore $\nabla\Phi_V$, and the quadratic terminal product sends it to $2E_V$. Once the metrics agree, the two potentials have equal differentials. They differ by a constant on the connected orbit. Both have weight one under positive scalar dilation, which forces that constant to vanish. $\square$

Theorem 6.3. *Let $\mathfrak{g}$ be complex simple and $\mathcal{O} \neq \{0\}$ nilpotent. Its normalized closure Y admits a homogeneous hyperkähler potential package of scalar weight one on the big open $\mathcal{O}$. For the representation of Proposition 6.2, the potential is the pullback of the matrix half-norm by the finite map $\rho \circ \nu$, the metric is $c_\rho g_{\text{Kt}}$, and the form is the displayed scaled KKS form. The potential extends continuously and properly to Y , and its top Monge–Ampère measure is the zero extension of the hyperkähler volume from $\mathcal{O}$. On the free scalar quotient, the local weights $2 \log(\Phi \circ s)$ give a globally locally bounded continuous plurisubharmonic metric on H .*

The finite map supplies one function on the whole normalization. In particular, its boundary values and the quotient metric are controlled by the same function that gives the smooth orbit metric.

Proof. We construct the finite map and its global potential first. We then identify the tensors on the open orbit, prove the boundary-mass assertion on a resolution, and descend the logarithm to the scalar quotient.

Step 1. In this step, we construct the finite map of closures and pull back its global weight-one potential. Put $X = \overline{\mathcal{O}}_{\text{red}}$ and let $\nu : Y \rightarrow X$ be its normalization. Choose $x_0 \in \mathcal{O}$ and set

$$\mathcal{O}_V = \text{SL}(V) \cdot \rho(x_0), \quad Y_V = \overline{\mathcal{O}_V}_{\text{red}}.$$

The image under ρ of an $\mathfrak{sl}_2$ -triple containing x_0 acts on V as an $\mathfrak{sl}_2$ -representation. Its raising operator $\rho(x_0)$ is nilpotent, and is nonzero by injectivity. It is traceless as proved above. The $\text{GL}(V)$ - and $\text{SL}(V)$ -orbits coincide: multiplying a conjugating matrix q by a scalar c with $c^{\dim V} \det q = 1$ does not change its conjugation action. Representation equivariance therefore sends $\mathcal{O}$ into $\mathcal{O}_V$ and X into Y_V .

The injective linear map ρ identifies $\mathfrak{g}$ with a closed linear subspace of $\mathfrak{sl}(V)$. Its restriction $\iota : X \hookrightarrow Y_V$ is a closed immersion. Thus

$$\psi = \rho \circ \nu = \iota \circ \nu : Y \longrightarrow Y_V$$

is finite, as the composite of a finite normalization and a closed immersion. In particular it is proper. It is scalar-equivariant by Proposition 3.4 and linearity of ρ . That proposition also gives the unique point y_0 over zero, so

$$\psi^{-1}(0) = \{y_0\}.$$

Let Φ_V be the global matrix potential of Theorem 4.6, using the chosen invariant Hermitian form on V , and define

$$\Phi = \Phi_V \circ \psi.$$

It is continuous, nonnegative, and plurisubharmonic by holomorphic pullback, with zero set exactly $\{y_0\}$. Its sublevel sets are

$$\{\Phi \leq C\} = \psi^{-1}(\{\Phi_V \leq C\}) \quad (C \geq 0),$$

and hence compact. Its extended logarithm is also plurisubharmonic. Indeed, for a holomorphic disc into Y the composition with ψ either lies at zero, when the logarithm is identically $-\infty$, or pulls back the plurisubharmonic function $\log \Phi_V$.

The full modulus homogeneity can be read directly from the matrix quotient. Multiplying every arrow by $z \in \mathbb{C}^*$ multiplies the complex moment-map components by z^2 and the real components by $|z|^2$, so preserves their simultaneous zero level. The terminal product is multiplied by z^2 and the half-norm by $|z|^2$. For $t \in \mathbb{C}^*$ choose $z^2 = t$. Both choices give the same terminal

product and the same norm, and the whole-quotient identification gives $\Phi_V(tA) = |t|\Phi_V(A)$. Equivariance of ψ yields

$$(6.3) \quad \Phi(ty) = |t|\Phi(y).$$

Step 2. In this step, we identify the pulled-back hyperkähler tensors on the open orbit. By Proposition 3.5, normalization identifies $\nu^{-1}(\mathcal{O})$ with $\mathcal{O}$, and this is a scalar-invariant smooth open with complement of codimension at least two. On this open, Proposition 6.2 identifies ψ with a triholomorphic immersion into the matrix quotient and gives

$$g = \psi^* g_Q = c_\rho g_{K^r}, \quad \omega_I = \psi^* \omega_{I,V} = \text{dd}^c \Phi, \quad \Omega = \psi^* \Omega_V = c_\rho \Omega_{\mathfrak{g}}.$$

The source complex structures are those of its trajectory space. For example, $d\psi \circ J = J_V \circ d\psi$ implies

$$(\psi^* \omega_{J,V})(v, w) = g_Q(d\psi(Jv), d\psi(w)) = g(Jv, w),$$

and the corresponding identity for K shows $\Omega = \omega_J + i\omega_K$. The potential is smooth and positive on $\mathcal{O}$. Formula (6.2) makes Ω algebraic, closed, and nondegenerate. It also gives

$$(t^* \Omega)_x([\xi, x], [\eta, x]) = -c_\rho b(tx, [\xi, \eta]) = t\Omega_x([\xi, x], [\eta, x]).$$

Its unique reflexive extension has the same weight by Proposition 3.5; equality of these extensions is determined on the big open. Together with (6.3), these statements verify all parts of Definition 2.4 with scalar weight one.

Step 3. In this step, we prove that the global Monge–Ampère measure is the zero extension of the orbit volume. Write $\dim_{\mathbb{C}} Y = 2n$. Lemma 4.7 gives on $\mathcal{O}$

$$(6.4) \quad \frac{(\text{dd}^c \Phi)^{2n}}{(2n)!} = \frac{\Omega^n \wedge \overline{\Omega}^n}{4^n (n!)^2}.$$

To apply the noncharging theorem on a smooth space, take a proper resolution $\pi : \tilde{Y} \rightarrow Y$ that is an isomorphism over $\mathcal{O}$. The function $\tilde{\Phi} = \Phi \circ \pi$ is continuous plurisubharmonic, hence locally bounded. Its Bedford–Taylor top measure is locally finite. The set $D = \pi^{-1}(Y \setminus \mathcal{O})$ is a proper analytic subset of the smooth space $\tilde{Y}$. It is locally pluripolar, so Theorem 4.5, specifically [DS10, Appendix A, Corollary A.32], gives

$$(\text{dd}^c \tilde{\Phi})^{2n}(D) = 0.$$

The pushforward is the Bedford–Taylor measure of Φ on Y . For completeness, locally embed Y in a complex manifold and use the recursive definition

$$(\text{dd}^c \Phi)^j \wedge [Y] = \text{dd}^c(\Phi(\text{dd}^c \Phi)^{j-1} \wedge [Y]).$$

Proper pushforward commutes with dd^c and with multiplication by the continuous function Φ pulled back to $\tilde{Y}$. Starting with $\pi_*[\tilde{Y}] = [Y]$ and inducting in this recursion gives

$$\pi_*((\text{dd}^c \tilde{\Phi})^j \wedge [\tilde{Y}]) = (\text{dd}^c \Phi)^j \wedge [Y].$$

Properness of π makes the top pushforward locally finite, and the preceding noncharging assertion gives it zero mass on $Y \setminus \mathcal{O}$. On $\mathcal{O}$, the map π is an isomorphism and (6.4) identifies this measure divided by $(2n)!$ with the hyperkähler volume. Thus (6.4) holds globally when the right-hand side is extended by zero.

Step 4. In this step, we descend the logarithm to a continuous locally bounded plurisubharmonic metric on H . Set $U = Y \setminus \{y_0\}$ and $F = \text{Proj } \mathbb{C}[Y]$. By Proposition 3.4, the scalar action on U is effective and free and its divisorial stabilizer boundary is zero. There are local holomorphic sections of $U \rightarrow F$ also over the boundary of the orbit quotient. To see this explicitly, pull back linear coordinate functions on $\mathfrak{g}$ along ν . Their common zero set is $\{y_0\}$, so their nonvanishing sets cover U . Each such function ℓ has weight one, and

$$y \mapsto \ell(y)^{-1}y$$

is invariant on $\{\ell \neq 0\}$ and identifies its quotient with the slice $\{\ell = 1\}$. This gives the required section on the corresponding chart of F .

For any local holomorphic section s define

$$\psi_s = 2 \log(\Phi \circ s).$$

These are plurisubharmonic by the global logarithmic plurisubharmonicity already proved. Since s takes values in U , $\Phi \circ s$ is strictly positive and continuous. It is bounded above and away from zero after shrinking around any base point; thus ψ_s is continuous and locally bounded there, including points outside the orbit quotient. If $s' = fs$ for a nowhere-zero holomorphic function f , then

$$\psi_{s'} = \psi_s + 2 \log |f|.$$

For the weight-one polarization H of Definition 2.2, the corresponding local frames satisfy $e_{s'} = f^{-1}e_s$. The displayed identity is therefore exactly the transition rule for squared frame norms $\|e_s\|^2 = e^{-\psi_s}$. It defines the asserted global Hermitian metric, with differential curvature $\text{dd}^c \psi_s \geq 0$ and integral first-Chern current $(2\pi)^{-1} \text{dd}^c \psi_s$ in the convention of Section 2. $\square$

7. FIXED LIFTS AND THE NATURAL TYPE-A MAP

We retain the positive framing and the trajectory conventions of Section 5. We now turn to orbit-slice intersections. The next three sections construct the type-A slice potential needed for faithful pullback. We identify the regular invariant ADHM quotient with the entire prescribed orbit-slice intersection, and then compare its complex structures with Kronheimer's in every direction. The endpoint representations determine the lift to the ADHM gauge space. The comparison of complex structures uses right quaternionic rotations together with an explicit correction at the positive endpoint.

7.1. Regular data and the fixed-lift quotient. For a pair of endomorphisms of a vector space V , we write $B = (B_1, B_2) : V \rightarrow V \otimes \mathbb{C}^2$, with respect to the standard basis of $\mathbb{C}^2$. The symbols P and Q below denote the two framing maps.

Definition 7.1. Fix an integer $r \geq 2$, nonnegative integers $d_1, \dots, d_{r-1}$, and decreasing nonnegative integers $A_1, \dots, A_r$ with

$$N = \sum_{i=1}^{r-1} id_i = \sum_{i=1}^r A_i \geq 2, \quad D_i = \sum_{j=i}^{r-1} d_j, \quad D_r = A_{r+1} = 0.$$

Let $e \in \mathfrak{sl}_N$ have Jordan type $1^{d_1} \dots (r-1)^{d_{r-1}}$, and let $\mathcal{O}_A$ have column lengths A_i . Assume $e \in \overline{\mathcal{O}_A}$ and $L = \mathcal{O}_A \cap \mathcal{S}_e$ has positive dimension. Put

$$v_i = \sum_{j=1}^i (D_j - A_j), \quad R_i = \text{Sym}^{i-1} \mathbb{C}^2, \quad K_+ = \bigoplus_{i=1}^{r-1} \mathbb{C}^{v_i} \otimes R_i,$$

$$W_+ = \bigoplus_{i=1}^{r-1} \mathbb{C}^{d_i} \otimes R_i, \quad W_- = \bigoplus_{i=1}^r \mathbb{C}^{A_i - A_{i+1}} \otimes R_i.$$

Choose endpoint homomorphisms $p_{\pm} : \text{SU}(2) \rightarrow \text{SU}(N)$ with these modules, with positive complex triple $(f, -h, e)$ and the corresponding reversed negative triple. Write λ for the action on K_+ and $C_{\lambda} = Z_{\text{U}(K_+)}(\lambda(\text{SU}(2)))$. The prescribed-lift regular ADHM locus $\mathcal{E}_{\lambda}$ consists of $(B_1, B_2, P, Q) \in \text{End}(K_+)^2 \oplus \text{Hom}(W_+, K_+) \oplus \text{Hom}(K_+, W_+)$ satisfying

$$[B_1, B_2] + PQ = 0, \quad [B_1, B_1^*] + [B_2, B_2^*] + PP^* - Q^*Q = 0,$$

the fixed-lift relations

$$B\lambda(a) = (\lambda(a) \otimes a)B, \quad \lambda(a)P = Pp_+(a), \quad p_+(a)Q = Q\lambda(a),$$

and, at every $x = (x_1, x_2) \in \mathbb{C}^2$, injectivity and surjectivity respectively of

$$z \mapsto ((B_1 - x_1)z, (B_2 - x_2)z, Qz), \quad (z_1, z_2, w) \mapsto -(B_2 - x_2)z_1 + (B_1 - x_1)z_2 + Pw.$$

Set $\mathcal{R}_\lambda = \mathcal{E}_\lambda / C_\lambda$. The *natural map* $v : \mathcal{R}_\lambda \rightarrow L$ is framed ADHM, followed by invariant radial reduction with terminal temporal normalization, followed by the positive-normalized Kronheimer chart.

Both moment equations and both monad regularity conditions are part of this definition. We equip the matrix space with its unscaled Hilbert–Schmidt metric. The framed correspondence uses the following normalization [Mac91, Theorem 1.1 and the following paragraph, Proposition 2.4, and Theorems 2.5, 3.1–3.2].

Theorem 7.2. *On the smooth regular framed locus, the ADHM correspondence with framed instantons is a hyperkähler isometry. For the invariant form $-\text{Tr}$ the instanton metric is normalized by $(2\pi^2)^{-1}$, and the second-moment potential pulls back to*

$$\frac{1}{2}(\|B_1\|_{\text{HS}}^2 + \|B_2\|_{\text{HS}}^2 + \|P\|_{\text{HS}}^2 + \|Q\|_{\text{HS}}^2).$$

The instanton is the middle cohomology of the regular monad, equipped with its projected connection and frame at infinity.

Explicitly, if $\mathcal{A}$ is a framed instanton and a, b are its smooth Coulomb tangent representatives, these conventions give

$$(7.1) \quad g_{\text{inst}}(a, b) = \frac{1}{2\pi^2} \int_{\mathbb{R}^4} \sum_{\mu=1}^4 -\text{Tr}(a_\mu b_\mu) \, \text{dvol}, \quad m_2(\mathcal{A}) = \frac{1}{16\pi^2} \int_{\mathbb{R}^4} |x|^2 \text{Tr}(F_{\mathcal{A}} \wedge F_{\mathcal{A}}).$$

The half-norm identity is Maciocia’s journal Theorem 3.1; the isometry is his Theorem 3.2, both on p. 476. We write Φ for the displayed half-norm on a regular quotient, so $m_2 \circ p_\lambda = \Phi$ when p_λ denotes ADHM in the direction from matrices to instantons.

The sign and sphere-volume factor can also be checked directly. For an orthogonal complex structure I_0 on $\mathbb{R}^4$, let $\omega_0(\xi, \zeta) = \langle I_0 \xi, \zeta \rangle$ and let I_0 act on one-forms by $a \mapsto a \circ I_0$. In an adapted orthonormal coframe,

$$\text{Tr}(a \wedge I_0 b) \wedge \omega_0 = - \sum_{\mu=1}^4 \text{Tr}(a_\mu b_\mu) \, \text{dvol}.$$

Thus the normalized instanton two-form gives the positive metric in (7.1). In logarithmic polar coordinates $x = e^t a_0$, with $a_0 \in S^3$, the Euclidean metric is $e^{2t}(dt^2 + g_{S^3})$. A one-form squared norm contributes e^{-2t} and the volume contributes e^{4t} . Consequently, for a smooth one-form with sphere-independent coefficients $b_\mu(t)$ in the orthonormal cylinder coframe,

$$\frac{1}{2\pi^2} \int_{\mathbb{R}^4} |b|^2 \, \text{dvol} = \int_{\mathbb{R}} e^{2t} \sum_{\mu=0}^3 |b_\mu(t)|^2 \, dt.$$

Here $\text{vol}(S^3) = 2\pi^2$, as follows by writing the Gaussian integral π^2 on $\mathbb{R}^4$ in polar coordinates. This calculation applies to the displayed smooth representatives. The moduli-space metric itself is supplied by Theorem 7.2.

We next describe the spatial symmetry on the actual regular quotient.

Lemma 7.3. *Let V, W be Hermitian spaces with $\dim V \geq 1$ and $\dim W \geq 2$, and let $p_+ : \text{SU}(2) \rightarrow \text{SU}(W)$ be a homomorphism. Let $\mathcal{Z}$ be the simultaneous-zero regular ADHM locus with V, W in place of K_+, W_+ in Definition 7.1. For $a \in \text{SU}(2)$ define*

$$a \cdot (B_1, B_2, P, Q) = \left(\sum_{\beta=1}^2 a_{1\beta} B_\beta, \sum_{\beta=1}^2 a_{2\beta} B_\beta, P p_+(a)^{-1}, p_+(a) Q \right).$$

Then this action preserves $\mathcal{Z}$, commutes with scalar multiplication and with the free gauge action

$$g \cdot (B_1, B_2, P, Q) = (gB_1g^{-1}, gB_2g^{-1}, gP, Qg^{-1}), \quad g \in \mathrm{U}(V),$$

and induces hyperkähler isometries of $\mathcal{Z}/\mathrm{U}(V)$. Under framed ADHM it is inverse pullback by $x \mapsto ax$, with the pulled-back frame composed with $p_+(a)^{-1}$. For a prescribed lift $\eta : \mathrm{SU}(2) \rightarrow \mathrm{U}(V)$, the equation $a \cdot z = \eta(a)^{-1} \cdot z$ is exactly the fixed-lift system in Definition 7.1, with η replacing λ .

The framing correction in this action keeps track of the specified representation at infinity.

Proof. We check preservation of the equations, framed equivariance, and freeness.

Step 1. In this step, we verify preservation of the moment equations and monad regularity. On the pair (B_1, B_2) , left quaternionic multiplication commutes with the right quaternionic complex structures. On (P, Q^*) the two framing actions commute with the quaternionic pairing because $p_+(a)$ is unitary. The action therefore preserves the metric and each complex structure. Gauge and scalar commutation follow by substitution. Since $\det a = 1$ and a is unitary,

$$[B'_1, B'_2] = [B_1, B_2], \quad \sum_{\alpha=1}^2 [B'_\alpha, (B'_\alpha)^*] = \sum_{\alpha=1}^2 [B_\alpha, B_\alpha^*].$$

The products PQ, PP^*, Q^*Q are also preserved, so both moment equations are preserved.

Write α_x^z and β_x^z for the first and second monad maps. Let U_a act by a on $V \oplus V$ and by $p_+(a)$ on W . Direct substitution gives

$$(7.2) \quad U_a \alpha_x^z = \alpha_{ax}^{a \cdot z}, \quad \beta_{ax}^{a \cdot z} U_a = \beta_x^z.$$

Both injectivity and surjectivity are preserved because U_a is unitary.

Step 2. In this step, we identify the monad action on the projected connection and its frame at infinity. The complex equation gives $\beta_x^z \alpha_x^z = 0$. Its harmonic cohomology is

$$\mathcal{H}_z(x) = \ker \beta_x^z \cap \ker (\alpha_x^z)^*.$$

Taking adjoints in (7.2) shows that $U_a \mathcal{H}_z(x) = \mathcal{H}_{a \cdot z}(ax)$. If $\Pi_z(x)$ is the orthogonal projection to this subspace, then

$$\Pi_{a \cdot z}(ax) U_a = U_a \Pi_z(x).$$

The constant unitary map U_a therefore intertwines the projected connections of Theorem 7.2. At infinity it acts on the W summand by $p_+(a)$, which gives exactly the stated framing correction. Rearranging $a \cdot z = \eta(a)^{-1} \cdot z$ gives the three fixed-lift equations.

Step 3. In this step, we prove freeness of the gauge action and descend the spatial hyperkähler isometries. Suppose that $g \in \mathrm{U}(V)$ fixes z . It commutes with both B_α and satisfies $Qg = Q$. If g has an eigenspace F with eigenvalue different from 1, then F is invariant under both B_α and $Q|_F = 0$. The complex moment equation implies $[B_1, B_2]|_F = 0$. Two commuting endomorphisms of the nonzero complex space F have a common eigenvector w ; for their eigenvalues x_1, x_2 , one has $\alpha_x^z(w) = 0$. This contradicts regularity. A unitary map is diagonalizable, so $g = 1$. The quotient is the smooth regular quotient in Theorem 7.2, and the commuting hyperkähler isometries descend to it. $\square$

Lemma 7.4. *Let $\mathrm{SU}(2)$ and a compact group $G_c = \mathrm{U}(V)$ act smoothly and commute on a manifold Z . Assume that the G_c -action is free, and that its smooth quotient $\mathcal{Q} = Z/G_c$ is hyperkähler with the induced $\mathrm{SU}(2)$ -action preserving the metric and each complex structure. For a homomorphism $\eta : \mathrm{SU}(2) \rightarrow G_c$, put*

$$Z_\eta = \{z : a \cdot z = \eta(a)^{-1} \cdot z \text{ for all } a \in \mathrm{SU}(2)\}, \quad C_\eta = Z_{G_c}(\eta(\mathrm{SU}(2))).$$

Then Z_η/C_η identifies smoothly with a union of connected components of $\mathcal{Q}^{\mathrm{SU}(2)}$. These components inherit the metric, complex structures and Kähler forms by restriction. The lift

of a fixed quotient point is unique up to G_c -conjugacy, and two representatives with the specified lift η differ by an element of C_η .

Thus prescribing a lift determines the gauge group of its sector.

Proof. We first identify the equivalence relation and then show that lift type is locally constant.

Step 1. In this step, we construct the lift of a fixed quotient point and identify the centralizer equivalence relation. For a fixed quotient point represented by z , there is a unique $k_z(a) \in G_c$ with $a \cdot z = k_z(a) \cdot z$. The division map of the principal G_c -bundle is smooth, so k_z is smooth. Commutation of the actions gives

$$k_z(a_1 a_2) = k_z(a_2) k_z(a_1).$$

Hence $\eta_z(a) = k_z(a)^{-1}$ is a homomorphism. Replacing z by $g \cdot z$ conjugates η_z by g . If both z and $g \cdot z$ have the specified lift η , freeness gives $g\eta(a) = \eta(a)g$ for every a . This proves the centralizer equivalence relation and its injectivity into the fixed quotient.

Step 2. In this step, we prove local constancy of the lift type and smoothness of the identification with the fixed locus. A local section of $Z \rightarrow \mathcal{Q}$ gives a smooth family of lift homomorphisms η_y for y in the fixed locus. If χ_j is the character of the j -dimensional irreducible representation, its multiplicity is

$$m_j(y) = \int_{\mathrm{SU}(2)} \mathrm{Tr}(\eta_y(a)) \overline{\chi_j(a)} \, da,$$

where Haar measure has mass one. This is a continuous integer and therefore is locally constant. Only $j \leq \dim V$ can occur. These multiplicities determine the conjugacy class, so each sector is open and closed in the fixed locus.

The identification is smooth in both directions. To see this locally, fix a lift η at a base point and take nearby η_y . The operator

$$F_y = \int_{\mathrm{SU}(2)} \eta_y(a) \eta(a)^{-1} \, da$$

intertwines η and η_y and is invertible near the base point, where it is the identity. Its unitary polar factor is a smooth unitary intertwiner. Adjusting the local section by its inverse gives local representatives in Z_η , unique modulo C_η .

Step 3. In this step, we verify that the fixed components inherit the restricted hyperkähler tensors. At a fixed point y , an equivariant exponential chart identifies the fixed locus with $(T_y \mathcal{Q})^{\mathrm{SU}(2)}$. This subspace is preserved by each complex structure, since each commutes with the action. Tangential projection of the ambient Levi-Civita connection preserves the restricted complex structures: the ambient structures are parallel and preserve both the tangent and normal spaces. The restrictions are therefore hyperkähler. Finally, an element normalizing the image of η but not centralizing it changes the specified homomorphism. The equivalence relation already proved is precisely that of C_η . $\square$

7.2. Endpoint modules and coverage.

Proposition 7.5. *For the data of Definition 7.1, all v_i are nonnegative, $K_+ \neq 0$, $\mathcal{E}_\lambda$ is nonempty, and C_λ acts freely. Its framed instanton image consists of the full invariant sector with endpoint modules W_- , W_+ , and the natural map v is a diffeomorphism onto L . Every invariant framed instanton with these endpoints has lift conjugate to λ . The endpoint modules obey*

$$[W_-] = [W_+] + [K_+ \otimes R_2] - 2[K_+], \quad d_i + v_{i-1} + v_{i+1} - 2v_i = A_i - A_{i+1},$$

where omitted indices are zero. The restricted half-norm is the hyperkähler potential on this regular quotient; with the raw scalar generator E , it satisfies $\nabla \Phi = E$ and $\Phi = \frac{1}{2}g(E, E)$ and has weight two. The quotient is by the centralizer, with the positive frame retained and the negative frame forgotten.

The representation at the negative endpoint fixes the gauge module uniquely. In particular, coverage here includes every component of L .

Proof. We prove nonnegativity, determine every possible lift from the endpoint monad, and use the framed correspondences to obtain coverage. We then restrict the quotient potential.

Step 1. In this step, we prove nonnegativity of the dimension vector and $K_+ \neq 0$. For $e_A \in \mathcal{O}_A$, the kernel dimensions of powers are

$$\dim \ker e^i = \sum_{j=1}^i D_j, \quad \dim \ker e_A^i = \sum_{j=1}^i A_j.$$

Kernel dimensions can increase under specialization. Since $e \in \overline{\mathcal{O}_A}$, their differences v_i are nonnegative. Extending the sequences by zero, we have

$$v_i - v_{i-1} = D_i - A_i \quad (1 \leq i < r), \quad v_{r-1} = A_r, \quad v_0 = v_r = v_{r+1} = d_r = 0.$$

For $i < r$, substitution of $d_i = D_i - D_{i+1}$ gives

$$(7.3) \quad d_i + v_{i-1} + v_{i+1} - 2v_i = A_i - A_{i+1}.$$

For $i = r$, the same identity is $v_{r-1} = A_r$; beyond r both sides vanish. If $K_+ = 0$, all v_i vanish, whence $D_i = A_i$ and $A_r = 0$. The two nilpotent orbits would coincide. Transversality and the dimension formula of Theorem 3.1 would then give $\dim L = 0$, contrary to the hypothesis. Thus $\dim K_+ \geq 1$.

Step 2. In this step, we determine every lift from the equivariant monad and its endpoint modules. By Lemma 7.3, an invariant framed instanton gives a fixed point of the regular ADHM quotient. Its lift exists and is unique up to unitary conjugacy by Lemma 7.4. Write its gauge module as

$$V_\eta = \bigoplus_{i \geq 1} \mathbb{C}^{b_i} \otimes R_i, \quad b_0 = 0,$$

with finite support. At the fixed point $0 \in \mathbb{R}^4$, the monad is

$$0 \longrightarrow V_\eta \xrightarrow{\alpha_0} (V_\eta \otimes R_2) \oplus W_+ \xrightarrow{\beta_0} V_\eta \longrightarrow 0.$$

Its maps are equivariant. This follows either by restricting (7.2) and using the lift, or directly from the three intertwining equations. Regularity says that α_0 is injective and β_0 is surjective. The middle cohomology is the actual negative endpoint fiber. The two exact sequences for $\ker \beta_0$ and $\ker \beta_0 / \text{im } \alpha_0$ give

$$(7.4) \quad [W_-] = [W_+] + [V_\eta \otimes R_2] - 2[V_\eta].$$

Using $R_i \otimes R_2 = R_{i-1} \oplus R_{i+1}$, with $R_0 = 0$, its R_i coefficient is

$$d_i + b_{i-1} + b_{i+1} - 2b_i = A_i - A_{i+1} \quad (i \geq 1).$$

The solution with $b_0 = 0$ and finite support is unique. Indeed, the difference w_i of two solutions satisfies $w_{i+1} - w_i = w_i - w_{i-1}$, so its first differences are constant. They are eventually zero, and $w_0 = 0$, hence $w_i = 0$ for all i . Equation (7.3) supplies the solution $b_i = v_i$. Therefore $V_\eta \cong K_+$ and η is conjugate to λ .

Conversely, every datum of $\mathcal{E}_\lambda$ has this same exact monad. Its negative fiber is an honest unitary representation with R_i multiplicity $A_i - A_{i+1}$, so it is W_- . The raising nilpotent on R_i has a single Jordan block of size i . The resulting column lengths are

$$\sum_{j \geq i} (A_j - A_{j+1}) = A_i.$$

This identifies the endpoint orbit directly from its fiber.

Step 3. In this step, we prove that the framed correspondences give a diffeomorphism onto the whole intersection L . In the positive complex triple, $X_+ = f$, $H_+ = -h$ and $Y_+ = e$.

Consequently the positive slice in Theorem 5.6 is $Y_+ + \mathfrak{sl}_N^{X_+} = e + \mathfrak{sl}_N^f$. Choose the reversed triple $(f_A, -h_A, e_A)$ at the negative end; its orbit is $\mathcal{O}_A$. Thus that theorem and Proposition 5.7 identify every point of L with a smooth invariant instanton whose positive frame is retained and whose negative frame is forgotten. Framed ADHM then supplies a regular datum. The preceding uniqueness shows that a unitary change of representative puts its lift equal to λ . This proves nonemptiness and surjectivity.

The reverse construction lands in L by the monad computation. If two data give the same point, the framed ADHM and radial correspondences give a unitary gauge between them; Lemma 7.4 puts that gauge in C_λ . Hence the map is injective. Each correspondence and its inverse is smooth, using Theorem 7.2, Proposition 5.7, Theorem 5.6, and the local identification in Lemma 7.4. This proves the diffeomorphism and includes all components. Schur's lemma gives

$$C_\lambda \cong \prod_{i=1}^{r-1} \mathrm{U}(\mathbb{C}^{v_i}),$$

acting freely by Lemma 7.3.

Step 4. In this step, we verify the restricted half-norm potential, its Euler identities, and its spatial dilation convention. The two lemmas show that $\mathcal{R}_\lambda$ is a union of fixed components of the smooth regular ADHM quotient, with the restricted hyperkähler tensors. Its potential is the restriction of Φ from Theorem 7.2. For $s > 0$, multiplication of every matrix by s preserves the quadratic moment equations and the lift equations. Moreover

$$\alpha_x^{sz} = s\alpha_{x/s}^z, \quad \beta_x^{sz} = s\beta_{x/s}^z,$$

so it preserves regularity. It scales the flat metric and each Kähler form by s^2 , and the same identities hold on the quotient. At a zero-level datum z , the radial vector z is tangent to the level set and orthogonal to gauge orbits, since the half-norm is gauge invariant. It is therefore a horizontal representative of E . For every horizontal tangent ξ ,

$$d\Phi(\xi) = \langle z, \xi \rangle = g(E, \xi), \quad g(E, E) = \|z\|^2 = 2\Phi.$$

The scalar action commutes with the spatial symmetry, so E is tangent to each fixed component it meets. Restricting these identities proves $\nabla\Phi = E$ and $\Phi = \frac{1}{2}g(E, E)$ on $\mathcal{R}_\lambda$. For each restricted complex structure, restriction also commutes with dd^c , so Φ is a common hyperkähler potential. Its scalar weight is two.

Finally, the monad equalities just displayed identify the harmonic cohomology of sz at x with that of z at x/s . They identify its projected connection and the limiting W_+ frame as well. Thus this scalar action is precisely inverse spatial pullback $\mathcal{A} \mapsto (\delta_s^{-1})^*\mathcal{A}$, where $\delta_s(x) = sx$. This records the dilation convention with the same normalization as (7.1). $\square$

The direct module calculation is also needed to interpret the partition notation in the chosen-complex comparison. The following small example isolates the issue in the printed formulas.

Lemma 7.6. *For the one-vertex quiver with gauge dimension 1 and framing dimension 4, Nakajima's printed formulas (8.2)–(8.3) give $\lambda = (1, 1, 1, 1)$ and $\mu = (3, 1)$. The regular simultaneous-zero quotient, through the invariant QP , is the nilpotent orbit of Jordan-block type $(2, 1, 1)$ in $\mathfrak{sl}_4$. Thus the printed formulas, the accompanying Jordan-cell sentence, and Theorem 8.4 cannot all have the literal standard row-partition interpretation in this example.*

Here λ and μ are the printed partition symbols; the lift homomorphism continues to be specified by Definition 7.1 outside this example.

Proof. We identify the regular quotient and then compare its Jordan type with the printed partitions.

Step 1. In this step, we identify the regular quotient with the orbit of Jordan type $(2, 1, 1)$. There are no arrow maps. Write $P : \mathbb{C}^4 \rightarrow \mathbb{C}$ and $Q : \mathbb{C} \rightarrow \mathbb{C}^4$. The simultaneous moment equations are

$$PQ = 0, \quad PP^* = Q^*Q.$$

Regularity requires P to be surjective and Q to be injective, hence both are nonzero. The invariant $X = QP$ satisfies

$$X^2 = Q(PQ)P = 0, \quad \text{rank } X = 1, \quad \text{Tr } X = \text{Tr}(PQ) = 0.$$

It therefore has one block of size two and two blocks of size one. Conversely, factor any nonzero rank-one square-zero X as QP through its one-dimensional image. Then $PQ = 0$, and its factorizations form a single $\mathbb{C}^*$ -orbit under $(P, Q) \mapsto (cP, Qc^{-1})$. Every such orbit has a real-zero representative: choose

$$|c|^4 = \frac{\|Q\|^2}{\|P\|^2}.$$

The remaining freedom is precisely $U(1)$. The factorizations can be chosen holomorphically on each patch where a fixed entry of X is nonzero, so this also gives the quotient's chosen-complex identification with the orbit of type $(2, 1, 1)$.

Step 2. In this step, we compare the quotient's Jordan type with the printed partition formulas. For the one-vertex Cartan matrix (2), Nakajima's variable is $u_1 = 4 - 2 \cdot 1 = 2$. Equations (8.2)–(8.3) on printed p. 395 give

$$\lambda = (1, 1, 1, 1), \quad \mu = (1 + 2, 1) = (3, 1).$$

The Jordan-cell sentence and Theorem 8.4 occur on printed p. 396 [Nak94, (8.2)–(8.3) and Theorem 8.4, pp. 395–396]. The printed zero-orbit slice is all of $\text{End}(\mathbb{C}^4)$; the trace calculation above justifies restriction to $\mathfrak{sl}_4$. The orbit of row type $(3, 1)$ has nilpotency index three and dimension $16 - (2^2 + 1^2 + 1^2) = 10$, whereas the orbit obtained above has dimension $16 - (3^2 + 1^2) = 6$. Thus the literal row reading is incompatible with this quotient. In this example, $(3, 1)$ is exactly its column-length tuple. $\square$

The example gives no general transpose rule for the printed formulas. For general data we use the actual endpoint modules, as determined by (7.4) and (7.3).

7.3. The natural map and its rotations. Let $\mathcal{I}_\lambda$ denote the invariant framed instanton sector identified in Proposition 7.5, and put $\mathcal{M}_\lambda = \mathcal{M}(p_-, p_+)$. Denote framed ADHM by $p_\lambda : \mathcal{R}_\lambda \rightarrow \mathcal{I}_\lambda$, radial reduction with terminal temporal normalization by C_{rad} , and the distinguished positive-normalized endpoint chart by χ_i . Thus

$$(7.5) \quad \mathcal{R}_\lambda \xrightarrow{p_\lambda} \mathcal{I}_\lambda \xrightarrow{C_{\text{rad}}} \mathcal{M}_\lambda \xrightarrow{\chi_i} L, \quad v = \chi_i \circ C_{\text{rad}} \circ p_\lambda.$$

Here (i, j, k) labels the distinguished oriented imaginary-quaternion directions, with the signed Lie-algebra convention of Definition 5.1.

The endpoint calculation identifies this composite with the one used by Nakajima. His gauge and framing multiplicity spaces are $\mathbb{C}^{v_i}$ and $\mathbb{C}^{d_i}$, respectively. Decomposing the B intertwiners by $R_i \otimes R_2 = R_{i-1} \oplus R_{i+1}$ gives the two arrows between adjacent multiplicity spaces; P, Q give the framing pairs. The compact gauge group is their product C_λ . The negative fiber has multiplicities $d_i + v_{i-1} + v_{i+1} - 2v_i = A_i - A_{i+1}$. Thus its nilpotent orbit is $\mathcal{O}_A$, and the positive fiber gives $\mathcal{S}_e$ in the signed convention already checked. The proof on p. 396 follows precisely the three arrows in (7.5). We use its chosen-complex conclusion in this endpoint formulation [Nak94, Theorem 8.4 and proof, p. 396].

Theorem 7.7. *For the regular invariant ADHM sector with its endpoint modules identified by the equivariant monad, the natural ADHM–framed-instanton–radial–Kronheimer map is a biholomorphism in the distinguished complex structure onto the negative orbit intersected with the positive slice.*

This supplies the distinguished complex structure for the literal map v . We now extend the comparison to the full sphere of complex structures.

Theorem 7.8. *For the data of Definition 7.1, the literal natural map $v : \mathcal{R}_\lambda \rightarrow L$ intertwines the ADHM complex structures and Kronheimer's natural complex structures in every twistor direction. It is equivariant for complex scalar multiplication of ADHM data and the raw Kazhdan action. In particular, transporting the ADHM metric through v gives the natural twistor triple on L .*

The metric in the last assertion is the transported ADHM metric. The assertion identifies its complex structures with the natural ones.

Proof. We rotate the framed instantons, compute the correction at the positive endpoint, and compare the resulting complex charts. Conjugating the distinguished-complex identity then gives every twistor direction.

Write I_q^{ADHM} for the source complex structures and I_q^{Kro} for the trajectory complex structures transported to L by χ_i . Theorem 7.7 gives

$$(7.6) \quad dv I_i^{\text{ADHM}} = I_i^{\text{Kro}} dv.$$

Step 1. In this step, we define right rotations on the prescribed-lift quotient and compute their action on the source complex structures. Fix $u \in \text{Sp}(1)$ and set

$$\rho_u(x) = xu^{-1}, \quad R = \text{Ad}_{u^{-1}}, \quad q = Ri = u^{-1}iu.$$

The spatial map ρ_u is an orientation-preserving Euclidean isometry, commutes with left $\text{SU}(2)$ multiplication, and fixes 0 and ∞ . Pullback, with the pulled-back frame at infinity, defines a map $\mathcal{R}_u^{\text{inst}}$ on framed instantons: it preserves ASD connections, and a based gauge pulls back to a based gauge. Commutation with the left action preserves its endpoint representations, with the negative one understood up to conjugacy. By Proposition 7.5, the unique lift class remains λ . Therefore

$$\mathcal{R}_u^{\text{ADHM}} = p_\lambda^{-1} \circ \mathcal{R}_u^{\text{inst}} \circ p_\lambda$$

is a diffeomorphism of the actual centralizer quotient $\mathcal{R}_\lambda$. Right pullback conjugates the complex direction i to $u^{-1}iu$. On one-forms this follows from $d\rho_u \circ J_q = J_i \circ d\rho_u$, where J_q is right multiplication by q on $\mathbb{H}$. The ADHM isometry of Theorem 7.2 consequently gives

$$(7.7) \quad d\mathcal{R}_u^{\text{ADHM}} I_i^{\text{ADHM}} = I_q^{\text{ADHM}} d\mathcal{R}_u^{\text{ADHM}}.$$

Step 2. In this step, we compute the positive-frame correction and prove covariance of radial reduction. Write $x = e^t a$ with $a \in \text{SU}(2)$ and set $\theta = a^{-1}da$. The angular part of a terminal temporal representative is $T(t)(\theta)$, where $T(t) : \text{Im } \mathbb{H} \rightarrow \mathfrak{su}(N)$. Its endpoint limits are dp_+ and $\text{Ad}_g dp_-$, respectively. Since

$$(au^{-1})^{-1}d(au^{-1}) = u\theta u^{-1}, \quad \rho_u^* \theta = R^{-1}\theta,$$

bare pullback sends T to $T \circ R^{-1}$. Put

$$(7.8) \quad c = p_+(u^{-1}), \quad S_u(T)(X) = \text{Ad}_c(T(R^{-1}X)).$$

The equivariance of the differentials of $p_\pm$ computes the two limits:

$$\begin{aligned} \lim_{t \rightarrow +\infty} S_u(T) &= \text{Ad}_c dp_+ \circ \text{Ad}_u = \text{Ad}_{c p_+(u)} dp_+ = dp_+, \\ \lim_{t \rightarrow -\infty} S_u(T) &= \text{Ad}_{cg} dp_- \circ \text{Ad}_u = \text{Ad}_{cgp_-(u)} dp_-. \end{aligned}$$

Thus the positive endpoint is restored exactly, while the forgotten negative-frame coordinate becomes $cgp_-(u)$. Replacing g by a representative differing by the negative centralizer gives the same negative endpoint. These constant unitary transformations preserve the equation and every rate-two derivative estimate. They preserve temporal gauge as well.

The correction has a direct framing interpretation. In the invariant radial trivialization, the frame near infinity is obtained from its fixed fiber frame by the factor $p_+(a)$. Right pullback replaces it by $p_+(au^{-1}) = p_+(a)c$. Expressing the pulled-back connection in the trivialization determined by the retained positive frame therefore conjugates its coefficients by c . Uniqueness of terminal temporal normalization gives

$$(7.9) \quad C_{\text{rad}} \circ \mathcal{R}_u^{\text{inst}} = S_u \circ C_{\text{rad}}.$$

The inverse is the corresponding rotation for u^{-1} , so S_u is a diffeomorphism of $\mathcal{M}_\lambda$.

Step 3. In this step, we compare the oriented positive normal forms and prove the inverse-composition identity for their charts. Complete q to the rotated oriented frame $(q, q_2, q_3) = (Ri, Rj, Rk)$. For a covariant representative with temporal component T_0 , put

$$\alpha_q(T) = \frac{T_0 + \sqrt{-1}T(q)}{2}, \quad \beta_q(T) = \frac{T(q_2) + \sqrt{-1}T(q_3)}{2}.$$

Under S_u , the temporal component changes by Ad_c . The positive complex triple for this frame satisfies

$$(H_{+,q}, X_{+,q}, Y_{+,q}) = \text{Ad}_c(H_{+,i}, X_{+,i}, Y_{+,i}).$$

By Theorem 5.6, there is a unique positive-normalizing complex gauge $g_q(t) \rightarrow 1$ which puts $2\alpha_q$ equal to $H_{+,q}$ and puts the transformed $\beta_q(0)$ in $Y_{+,q} + \mathfrak{sl}_N^{X_{+,q}}$. Denote this raw oriented chart by $\hat{\chi}_q$. Its negative orbit is $\mathcal{O}_A$, which is invariant under Ad_c , and its positive slice is $\text{Ad}_c \mathcal{S}_e$. Hence

$$\chi_q = \text{Ad}_{c^{-1}} \circ \hat{\chi}_q$$

has values in the fixed L . Theorem 5.6 also says that χ_q is a biholomorphism for the q -complex structure on the trajectory space and the ordinary complex structure on L .

Inverting (7.8) gives $S_u^{-1}T(X) = \text{Ad}_{c^{-1}} T(RX)$, and hence

$$(7.10) \quad \alpha_i(S_u^{-1}T) = \text{Ad}_{c^{-1}} \alpha_q(T), \quad \beta_i(S_u^{-1}T) = \text{Ad}_{c^{-1}} \beta_q(T).$$

Recall that a complex gauge acts by

$$\alpha \mapsto \text{Ad}_g \alpha - \frac{1}{2} g' g^{-1}, \quad \beta \mapsto \text{Ad}_g \beta.$$

Since c is constant, $c^{-1}g_q c$ tends to 1 at the positive end and puts the pair in (7.10) into the distinguished positive normal form. Uniqueness in Theorem 5.6 identifies it with the i -normalizing gauge. Evaluating the transformed β at time zero therefore gives the precise identity

$$(7.11) \quad \chi_q = \chi_i \circ S_u^{-1}.$$

This uses the finite positive normal form of [Kro90, Lemma 11, pp. 484–485]; orbit membership uses the two-ended classification already included in Theorem 5.6.

The charts on both sides of (7.11) are holomorphic for their indicated complex structures. Thus S_u is holomorphic from the i -structure to the q -structure. On L , set $\mathcal{K}_u = \chi_i \circ S_u \circ \chi_i^{-1}$. We obtain

$$(7.12) \quad d\mathcal{K}_u I_i^{\text{Kro}} = I_q^{\text{Kro}} d\mathcal{K}_u, \quad v \circ \mathcal{R}_u^{\text{ADHM}} = \mathcal{K}_u \circ v,$$

where the second identity follows from (7.5) and (7.9).

Step 4. In this step, we prove that the natural map intertwines the complex structures in every twistor direction. For $y \in \mathcal{R}_\lambda$ put $x = \mathcal{R}_u^{\text{ADHM}}(y)$. Differentiating the second identity in (7.12), and using (7.7), (7.6), and the first identity in (7.12), yields

$$\begin{aligned} dv_x I_q^{\text{ADHM}} d\mathcal{R}_{u,y}^{\text{ADHM}} &= dv_x d\mathcal{R}_{u,y}^{\text{ADHM}} I_i^{\text{ADHM}} \\ &= d\mathcal{K}_{u,v(y)} dv_y I_i^{\text{ADHM}} \\ &= d\mathcal{K}_{u,v(y)} I_i^{\text{Kro}} dv_y \\ &= I_q^{\text{Kro}} d\mathcal{K}_{u,v(y)} dv_y \end{aligned}$$

$$= I_q^{\text{Kro}} dv_x d\mathcal{R}_{u,y}^{\text{ADHM}}.$$

Cancel the invertible last factor. Every unit imaginary quaternion has the form $u^{-1}iu$, so $dv I_q^{\text{ADHM}} = I_q^{\text{Kro}} dv$ for every q .

Step 5. In this step, we prove equivariance for complex scalars and identify the transported twistor triple. The last paragraph of Proposition 7.5 identifies $z \mapsto sz$, for $s > 0$, with inverse spatial dilation. By Proposition 5.8, radial reduction sends it to $T(t) \mapsto T(t - \log s)$, and the positive-normalized chart sends that translation to κ_s . Hence $v(sz) = \kappa_s v(z)$ for $s > 0$. Complex scalar multiplication also preserves the regular simultaneous zero locus: the complex equation scales by the square of the scalar, the real equation by its absolute square, and the monad maps have the same identities with x/s as above. It descends holomorphically in the distinguished complex structure. The map v is holomorphic by Theorem 7.7. For fixed z , the two maps $s \mapsto v(sz)$ and $s \mapsto \kappa_s v(z)$ are therefore holomorphic on $\mathbb{C}^*$ and agree on $\mathbb{R}_{>0}$. The identity theorem gives their equality on all of $\mathbb{C}^*$.

Transporting the ADHM metric through the diffeomorphism v transports its hyperkähler triple. The all-direction identity just proved identifies that triple with Kronheimer's natural complex structures. $\square$

8. THE WHOLE AFFINE TYPE-A MODEL

We retain the dominant data and the prescribed lift of Definition 7.1. We first identify the entire affine quiver quotient with the slice through Maffei's map. We then compare the quaternionic linear models, identify the regular simultaneous-zero quotient with the whole smooth leaf, and extend its physical potential across the affine boundary. The following section transports this potential to natural coordinates.

Definition 8.1. For the dominant data of Definition 7.1, set $V_i = \mathbb{C}^{v_i}$ and $D_i^{\text{vec}} = \mathbb{C}^{d_i}$. Let $E(d, v)$ be the doubled framed type-A representation space with arrows $V_i \rightleftharpoons V_{i+1}$ and framing pairs $D_i^{\text{vec}} \rightleftharpoons V_i$. Let Λ be its zero complex moment fiber and $G_v = \prod_i \text{GL}(V_i)$. Write

$$\mathfrak{M}^0 = \text{Spec } \mathbb{C}[\Lambda]^{G_v}, \quad \mathfrak{M} = \Lambda^{\text{st}}/G_v, \quad \mathfrak{M}^1 = \pi(\mathfrak{M})_{\text{red}},$$

where standard positive stability means generation by the framing images, and $\pi : \mathfrak{M} \rightarrow \mathfrak{M}^0$ is the projective quotient morphism. Put $X_A = (\overline{\mathcal{O}_A} \cap \mathcal{S}_e)_{\text{red}}$.

The distinction between $\mathfrak{M}^0$ and $\mathfrak{M}^1$ is needed even though they will coincide for these dominant data. The algebraic comparison initially concerns the reduced image of the projective map. We use Maffei's theorem in its published numbering [Maf05, Theorem 8]. The density assertion below is the consequence of the stratum-dimension argument in the proof of [Maf02, Lemma 55(1), p. 685].

Theorem 8.2. *For type-A data with nonnegative endpoint tuple and nonempty orbit-slice intersection, Maffei's map identifies the reduced image $\mathfrak{M}^1$ algebraically with the full reduced orbit-slice intersection, compatibly with the stable quiver variety and partial Springer-Slodowy resolution. For dominant data, the complex moment fiber is a complete intersection and its positive-stable locus is dense.*

Dominance is the hypothesis that makes the stable locus meet every irreducible component of the complex moment fiber. It supplies more than nonemptiness of the stable quotient.

The symplectic comparison in the next statement has the range of [Los12, Section 4.3, Case 2, Proposition 4.6.2 and Lemmas 4.6.3–4.6.4]. The local quotient assertion is the one in [Gin09, Theorem 2.2.4 and Corollary 2.2.6], and the projective Poisson assertion is [Gin09, Theorem 5.2.2(i)].

Theorem 8.3. *In Losev's positive type-A parabolic Slodowy setup, the comparison with the quiver resolution is algebraic, equivariant for the stated contracting action, and symplectic with the stated normalization. The standard morphism from the stable Nakajima variety to its affine*

quotient is projective and Poisson. On a smooth saturated invariant affine stable neighborhood where the reductive action is free on all semistable points, the geometric quotient is smooth and the quotient map is a principal bundle.

We will use the local assertion only after constructing such a neighborhood. The grading argument below also treats zero-dimensional blocks; it does not require an extension of the positive-data symplectic comparison.

Finally, we use the smooth-locus criterion of [BS21, Remark 1.1 and Theorem 1.15], including the $\alpha \in \Sigma_{\lambda, \theta}$ specialization immediately preceding that theorem in Section 1.4 of the published article.

Theorem 8.4. *A framed quiver quotient can be expressed as an unframed quotient by adjoining a dimension-one framing vertex. A point of a reduced quiver variety is smooth exactly when its representation is canonically polystable. If its dimension vector belongs to $\Sigma_{\lambda, \theta}$, canonical polystability reduces to stability; at zero parameters this applies when a simple representation of that dimension vector exists.*

Here the existence of a simple representation is an additional premise. We will establish it before using the criterion to cover the whole smooth locus.

Proposition 8.5. *For the dominant data of Definition 7.1, X_A is normal and irreducible, $L = (X_A)_{\text{reg}}$ is connected, and $X_A \setminus L$ has codimension at least two. The reduced affine varieties $\mathfrak{M}^0$ and $\mathfrak{M}^1$ coincide. The whole-affine Maffei isomorphism $\phi : \mathfrak{M}^0 \rightarrow X_A$ intertwines uniform scalar multiplication of all quiver coordinates with the raw Kazhdan action, including data with zero-dimensional blocks.*

The proposition supplies the affine space on which the potential will be defined. Its regular-locus assertion concerns the slice itself; the identification with regular ADHM data will be proved separately.

Proof. We prove the geometric assertions by smooth base change, identify the affine quotient by stable density, and check equivariance in the augmented block construction.

Step 1. In this step, we prove normality and irreducibility of X_A and identify its smooth locus with L . Put $Y_A = \mathcal{O}_A$. Theorem 4.4 gives normality of Y_A . Indeed, a GL_N -conjugating matrix may be multiplied by a scalar to have determinant one, so the nilpotent GL_N - and SL_N -orbits coincide. The same theorem gives symplectic singularities of Y_A . Thus the KKS form extends as a nondegenerate holomorphic form on $(Y_A)_{\text{reg}}$. Its inverse agrees with the Lie–Poisson bivector on $\mathcal{O}_A$, hence everywhere on $(Y_A)_{\text{reg}}$ by equality on a dense open. The latter bivector has rank $\dim \mathcal{O}_x$ at x . Every boundary orbit has dimension smaller than $\dim Y_A$, so no boundary point is smooth. Consequently

$$(Y_A)_{\text{reg}} = \mathcal{O}_A.$$

Consider the action morphism

$$a : \text{SL}_N \times \mathcal{S}_e \longrightarrow \mathfrak{sl}_N, \quad (g, x) \longmapsto \text{Ad}(g)x.$$

At $(1, e)$ its differential is $(\xi, w) \mapsto [\xi, e] + w$. The decomposition $\mathfrak{sl}_N = [\mathfrak{sl}_N, e] \oplus \mathfrak{sl}_N^f$ makes this differential surjective. The morphism is therefore smooth there. On the h -weight- j part of $\mathfrak{sl}_N^f$ one has $j \leq 0$, and

$$\kappa_t \left(e + \sum_{j \leq 0} w_j \right) = e + \sum_{j \leq 0} t^{2-j} w_j.$$

This extends to $t = 0$ and contracts the slice to e . It preserves X_A , since conjugation and nonzero scalar multiplication preserve nilpotent orbit closures. With

$$F_t(g, x) = (t^{-h} g t^h, \kappa_t x), \quad K_t(y) = t^2 \text{Ad}(t^{-h})y,$$

we have $a \circ F_t = K_t \circ a$. Smoothness is invariant under these automorphisms. Contracting x towards e proves smoothness at $(1, x)$ for every $x \in \mathcal{S}_e$, and left translation proves smoothness at every (g, x) .

The scheme-theoretic inverse image of Y_A is $\mathrm{SL}_N \times (Y_A \cap \mathcal{S}_e)$. Smooth base change from the normal variety Y_A makes this inverse image normal, hence reduced. Normality and reducedness descend along the smooth surjective projection to $Y_A \cap \mathcal{S}_e$. In particular the intersection is already reduced and X_A is normal. Contraction makes X_A connected: an extended scalar orbit joins each point to e and cannot meet two disjoint open-and-closed components. Irreducible components of a normal variety are disjoint and open and closed, so X_A is irreducible.

Smooth local ascent and descent of regularity for $\mathrm{SL}_N \times X_A \rightarrow Y_A$, at $(1, x)$, give

$$x \in (X_A)_{\mathrm{reg}} \iff x \in (Y_A)_{\mathrm{reg}}.$$

Thus $(X_A)_{\mathrm{reg}} = \mathcal{O}_A \cap \mathcal{S}_e = L$. It is a nonempty open subset of an irreducible variety, so it is irreducible and connected. Normality gives $\mathrm{codim}_{X_A}(X_A \setminus L) \geq 2$.

Step 2. In this step, we use stable density to identify the whole reduced affine quotient with the slice. With $v_0 = v_r = 0$, the defining prefix sums give

$$d_i - 2v_i + v_{i-1} + v_{i+1} = A_i - A_{i+1} \geq 0.$$

These are the simple-coroot coordinates of the framing-minus-dimension weight. Also $D_i - v_i + v_{i-1} = A_i$ and $v_{r-1} = A_r$, so the endpoint tuple in Theorem 8.2 is precisely A . That theorem gives a complete intersection Λ and a stable locus dense in every irreducible component.

The stable locus is smooth. To see this directly, an infinitesimal gauge stabilizer commutes with every arrow and kills every framing image. Generation from these images makes it zero. For the constant complex symplectic form Ω_E on $E(d, v)$,

$$\langle d\mu_{\mathbb{C}, z}(w), \xi \rangle = \Omega_E(\xi_z^\#, w).$$

Nondegeneracy of Ω_E then makes $d\mu_{\mathbb{C}, z}$ surjective. Thus Λ is generically reduced on every component. A complete intersection is Cohen–Macaulay and has no embedded associated components; generic reducedness therefore implies reducedness. Its invariant ring, and hence $\mathfrak{M}^0$, is reduced.

Let $q_0 : \Lambda \rightarrow \mathfrak{M}^0$ be the affine quotient. Every nonempty open subset of $\mathfrak{M}^0$ has a nonempty open inverse image, which meets Λ^{st} . Hence $q_0(\Lambda^{\mathrm{st}})$ is dense. It is contained in the closed image of the projective morphism π . Thus $\mathfrak{M}^1 = \mathfrak{M}^0$ as closed sets, and reducedness makes this equality an equality of reduced affine varieties. Theorem 8.2 now gives the whole-affine isomorphism $\phi : \mathfrak{M}^0 \rightarrow X_A$.

Step 3. In this step, we prove raw Kazhdan equivariance from the augmented block weights, including zero-dimensional blocks. We describe the weights in the construction of the map in Theorem 8.2. Its augmented gauge spaces are

$$\tilde{V}_i = V_i \oplus D'_i, \quad D'_i = \bigoplus_{j=i+1}^{r-1} \bigoplus_{q=1}^{j-i} D_j^{(q)},$$

where $D_j^{(q)}$ is a copy of D_j^{vec} ; at $i = 0$ we put $V_0 = 0$, so $\tilde{V}_0 = W_+$. Write $\mathcal{F} : \Lambda \rightarrow \mathcal{T}$ for Maffei's augmented block map. Its distinguished blocks are the original arrows and framing maps. The remaining blocks are uniquely determined by the moment equations and transversality equations in that construction [Maf05, Theorem 8, Sections 3.1–3.3, Definition 16, Lemmas 17–19, and Definition 20].

Give V_i weight zero and give $D_j^{(q)} \subset D'_i$ weight $j - i + 1 - 2q$. Let $\gamma_i(t)$ be the resulting one-parameter group. On the augmented arrows define

$$\tilde{a}_i \mapsto t^{-1} \gamma_{i+1}(t) \tilde{a}_i \gamma_i(t)^{-1}, \quad \tilde{b}_i \mapsto t^{-1} \gamma_i(t) \tilde{b}_i \gamma_{i+1}(t)^{-1}.$$

These are the block weights in the proof of [Los12, Lemma 4.6.3]. For completeness, the prescribed identity block $D_j^{(q+1)} \rightarrow D_j^{(q)}$ of $\tilde{a}_i$ has target weight minus source weight equal to 1, as does the identity block $D_j^{(q)} \rightarrow D_j^{(q)}$ of $\tilde{b}_i$. The factor t^{-1} therefore fixes these blocks. Prescribed zero blocks remain zero. On D'_i the raising and lowering operators e_i, f_i have weights 2, -2 . Consequently the equations

$$[\text{pr}_{D'_i} \tilde{b}_i \tilde{a}_i|_{D'_i} - e_i, f_i] = 0$$

are preserved, as are the moment equations, whose two sides transform by t^{-2} and conjugation. This verifies preservation of $\mathcal{T}$. The distinguished framing blocks involve $D_{i+1}^{(1)}$, whose weight is zero, and the other distinguished blocks join the V_i . Every original coordinate is thus multiplied by t^{-1} . Uniqueness in the augmented construction gives

$$t \star \mathcal{F}(z) = \mathcal{F}(t^{-1}z).$$

The augmented endomorphism on W_+ transforms as $t^{-2} \text{Ad}(\gamma_0(t))$, and $\gamma_0(t) = t^h$. Descending and putting $s = t^{-1}$ gives

$$\phi([sz]) = s^2 \text{Ad}(s^{-h})\phi([z]) = \kappa_s \phi([z]).$$

All identities are identities of block maps. A zero-dimensional block removes its identities and leaves the others unchanged. No division by a gauge dimension or a parabolic block size occurs. This proves the asserted grading for all the stated dominant data. $\square$

We next fix the linear coordinates, including their metric normalization. Write a point of $E(d, v)$ as

$$(\mathbf{a}_i, \mathbf{b}_i, p_i, q_i), \quad \mathbf{a}_i : V_i \rightarrow V_{i+1}, \quad \mathbf{b}_i : V_{i+1} \rightarrow V_i, \quad p_i : D_i^{\text{vec}} \rightarrow V_i, \quad q_i : V_i \rightarrow D_i^{\text{vec}}.$$

We use the sum of Hilbert–Schmidt squares, I given by multiplication by $\sqrt{-1}$, and

$$J_E(\mathbf{a}_i, \mathbf{b}_i, p_i, q_i) = (-\mathbf{b}_i^*, \mathbf{a}_i^*, -q_i^*, p_i^*).$$

Let $\mathcal{A}_\lambda$ be the linear ADHM space satisfying the fixed-lift relations of Definition 7.1, before imposing moment equations or regularity. Its compact gauge group is $C_\lambda = \prod_i \text{U}(V_i)$ and its complexification is G_v . On this space use

$$J_{\text{ADHM}}(B_1, B_2, P, Q) = (-B_2^*, B_1^*, -Q^*, P^*).$$

Write μ_E and μ_λ for the three moment maps on these spaces. They take values in the Lie algebra dual, vanish at zero, and satisfy $d\langle \mu_a, \xi \rangle = \iota_{\xi\#} \omega_a$.

Lemma 8.6. *For the dominant data of Definition 7.1, there are isometric equivariant inclusions*

$$T_i^+ : R_i \longrightarrow R_{i+1} \otimes R_2, \quad T_{i+1}^- : R_{i+1} \longrightarrow R_i \otimes R_2$$

whose phases make the block map $S : E(d, v) \rightarrow \mathcal{A}_\lambda$,

$$\begin{aligned} \mathbf{a}_i &\longmapsto i^{-1/2} \mathbf{a}_i \otimes T_i^+, & \mathbf{b}_i &\longmapsto (i+1)^{-1/2} \mathbf{b}_i \otimes T_{i+1}^-, \\ p_i &\longmapsto i^{-1/2} p_i \otimes 1_{R_i}, & q_i &\longmapsto i^{-1/2} q_i \otimes 1_{R_i}, \end{aligned}$$

a C_λ -equivariant linear hyperkähler isometry. It preserves the half-squared norm and satisfies $\mu_\lambda \circ S = \mu_E$ for all three dual-valued moment maps. It identifies the quiver simultaneous zero set with the ambient ADHM simultaneous zero set in $\mathcal{A}_\lambda$.

The normalization concerns each multiplicity space separately. In particular, an isometric intertwiner of i -dimensional representations has Hilbert–Schmidt square i , rather than 1.

Proof. We compute the norm factors, choose quaternionically compatible phases, and then compare the moment maps.

Step 1. In this step, we compute the norm factors that make S a complex-linear isometric bijection. Clebsch–Gordan gives $R_j \otimes R_2 = R_{j+1} \oplus R_{j-1}$, with $R_0 = 0$. Schur’s lemma

therefore decomposes $\text{Hom}_{\text{SU}(2)}(K_+, K_+ \otimes R_2)$ into exactly the adjacent-arrow blocks above. It decomposes the two framing spaces into $\text{Hom}(D_i^{\text{vec}}, V_i) \otimes 1_{R_i}$ and their reverses. Every adjacent intertwiner line is one-dimensional. A nonzero intertwiner has T^*T a positive scalar, so it can be normalized to an isometric inclusion, unique up to phase.

Tensor-product orthonormal bases give $\|F \otimes T\|_{\text{HS}}^2 = \|F\|_{\text{HS}}^2 \|T\|_{\text{HS}}^2$. Since the inclusions have domains of dimensions i and $i+1$,

$$\begin{aligned} \|a_i \otimes T_i^+\|_{\text{HS}}^2 &= i \|a_i\|_{\text{HS}}^2, & \|b_i \otimes T_{i+1}^-\|_{\text{HS}}^2 &= (i+1) \|b_i\|_{\text{HS}}^2, \\ \|p_i \otimes 1_{R_i}\|_{\text{HS}}^2 &= i \|p_i\|_{\text{HS}}^2, & \|q_i \otimes 1_{R_i}\|_{\text{HS}}^2 &= i \|q_i\|_{\text{HS}}^2. \end{aligned}$$

Distinct blocks are orthogonal. The displayed factors in S thus make it a complex-linear isometric bijection.

Step 2. In this step, we choose the reverse phases so that S intertwines the quaternionic structures. The external $\text{SU}(2)$ -action commutes with J_{ADHM} , so the fixed space is quaternionic. On an adjacent pair, initially chosen isometric intertwiners satisfy

$$J_{\text{ADHM}}(a_i \otimes T_i^+) = c_i a_i^* \otimes T_{i+1}^-.$$

Isometry and the preceding norm factors give $|c_i| = \sqrt{i/(i+1)}$. Choose the reverse phase so that $c_i = \sqrt{i/(i+1)}$. The identity $J_{\text{ADHM}}^2 = -1$ then gives

$$J_{\text{ADHM}}(b_i \otimes T_{i+1}^-) = -\sqrt{(i+1)/i} b_i^* \otimes T_i^+.$$

After inserting the square-root normalizations, these are exactly the two arrow rules of J_E . On the framing blocks the formula for J_{ADHM} sends $i^{-1/2} p_i \otimes 1$ to $i^{-1/2} p_i^* \otimes 1$ in the reverse block, and sends $i^{-1/2} q_i \otimes 1$ to $-i^{-1/2} q_i^* \otimes 1$ in the forward block. Thus $SJ_E = J_{\text{ADHM}}S$. Complex linearity now gives compatibility with all three complex structures and Kähler forms.

Step 3. In this step, we prove equivariance and equality of the dual-valued moment maps and their simultaneous zero sets. The block action of (g_i) is

$$a_i \mapsto g_{i+1} a_i g_i^{-1}, \quad b_i \mapsto g_i b_i g_{i+1}^{-1}, \quad p_i \mapsto g_i p_i, \quad q_i \mapsto q_i g_i^{-1}.$$

It makes S equivariant. For every $\xi \in \text{Lie } C_\lambda$ and every component of the moment map,

$$d\langle \mu_{\lambda,a} \circ S, \xi \rangle = S^*(\iota_{\xi\#} \omega_{\lambda,a}) = \iota_{\xi\#} \omega_{E,a} = d\langle \mu_{E,a}, \xi \rangle.$$

Both functions vanish at zero, which removes the constant of integration. Hence $\mu_\lambda \circ S = \mu_E$.

The ambient ADHM moments of a fixed datum commute with λ . They therefore belong to the centralizer algebra, and vanishing of their restricted dual-valued moments is equivalent to their full vanishing. More explicitly,

$$-\text{Tr}_{V_i \otimes R_i}((\xi \otimes 1)(\eta \otimes 1)) = i(-\text{Tr}_{V_i}(\xi\eta)).$$

A given covector is represented by i^{-1} times the usual quiver endomorphism when the ambient trace pairing is used. Its zero set is unchanged. This proves the last assertion as well as preservation of the half-norm. $\square$

Definition 8.7. A point (a_i, b_i, p_i, q_i) of $E(d, v)$ is *costable* if the only collection $U_i \subseteq V_i$ preserved by all arrows and satisfying $U_i \subseteq \ker q_i$ is the zero collection. An ADHM datum is *stable* if no proper B_1, B_2 -invariant subspace contains $P(W_+)$, and is *costable* if no nonzero such subspace lies in $\ker Q$. Write Λ^{sc} for the locus in Λ that is both quiver-stable and costable.

Lemma 8.8. For the data of Definition 7.1, the map S of Lemma 8.6 identifies quiver stability and costability with the corresponding ADHM conditions. On the complex moment-zero fiber, ADHM regularity is equivalent to stability and costability together. On the simultaneous real and complex zero fiber, stability, costability and regularity are equivalent, and every regular datum has trivial compact and complex gauge stabilizers.

The real equation is essential for replacing the pair of stability conditions by either one of them.

Proof. We compare invariant subspaces, prove the monad criterion, and use the real moment equation to relate the two stability conditions.

Step 1. In this step, we identify the quiver and ADHM stability conditions through their multiplicity spaces. The smallest B_1, B_2 -invariant subspace containing $P(W_+)$ is the span of words in B_1, B_2 applied to $P(W_+)$. Covariance of the pair (B_1, B_2) and equivariance of P make this subspace $SU(2)$ -invariant. It has the form $\bigoplus_i U_i \otimes R_i$. Under S , invariance and containment of $P(W_+)$ become exactly arrow-invariance and containment of every $p_i(D_i^{\text{vec}})$. The largest invariant subspace in $\ker Q$, obtained by summing all such subspaces, is likewise $SU(2)$ -invariant. Its multiplicity spaces are exactly the arrow-invariant collections in $\ker q_i$. This proves both comparisons.

Step 2. In this step, we prove that monad regularity is equivalent to stability and costability on the complex zero fiber. For the complex equation $[B_1, B_2] + PQ = 0$, write α_x, β_x for the two maps displayed in Definition 7.1. Failure of surjectivity of β_x gives a nonzero covector ℓ such that

$$\ell(B_j - x_j) = 0 \quad (j = 1, 2), \quad \ell P = 0.$$

Its kernel violates stability. Conversely, a proper invariant subspace containing $P(W_+)$ gives a quotient on which B_1, B_2 commute. A common left eigenvector on this nonzero quotient gives such an ℓ , and hence a failure of surjectivity. Failure of injectivity of α_x gives a common eigenline in $\ker Q$. Conversely, on an invariant nonzero subspace in $\ker Q$ the complex equation makes B_1, B_2 commute; a common eigenvector is in the kernel of some α_x . This proves the regularity criterion.

Step 3. In this step, we prove equivalence of the stability conditions at real zero and triviality of regular gauge stabilizers. Now impose the real equation. If a stable datum has an invariant subspace $U \subseteq \ker Q$, decompose $K_+ = U \oplus U^\perp$ and write $C_j : U^\perp \rightarrow U$ for the upper-right block of B_j . Taking the trace over U gives

$$0 = \sum_{j=1}^2 \|C_j\|_{\text{HS}}^2 + \|\text{pr}_U P\|_{\text{HS}}^2.$$

Thus $U^\perp$ is invariant and contains $P(W_+)$. Stability forces $U = 0$. Conversely, if $T \subsetneq K_+$ is invariant and contains $P(W_+)$, let $C'_j : T^\perp \rightarrow T$ be the upper-right blocks. The trace over $T^\perp$ gives

$$0 = - \sum_{j=1}^2 \|C'_j\|_{\text{HS}}^2 - \|Q|_{T^\perp}\|_{\text{HS}}^2.$$

Then $T^\perp$ is a nonzero invariant subspace in $\ker Q$, contradicting costability. Stability and costability coincide at real zero, and the complex criterion gives regularity.

Finally a complex gauge transformation fixing a stable quiver datum commutes with the arrows and fixes every framing image. It fixes all words in the arrows applied to those images, which generate every V_i , so it is the identity. This proves triviality for G_v and for its compact subgroup C_λ . Equivalently, the same word-span argument applies to the full ADHM gauge group on a stable datum. $\square$

We will use the extended quiver explicitly. Choose bases in D_i^{vec} and replace its framing by d_i arrows from a vertex ∞ of dimension one to vertex i , together with the reverse arrows. The extended dimension vector is

$$\alpha = (1, v_1, \dots, v_{r-1}).$$

The moment equation at ∞ follows by taking traces and summing the gauge-vertex equations. The effective base-change group $(\mathbb{C}^* \times G_v)/\mathbb{C}_{\text{diag}}^*$ is G_v : the class of $(c, (g_i))$ corresponds to

$(c^{-1}g_i)$. In particular the affine quotients agree, including their reduced structures. This is the construction in Theorem 8.4.

Lemma 8.9. *For the dominant data of Definition 7.1, Λ^{sc} is nonempty. Every point of Λ^{sc} has closed G_v -orbit and has a representative at real moment level zero, unique modulo C_λ . Consequently $\mathcal{R}_\lambda$ is nonempty and the extended dimension vector α admits a simple zero-parameter preprojective representation.*

This gives the existence premise for the smooth-locus criterion without using a prior identification of the physical map with the natural map.

Proof. We obtain a stable and costable point from density, prove closedness using the extended quiver, and then take its real-zero representative.

Step 1. In this step, we prove that the stable and costable loci meet in every irreducible component. Choose nondegenerate bilinear forms on each multiplicity space, and write $^\top$ for transpose. The algebraic automorphism

$$\sigma(\mathbf{a}_i, \mathbf{b}_i, p_i, q_i) = (-\mathbf{b}_i^\top, \mathbf{a}_i^\top, -q_i^\top, p_i^\top)$$

preserves the complex zero fiber. Indeed, writing its vertex equation as

$$\mathbf{a}_{i-1}\mathbf{b}_{i-1} - \mathbf{b}_i\mathbf{a}_i + p_iq_i = 0,$$

with missing arrows zero, the transformed left-hand side is the negative transpose of the original one. Taking annihilators of invariant collections shows that σ exchanges stability and costability. Stable density in Theorem 8.2 therefore gives dense costability in every irreducible component too. The two open dense sets meet in each component, proving $\Lambda^{\text{sc}} \neq \emptyset$.

Step 2. In this step, we identify stable and costable data with simple extended representations and prove closedness of their orbits. For any framed datum, simplicity of its extended representation is equivalent to stability and costability together. A subrepresentation with zero space at ∞ is an invariant collection in all $\ker q_i$, and costability makes it zero. A subrepresentation with full space at ∞ contains every framing image, and stability makes it the entire representation. These are the only possibilities at the dimension-one vertex.

The orbit of such a simple representation is closed. One can see the closedness criterion directly from one-parameter subgroups: if a one-parameter subgroup has a finite limit, its weight filtration is preserved by every arrow. Simplicity forces that filtration to have just one nonzero step, so all nonzero vertex spaces have one common weight. The subgroup then acts trivially on every arrow. There is no one-parameter degeneration outside the orbit. The affine Hilbert–Mumford criterion gives closedness. The effective group identification above makes this the original G_v -orbit.

Step 3. In this step, we obtain regular simultaneous-zero representatives by Kempf–Ness. Apply the affine Kempf–Ness homeomorphism of Theorem 4.1 to this closed orbit in the invariant closed cone $\Lambda \subset E(d, v)$. It meets the real zero level in one compact orbit. Stability and costability are unchanged by complex gauge transformations. The normalized map S and Lemma 8.8 therefore give a regular fixed-lift ADHM point at simultaneous zero. This proves both nonemptiness and the asserted simple-representation premise. $\square$

Lemma 8.10. *For the dominant data of Definition 7.1, the affine Kempf–Ness map and the normalized coordinates identify $\mathcal{R}_\lambda$ biholomorphically with $(\mathfrak{M}^0)_{\text{reg}}$. The restriction of ϕ then gives a biholomorphism $u : \mathcal{R}_\lambda \rightarrow L$.*

Here surjectivity uses the nonemptiness proved in Lemma 8.9. It is not a consequence of generic-stability freeness on the resolution.

Proof. We first verify local smoothness at a regular real-zero representative, then prove coverage and compatibility of complex structures.

Step 1. In this step, we construct saturated affine principal neighborhoods and prove smoothness at regular quotient points. Let $z \in \Lambda^{\text{sc}}$ be at real zero. Its complex orbit is closed by Kempf–Ness, and its complex stabilizer is trivial by Lemma 8.8. The moment differential identity used in Proposition 8.5 shows that Λ is smooth at z .

We construct the required saturated affine neighborhood. Let $U = \Lambda^{\text{sc}}$. This is invariant and open, every point of U is smooth, and its orbits are closed with trivial stabilizers by Lemmas 8.8 and 8.9. The fiber $q_0^{-1}(q_0(z))$ consists of the orbit $G_v z$ alone. Indeed, every orbit in that fiber has $G_v z$ in its closure; a different such orbit would have larger dimension than $\dim G_v$, since a proper boundary orbit has smaller dimension. This is impossible. Hence $q_0(z)$ is outside the closed set $q_0(\Lambda \setminus U)$; the image is closed because $\Lambda \setminus U$ is closed and invariant in an affine reductive quotient. Choose $f \in \mathbb{C}[\mathfrak{M}^0]$ with

$$f(q_0(z)) \neq 0, \quad f|_{q_0(\Lambda \setminus U)} = 0.$$

Then $\Lambda_f = q_0^{-1}((\mathfrak{M}^0)_f)$ is an invariant saturated affine principal neighborhood contained in U . It is smooth and every orbit is closed and free. For the trivial character all its points are semistable. The local assertion of Theorem 8.3 applies and gives a smooth quotient $(\mathfrak{M}^0)_f$ and a principal G_v -bundle. In particular $q_0(z)$ is smooth.

Step 2. In this step, we prove that every smooth affine point has a regular simultaneous-zero representative. By Lemma 8.9, a simple extended representation of dimension α exists. Thus $\alpha \in \Sigma_{0,0}$ in Theorem 8.4. At these parameters its canonical polystability condition is exactly simplicity. That theorem identifies all smooth points of the reduced affine quotient with simple representations. Simplicity is stability and costability by the preceding lemma. The closed orbit has a real-zero representative, unique up to C_λ , and Lemma 8.8 makes it ADHM-regular. This proves surjectivity onto $(\mathfrak{M}^0)_{\text{reg}}$; uniqueness in Kempf–Ness proves injectivity.

Step 3. In this step, we identify the quotient complex structures and obtain the biholomorphism u . On each principal neighborhood just constructed, at a real-zero representative let $\mathcal{V}_z = \text{Lie}(C_\lambda) \cdot z$ and let $\mathcal{H}_z$ be the orthogonal complement of $\mathcal{V}_z$ in the tangent space of the simultaneous-zero set. Equivariance of the moment maps at zero makes $\mathcal{V}_z$ isotropic for all three Kähler forms. Thus $\mathcal{V}_z, I\mathcal{V}_z, J\mathcal{V}_z, K\mathcal{V}_z$ are mutually orthogonal; freeness makes each have dimension $\dim C_\lambda$. The moment differential identities give $T_z \Lambda = (J\mathcal{V}_z \oplus K\mathcal{V}_z)^\perp$ and show that $\mathcal{H}_z$ is the quaternionic orthogonal complement of their fourfold sum. Consequently

$$T_z \Lambda = \mathcal{H}_z \oplus \mathcal{V}_z \oplus I\mathcal{V}_z \quad \text{as real vector spaces.}$$

Here $\mathcal{H}_z$ is invariant under I , and $\mathcal{V}_z \oplus I\mathcal{V}_z$ is the tangent to the complex gauge orbit. The differential of q_0 therefore identifies $\mathcal{H}_z$, with its quotient complex structure, complex-linearly with the tangent of the algebraic quotient. The induced bijection is a local biholomorphism and hence a biholomorphism. Composing it with the algebraic isomorphism ϕ and using Proposition 8.5 proves the assertion about u . $\square$

Theorem 8.11. *For the dominant data of Definition 7.1, normalized Clebsch–Gordan coordinates identify the fixed ADHM space with the standard doubled framed quiver space by a centralizer-equivariant linear hyperkähler isometry, preserving the dual-valued moment maps and half-norm. The affine Kempf–Ness map followed by ϕ restricts to a biholomorphism $u : \mathcal{R}_\lambda \rightarrow L$, and transports the quotient metric to a homogeneous hyperkähler potential package of raw weight two on X_A . Its global potential Φ_{phys} restricts to the physical ADHM half-norm and satisfies*

$$\text{dd}^c \Phi_{\text{phys}} = \omega_I, \quad \nabla_g \Phi_{\text{phys}} = E_{\text{raw}}, \quad \Phi_{\text{phys}} = \frac{1}{2} g(E_{\text{raw}}, E_{\text{raw}})$$

on L . The map u is equivariant and the transported holomorphic symplectic form is algebraic. The map u is not identified with the natural map v by this statement.

The potential on the singular affine variety is defined by the compact quotient. The smooth hyperkähler structure is transported through u on the whole leaf L .

Proof. The preceding lemmas supply the normalized linear comparison and the whole-leaf parametrization. We construct the global function on the affine quotient, verify its restriction and homogeneity, and descend the algebraic symplectic form.

Step 1. In this step, we construct the global potential on X_A and identify its normalization with the physical ADHM half-norm. Apply Proposition 4.2 to the closed invariant complex cone $\Lambda \subset E(d, v)$, the compact group C_λ , and its complexification G_v . The descended function ρ on $\mathfrak{M}^0$ is continuous, proper, nonnegative and plurisubharmonic, has its only zero at the quotient origin, and has scalar modulus weight two. Proposition 8.5 makes $\mathfrak{M}^0$ normal and positive-dimensional, so the same half-norm proposition gives plurisubharmonicity of $\log \rho$, with value $-\infty$ at the origin. Define

$$\Phi_{\text{phys}} = \rho \circ \phi^{-1} : X_A \longrightarrow [0, \infty).$$

The algebraic isomorphism ϕ preserves all these properties and sends the origin to e . The norm is precisely the physical ADHM norm: Lemma 8.6 proves

$$\frac{1}{2} \|z\|_E^2 = \frac{1}{2} (\|B_1\|_{\text{HS}}^2 + \|B_2\|_{\text{HS}}^2 + \|P\|_{\text{HS}}^2 + \|Q\|_{\text{HS}}^2), \quad (B_1, B_2, P, Q) = S(z).$$

Thus its restriction to $\mathcal{R}_\lambda$ is the same function used in Proposition 7.5.

Step 2. In this step, we transport the regular quotient metric and verify its raw Kazhdan generator identities. By Lemma 8.8, the compact centralizer acts freely on the regular simultaneous-zero locus. The linear hyperkähler isometry S identifies its reduction with the physical ADHM reduction, including horizontal tangent spaces. Transport this metric and its three complex structures by the biholomorphism u of Lemma 8.10. Its distinguished complex structure is the algebraic complex structure of L . Proposition 7.5 identifies the restricted half-norm as its hyperkähler potential and gives its gradient and half-squared-generator identities.

For precision, if $c \in \mathbb{C}^*$ multiplies every coordinate, the complex and real moments transform differently:

$$\mu_{\mathbb{C}}(cz) = c^2 \mu_{\mathbb{C}}(z), \quad \mu_{\mathbb{R}}(cz) = |c|^2 \mu_{\mathbb{R}}(z).$$

Both zero equations are preserved. Stability and costability are preserved too, since multiplying all arrows and framing maps by a nonzero scalar leaves their invariant-subspace conditions unchanged. The linear map S , the compact quotient map and the affine quotient map commute with this scalar action. The equivariance of ϕ in Proposition 8.5 therefore gives

$$u(c[z]) = \kappa_c u([z]), \quad \Phi_{\text{phys}}(\kappa_c x) = |c|^2 \Phi_{\text{phys}}(x).$$

The scalar logarithmic generator in Proposition 7.5 transports to E_{raw} . Its three identities become exactly the displayed identities in the theorem. In particular Φ_{phys} is smooth and positive on L .

Step 3. In this step, we descend the algebraic symplectic form and verify the remaining weight-two package properties. The constant algebraic two-form Ω_E restricts to a basic form on Λ^{sc} : contraction with a gauge vector is the restriction of $d\langle \mu_{\mathbb{C}}, \xi \rangle$, which vanishes on the moment fiber. On the saturated affine principal neighborhoods in the proof of Lemma 8.10, descent along the principal algebraic quotient gives an algebraic two-form. These local descents agree, since their pullbacks agree. They define the nondegenerate reduced form on $(\mathfrak{M}^0)_{\text{reg}}$. Transport by ϕ gives a closed algebraic form $\Omega = \omega_J + i\omega_K$ on L . The constant form has scalar weight two, so

$$\kappa_c^* \Omega = c^2 \Omega.$$

This argument uses the reduced quiver form itself and applies to the zero-block data as well.

The normal positive cone and codimension-two assertions follow from Proposition 8.5; its contraction formula also gives $\mathbb{C}[X_A]_0 = \mathbb{C}$. The point e is outside L : if $e \in \mathcal{O}_A$, transversality at e would give $\dim L = 0$, contrary to the hypothesis. Together with the global properties of Φ_{phys} and the metric and form just constructed, these verify every condition of Definition 2.4,

with raw weight two. If $\dim_{\mathbb{C}} X_A = 2m$, Lemma 4.7 further gives the same normalization on the whole leaf:

$$\frac{\omega_I^{2m}}{(2m)!} = \frac{\Omega^m \wedge \overline{\Omega}^m}{4^m (m!)^2}.$$

□

9. TRANSPORT TO NATURAL COORDINATES

We retain the dominant type-A data of Definition 7.1. The whole affine model supplies a potential in the coordinates of the map u , while the natural map v identifies the complex structures needed for faithful pullback. We compare these parametrizations by extending their discrepancy over the affine cone. The extension transports both the global potential and the algebraic holomorphic symplectic form into natural coordinates.

We first recall the scalar Hartogs theorem [BO23, Lemma 2.6].

Theorem 9.1. *Let X be a normal complex space and $A \subset X$ an analytic subset of codimension at least two. Every holomorphic function on $X \setminus A$ extends uniquely to a holomorphic function on X .*

This theorem extends the coordinate functions of a holomorphic map. A positive grading will make those extended functions algebraic.

Proposition 9.2. *Let $X = \operatorname{Spec} A$ be a normal irreducible affine variety with a positive integer grading, and let $L \subset X$ be an invariant open subset whose complement has codimension at least two. Every $\mathbb{C}^*$ -equivariant biholomorphism $\beta : L \rightarrow L$ extends uniquely to a graded algebraic automorphism $\mathcal{B} : X \rightarrow X$. If $u, v : R \rightarrow L$ are equivariant biholomorphisms and $\beta = v \circ u^{-1}$, then*

$$\mathcal{B}^{-1} \circ v = u, \quad (\Phi \circ \mathcal{B}^{-1}) \circ v = \Phi \circ u$$

for every function Φ on X . Inverse pullback through $\mathcal{B}$ preserves continuity, properness, plurisubharmonicity, logarithmic plurisubharmonicity, and homogeneity of a potential, and preserves algebraicity and weight of differential forms on L .

Thus the comparison on the big open determines a change of coordinates on the entire cone, including its vertex and singular locus.

Proof. We extend homogeneous coordinate functions by Hartogs, use their Taylor expansions at the vertex to prove algebraicity, and repeat the construction for the inverse. We then check the effect of inverse pullback.

Step 1. In this step, we extend the homogeneous coordinate functions by Hartogs while preserving their weights. Write $A = \bigoplus_{m \geq 0} A_m$, with $A_0 = \mathbb{C}$, and denote the vertex by x_0 . For $b \in A_m$, the function $b \circ \beta$ is holomorphic on L^{an} . Normality of X gives normality of X^{an} , and $X \setminus L$ has analytic codimension at least two. Theorem 9.1 therefore gives a unique holomorphic extension ϑ_b to X^{an} . On L , equivariance gives

$$\vartheta_b(tx) = b(\beta(tx)) = b(t\beta(x)) = t^m \vartheta_b(x).$$

For each fixed $t \in \mathbb{C}^*$, the two holomorphic functions in this identity agree on the dense open L . Uniqueness of extension gives the identity on all of X^{an} .

Step 2. In this step, we prove that a homogeneous holomorphic function on the positive cone is algebraic. Let ϑ be a holomorphic function on X^{an} satisfying $\vartheta(tx) = t^m \vartheta(x)$ for a fixed integer $m \geq 0$. Choose homogeneous algebra generators $a_1, \dots, a_k$ of positive weights $w_1, \dots, w_k$. They define a closed equivariant embedding

$$j : X \hookrightarrow \mathbb{C}^k, \quad t \cdot (z_1, \dots, z_k) = (t^{w_1} z_1, \dots, t^{w_k} z_k), \quad j(x_0) = 0.$$

If $A = \mathbb{C}$, the assertion is immediate, so we may assume $k > 0$. The germ of ϑ at x_0 is the restriction of an ambient holomorphic function. Choose such a representative on a polydisc about zero, with convergent Taylor expansion

$$\tilde{\vartheta}(z) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^k} c_\alpha z^\alpha.$$

The polydisc is invariant under the weighted circle action. On a smaller polydisc the expansion converges normally, so its weight- m circle projection can be integrated term by term:

$$(9.1) \quad \begin{aligned} p_m(z) &= \frac{1}{2\pi} \int_0^{2\pi} e^{-im\theta} \tilde{\vartheta}(e^{iw_1\theta} z_1, \dots, e^{iw_k\theta} z_k) d\theta \\ &= \sum_{\sum_j w_j \alpha_j = m} c_\alpha z^\alpha. \end{aligned}$$

Every surviving multi-index satisfies $0 \leq \alpha_j \leq m/w_j$. Consequently the sum in (9.1) is finite, and p_m is a weighted homogeneous polynomial of degree m .

On X near x_0 , the integrand restricts to $e^{-im\theta} \vartheta(e^{i\theta} x) = \vartheta(x)$. Hence $p_m \circ j = \vartheta$ there. For an arbitrary $x \in X$, positivity of the weights gives $j(sx) \rightarrow 0$ as $s \rightarrow 0$ through positive real numbers. Choose $s > 0$ so that sx belongs to the neighborhood of equality. Then

$$s^m \vartheta(x) = \vartheta(sx) = p_m(j(sx)) = s^m p_m(j(x)).$$

Canceling s^m proves $\vartheta = p_m \circ j$ on all of X . In particular, ϑ is the regular function represented by an element of A_m .

Step 3. In this step, we construct the graded extensions of β and β^{-1} and prove that they are inverse. Applying the preceding argument to ϑ_b gives a degree-preserving map on homogeneous functions. Extend it to A by finite homogeneous decomposition. Addition, multiplication and the unit are preserved because they are preserved by pullback through β on L , and regular functions agreeing on the dense open L agree on X . We obtain a graded $\mathbb{C}$ -algebra homomorphism

$$\mathcal{B}^* : A \longrightarrow A, \quad b \longmapsto \vartheta_b,$$

and thus an equivariant algebraic morphism $\mathcal{B} : X \rightarrow X$ restricting to β .

The inverse $\beta^{-1} : L \rightarrow L$ is also an equivariant biholomorphism. The same construction extends it to an equivariant algebraic morphism $\Gamma : X \rightarrow X$. Both $\Gamma \circ \mathcal{B}$ and $\mathcal{B} \circ \Gamma$ are the identity on L . The target X is separated, so their equalizers with the identity are closed; density of L and reducedness of X give

$$\Gamma \circ \mathcal{B} = \text{id}_X, \quad \mathcal{B} \circ \Gamma = \text{id}_X.$$

Thus $\Gamma = \mathcal{B}^{-1}$ and $\mathcal{B}$ is a graded algebraic automorphism. The same density argument proves uniqueness. Its preservation of the homogeneous maximal ideal shows that $\mathcal{B}(x_0) = x_0$.

Step 4. In this step, we verify the composition identities and preservation of the potential and differential-form data under inverse pullback. If $\beta = v \circ u^{-1}$, then $\mathcal{B}^{-1} \circ v = u$. Composing this identity with any function Φ gives the asserted equality on R . Put $\tilde{\Phi} = \Phi \circ \mathcal{B}^{-1}$. The map $\mathcal{B}$ is a homeomorphism and an analytic biholomorphism. Continuity is therefore preserved, and for every compact set C in the target of Φ ,

$$\tilde{\Phi}^{-1}(C) = \mathcal{B}(\Phi^{-1}(C)).$$

This proves preservation of properness. Biholomorphic pullback preserves plurisubharmonicity. For a nonnegative potential, $\log \tilde{\Phi} = (\log \Phi) \circ \mathcal{B}^{-1}$ also holds at zeros with the value $-\infty$, so logarithmic plurisubharmonicity is preserved as well. Nonnegativity and vanishing exactly at x_0 are preserved. If $\Phi(tx) = |t|^\ell \Phi(x)$, equivariance gives

$$\tilde{\Phi}(tx) = \Phi(t\mathcal{B}^{-1}x) = |t|^\ell \tilde{\Phi}(x).$$

Finally, $\mathcal{B}^{-1}|_L$ is algebraic. For an algebraic differential form η of weight ℓ on L , its pullback is algebraic and satisfies

$$t^*(\mathcal{B}^{-1}|_L)^*\eta = (\mathcal{B}^{-1}|_L)^*t^*\eta = t^\ell(\mathcal{B}^{-1}|_L)^*\eta.$$

This proves the assertions about forms. $\square$

We now apply this comparison to the two maps from the prescribed-lift quotient $\mathcal{R}_\lambda = \mathcal{E}_\lambda/C_\lambda$ of Definition 7.1. This is the regular simultaneous real and complex zero-level quotient, with its fixed lift and centralizer group. Its natural parametrization v retains the positive instanton frame and forgets the negative frame. The map u of Theorem 8.11 instead uses affine Kempf–Ness and the whole-affine Maffei isomorphism. Write

$$(g_{\mathcal{R}}, I_{\mathcal{R}}, J_{\mathcal{R}}, K_{\mathcal{R}}), \quad \Omega_{\mathcal{R}} = \omega_{\mathcal{R},J} + i\omega_{\mathcal{R},K}$$

for the ADHM quotient tensors. We denote their transports through u by (g_u, I_u, J_u, K_u) and Ω_u ; in particular,

$$g_u = (u^{-1})^*g_{\mathcal{R}}, \quad \Omega_u = (u^{-1})^*\Omega_{\mathcal{R}}.$$

Theorem 8.11 gives the global function Φ_{phys} and the algebraicity of Ω_u .

Theorem 9.3. *For the dominant type-A data, the natural parametrization $v : \mathcal{R}_\lambda \rightarrow L$ transports the ADHM metric to a homogeneous hyperkähler potential package of raw Kazhdan weight two on X_A , with the natural twistor triple on L . More precisely, if $\mathcal{B}$ extends $v \circ u^{-1}$, then*

$$\Phi_{\text{nat}} = \Phi_{\text{phys}} \circ \mathcal{B}^{-1}, \quad \Omega_{\text{nat}} = (\mathcal{B}^{-1}|_L)^*\Omega_u.$$

On L this package satisfies

$$\text{dd}^c \Phi_{\text{nat}} = \omega_I, \quad \nabla_g \Phi_{\text{nat}} = E_{\text{raw}}, \quad \Phi_{\text{nat}} = \frac{1}{2}g(E_{\text{raw}}, E_{\text{raw}}).$$

The metric in this statement is fixed by transport of the ADHM quotient metric through v . The global potential is determined by the same transport on the regular quotient and the automorphism of the affine cone.

Proof. We first check the hypotheses for algebraizing $v \circ u^{-1}$. We then transport the quotient tensors and verify the potential identities on their common source. Finally we check the global potential and the algebraic form.

Step 1. In this step, we prove equivariance of $v \circ u^{-1}$ and extend it to a graded algebraic automorphism of X_A . Proposition 8.5 gives normality and irreducibility of X_A , connectedness of $L = (X_A)_{\text{reg}}$, and $\text{codim}_{X_A}(X_A \setminus L) \geq 2$. The raw Kazhdan action is the positive integer grading of Theorem 8.11, and L is invariant. Theorems 8.11 and 7.8 give the two biholomorphisms from the same complex manifold $(\mathcal{R}_\lambda, I_{\mathcal{R}})$ to L .

Let σ_t be the action on $\mathcal{R}_\lambda$ induced by multiplication of all ADHM coordinates by $t \in \mathbb{C}^*$. This is a holomorphic action in $I_{\mathcal{R}}$, as also seen through the equivariant biholomorphism u . Proposition 5.8 fixes the parameter in the natural map: inverse spatial dilation corresponds to $T(t) \mapsto T(t - \log s)$ and then to κ_s , for $s > 0$. Thus the equivariance identities for the two parametrizations use the same positive-real parameter:

$$u(\sigma_s \zeta) = \kappa_s u(\zeta), \quad v(\sigma_s \zeta) = \kappa_s v(\zeta) \quad (s > 0, \zeta \in \mathcal{R}_\lambda).$$

The passage to complex parameters uses the identity theorem. For fixed ζ , both $t \mapsto v(\sigma_t \zeta)$ and $t \mapsto \kappa_t v(\zeta)$ are holomorphic maps from $\mathbb{C}^*$ into the affine variety X_A . Compose them with each coordinate function of an affine embedding of X_A . The resulting holomorphic functions agree for all $t > 0$, a set with a limit point in $\mathbb{C}^*$, and hence agree on $\mathbb{C}^*$. Coordinate functions separate points, so the two maps agree there. The same argument applies to u . Therefore $\beta = v \circ u^{-1}$ is a $\mathbb{C}^*$ -equivariant biholomorphism of L . Proposition 9.2 supplies its unique graded algebraic extension $\mathcal{B} : X_A \rightarrow X_A$ and the identity

$$(9.2) \quad \mathcal{B}^{-1} \circ v = u.$$

Step 2. In this step, we identify the transported hyperkähler tensors with the natural complex structures on L . Define $g = (v^{-1})^*g_{\mathcal{R}}$ and transport the complex structures by conjugation with the differential. For example, at $\zeta \in \mathcal{R}_\lambda$ set

$$I_{v(\zeta)} = dv_\zeta I_{\mathcal{R},\zeta} (dv_\zeta)^{-1},$$

and use the same formula for J and K . Transport through a diffeomorphism preserves the quaternion relations and the parallelism of these endomorphisms for the transported Levi-Civita connection. Thus (g, I, J, K) is hyperkähler. Theorem 7.8 identifies each of these complex structures with the corresponding natural Kronheimer complex structure on L . In particular, I is its algebraic complex structure.

The associated covariant forms satisfy

$$\omega_I = (v^{-1})^*\omega_{\mathcal{R},I}, \quad \omega_J = (v^{-1})^*\omega_{\mathcal{R},J}, \quad \omega_K = (v^{-1})^*\omega_{\mathcal{R},K}.$$

Since $v^{-1} = u^{-1} \circ \mathcal{B}^{-1}|_L$, these identities also give

$$g = (\mathcal{B}^{-1}|_L)^*g_u, \quad \omega_J + i\omega_K = (\mathcal{B}^{-1}|_L)^*\Omega_u = \Omega_{\text{nat}}.$$

Step 3. In this step, we verify the potential and raw Euler identities by pulling them back to the ADHM quotient. Let $E_{\mathcal{R}}$ be the logarithmic positive-real generator of σ , and let $\Phi_{\mathcal{R}} = \Phi_{\text{phys}} \circ u$. On a regular simultaneous-zero representative its value is

$$\Phi_{\mathcal{R}}([B_1, B_2, P, Q]) = \frac{1}{2}(\|B_1\|_{\text{HS}}^2 + \|B_2\|_{\text{HS}}^2 + \|P\|_{\text{HS}}^2 + \|Q\|_{\text{HS}}^2).$$

Theorem 8.11, pulled back by u , gives

$$\text{dd}^c \Phi_{\mathcal{R}} = \omega_{\mathcal{R},I}, \quad \nabla_{g_{\mathcal{R}}} \Phi_{\mathcal{R}} = E_{\mathcal{R}}, \quad \Phi_{\mathcal{R}} = \frac{1}{2}g_{\mathcal{R}}(E_{\mathcal{R}}, E_{\mathcal{R}}).$$

Define $\Phi_{\text{nat}} = \Phi_{\text{phys}} \circ \mathcal{B}^{-1}$ on X_A . Equation (9.2) gives the literal equality of functions on $\mathcal{R}_\lambda$

$$(9.3) \quad \Phi_{\text{nat}} \circ v = \Phi_{\text{phys}} \circ \mathcal{B}^{-1} \circ v = \Phi_{\text{phys}} \circ u = \Phi_{\mathcal{R}}.$$

Because v is holomorphic in the distinguished complex structure, with the fixed convention $\text{dd}^c = i\partial\bar{\partial}$ we have

$$v^*(\text{dd}^c \Phi_{\text{nat}}) = \text{dd}^c(\Phi_{\text{nat}} \circ v) = \omega_{\mathcal{R},I} = v^*\omega_I.$$

Pullback by the diffeomorphism v is injective, proving $\text{dd}^c \Phi_{\text{nat}} = \omega_I$.

Differentiating equivariance at $s = 1$ in logarithmic time gives $dv(E_{\mathcal{R}}) = E_{\text{raw}}$. For $\xi \in T_\zeta \mathcal{R}_\lambda$, equation (9.3) and the quotient gradient identity yield

$$\begin{aligned} d\Phi_{\text{nat}}|_{v(\zeta)}(dv_\zeta \xi) &= d\Phi_{\mathcal{R}}|_\zeta(\xi) \\ &= g_{\mathcal{R},\zeta}(E_{\mathcal{R}}, \xi) = g_{v(\zeta)}(E_{\text{raw}}, dv_\zeta \xi). \end{aligned}$$

Surjectivity of dv_ζ proves $\nabla_g \Phi_{\text{nat}} = E_{\text{raw}}$. The same equality of functions gives

$$\Phi_{\text{nat}}(v(\zeta)) = \frac{1}{2}g_{\mathcal{R},\zeta}(E_{\mathcal{R}}, E_{\mathcal{R}}) = \frac{1}{2}g_{v(\zeta)}(E_{\text{raw}}, E_{\text{raw}}).$$

These use the raw action throughout, so the generator and the factor $1/2$ have the same normalization as on the ADHM quotient.

Step 4. In this step, we verify the global properties, weights, and volume normalization of the natural-coordinate package. The global properties of Φ_{phys} are supplied by Theorem 8.11. Proposition 9.2 therefore shows that Φ_{nat} is continuous, proper and nonnegative on X_A , vanishes exactly at the vertex e , and is plurisubharmonic together with its logarithm, valued $-\infty$ at e . It is smooth and positive on L by (9.3). Equivariance of $\mathcal{B}$ gives, on the entire affine cone,

$$\Phi_{\text{nat}}(\kappa_t x) = \Phi_{\text{phys}}(\kappa_t \mathcal{B}^{-1} x) = |t|^2 \Phi_{\text{nat}}(x).$$

Since Ω_u is algebraic of raw weight two and $\mathcal{B}^{-1}|_L$ is algebraic and equivariant, Ω_{nat} is algebraic and

$$\kappa_t^* \Omega_{\text{nat}} = (\mathcal{B}^{-1}|_L)^* \kappa_t^* \Omega_u = t^2 \Omega_{\text{nat}}.$$

Closedness and nondegeneracy follow by pullback through the isomorphism $\mathcal{B}^{-1}|_L$. Together with the big-open property already checked, these verify the homogeneous potential package.

For completeness, if $\dim_{\mathbb{C}} X_A = 2n$, the transported positive hyperkähler structure also satisfies the fixed volume normalization by Lemma 4.7:

$$\frac{\omega_I^{2n}}{(2n)!} = \frac{\Omega_{\text{nat}}^n \wedge \overline{\Omega_{\text{nat}}}^n}{4^n (n!)^2}.$$

Thus the global function, the algebraic form and the natural twistor structure are obtained with the ADHM normalization. $\square$

10. FAITHFUL PULLBACK FOR GENERAL SLICES

We retain the normalization and raw Kazhdan action of Section 3. We first embed a component into the type-A slice determined by its dense orbit. The natural-coordinate potential on that slice then supplies a global potential on the normalization. The compatibility with the natural twistor structures comes from the faithful map of positive-framed trajectory spaces in Section 5.

Proposition 10.1. *Let X be a positive-dimensional component of a nilpotent orbit-slice intersection in a complex semisimple Lie algebra, with normalization $\nu : Z \rightarrow X$ and connected top stratum $L = X \cap \mathcal{O}_{\text{top}}$. Choose a faithful algebraic representation of the simply connected semisimple group on V , and write $\rho : \mathfrak{g} \hookrightarrow \mathfrak{sl}(V)$ for its differential. For $x_{\text{top}} \in L$, put*

$$X_A = \left(\overline{\text{SL}(V) \cdot \rho(x_{\text{top}})} \cap (\rho(e) + \mathfrak{sl}(V)^{\rho(f)}) \right)_{\text{red}}.$$

The represented orbit is independent of x_{top} . The map $\psi = \rho \circ \nu : Z \rightarrow X_A$ is finite, proper and raw-Kazhdan equivariant, its vertex fiber is a singleton, and it maps $M = \nu^{-1}(L)$ into the smooth top stratum of X_A . Its restriction to M is an immersion.

The choice of ambient orbit ensures that the entire smooth open M lands where the type-A hyperkähler metric is defined.

Proof. We identify the ambient orbit and slice, verify the finite equivariant map, and then locate its image on the smooth strata.

Step 1. In this step, we identify the ambient matrix orbit and prove that the represented component lies in its slice. Let ρ_G denote the chosen representation of the simply connected group G_{sc} , and set

$$e_A = \rho(e), \quad h_A = \rho(h), \quad f_A = \rho(f), \\ \mathcal{S}_A = e_A + \mathfrak{sl}(V)^{f_A}, \quad \mathcal{O}_{A,\text{top}} = \text{SL}(V) \cdot \rho(x_{\text{top}}).$$

The image of ρ_G lies in $\text{SL}(V)$: the derivative of $\det \rho_G$ is $x \mapsto \text{Tr}(\rho(x))$, which vanishes on $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, and G_{sc} is connected. For $a \in G_{\text{sc}}$ and $x \in \mathfrak{g}$ we have

$$\rho(\text{Ad}_a x) = \text{Ad}_{\rho_G(a)} \rho(x).$$

All points of L belong to $\mathcal{O}_{\text{top}}$, so their images belong to the single orbit $\mathcal{O}_{A,\text{top}}$. This proves its independence of x_{top} . The represented elements are nilpotent: an $\mathfrak{sl}_2$ -triple containing a nilpotent element acts on V as a sum of finite-dimensional $\mathfrak{sl}_2$ -modules, on each of which its raising element is nilpotent.

For $x \in X$, the slice equation gives

$$[\rho(x) - e_A, f_A] = \rho([x - e, f]) = 0.$$

Thus $\rho(L) \subset \mathcal{O}_{A,\text{top}} \cap \mathcal{S}_A$. By Proposition 3.5, $X = \overline{L}$. The injective linear map ρ is a closed immersion onto its image, and hence

$$\rho(X) = \overline{\rho(L)} \subset \overline{\mathcal{O}_{A,\text{top}}} \cap \mathcal{S}_A.$$

Since X is reduced, this inclusion factors through X_A .

Step 2. In this step, we prove finiteness and raw Kazhdan equivariance of ψ and identify its vertex fiber. The map ν is finite, and $\rho|_X : X \rightarrow X_A$ is a closed immersion. Their composite ψ is therefore finite and proper, with closed image $\rho(X)$. The homomorphism obtained by integrating the source triple is carried by ρ_G to the represented triple. Consequently its cocharacters satisfy

$$\rho(t^2 \operatorname{Ad}(t^{-h})x) = t^2 \operatorname{Ad}(t^{-h_A})\rho(x).$$

Proposition 3.4 gives the lifted action on Z and the equivariance of ν . Therefore

$$\psi \circ \kappa_t = \kappa_t^A \circ \psi.$$

Write z_0 for the vertex of Z . The same proposition and the injectivity of ρ give

$$\psi^{-1}(e_A) = \nu^{-1}(e) = \{z_0\}.$$

Step 3. In this step, we locate $\psi(M)$ in the smooth top stratum and prove that $\psi|_M$ is an immersion. Put $L_A = \mathcal{O}_{A,\text{top}} \cap \mathcal{S}_A$. We have $\psi(M) = \rho(L) \subset L_A$. In particular L_A has positive dimension. Also $e \in X$ by Proposition 3.4, so $e_A \in \overline{\mathcal{O}_{A,\text{top}}}$. The Jordan types of e_A and $\mathcal{O}_{A,\text{top}}$, padded with zeros, give the dominant data of Definition 7.1; the index bound there may be chosen larger than all their Jordan block sizes. Proposition 8.5 now identifies $L_A = (X_A)_{\text{reg}}$ and shows that it is connected. Finally, $\nu|_M : M \rightarrow L$ is an isomorphism by Proposition 3.5, and $\rho|_L$ has injective differential. Their composite into L_A is an immersion. $\square$

For the rest of the section, the subscript A denotes the natural type- A package of Theorem 9.3 on the envelope just constructed. In particular g_A , $\omega_{A,q}$, and $\Omega_{\text{nat},A}$ are defined on $L_A = (X_A)_{\text{reg}}$, while $\Phi_{\text{nat},A}$ is defined on all of X_A .

Theorem 10.2. *Let $\mathfrak{g}$ be complex simple, and let Z normalize a positive-dimensional component of a reduced nilpotent orbit-slice intersection. Then Z has a homogeneous hyperkähler potential package of raw Kazhdan weight two on the big open M of Proposition 3.5, with its natural twistor triple. For a faithful envelope $\psi : Z \rightarrow X_A$ as above, one such package is*

$$g_M = \psi^* g_A, \quad \omega_{M,q} = \psi^* \omega_{A,q}, \quad \Omega_M = \psi^* \Omega_{\text{nat},A}, \quad \Phi_Z = \Phi_{\text{nat},A} \circ \psi.$$

It satisfies $\nabla_{g_M} \Phi_Z = E_{\text{raw}}$ and $\Phi_Z = \frac{1}{2} g_M(E_{\text{raw}}, E_{\text{raw}})$ on M .

The natural complex structures on M are fixed by the trajectory construction. The faithful representation supplies a metric compatible with them and fixes the normalization of its holomorphic symplectic form.

Proof. We first identify $\psi|_M$ in the natural trajectory charts. We then verify the hyperkähler and Euler identities for the covariant pullbacks, and establish the global potential and algebraic form properties.

Step 1. In this step, we identify $\psi|_M$ in all natural trajectory charts, including their gauge and endpoint conditions. Write $\nu : Z \rightarrow X$ for the normalization map and let $\rho_G : G_{\text{sc}} \rightarrow \operatorname{SL}(V)$ be the faithful group representation defining $\psi = \rho \circ \nu$. Propositions 3.4 and 3.5 give the normal positive affine cone Z , its vertex z_0 , and its invariant smooth big open M , identified with the connected top leaf L . This open excludes z_0 : Theorem 3.1 gives dimension zero for the transverse leaf through e , whereas L has positive dimension. Choose a compact real form $K_c \subset G_{\text{sc}}$ compatible with the triple and a K_c -invariant Hermitian form on V . Then $\rho_c = \rho_G|_{K_c} : K_c \rightarrow \operatorname{U}(V)$ is faithful and $(d\rho_c)_{\mathbb{C}} = \rho$.

Choose the positive endpoint p_+ with complex triple $(f, -h, e)$ and a negative endpoint p_- representing $\mathcal{O}_{\text{top}}$. Theorem 5.6 identifies a connected component of the source positive-framed trajectory space with L . Its natural complex structures give integrable endomorphisms I_M, J_M, K_M on M satisfying the quaternion relations. The distinguished I_M is the algebraic complex structure. Apply Theorem 5.10 to K_c , ρ_c , and these two endpoints. For each twistor

direction q , choose an oriented decomposition realizing it, and write $\chi_{K_c, q}$ and $\chi_{V, q}$ for its source and target charts. The trajectory map F satisfies

$$\chi_{V, q} \circ F = \rho \circ \chi_{K_c, q}.$$

For the distinguished chart, this is exactly $\psi|_M = \rho \circ \nu|_M$. For the other directions the same identity applies to the corresponding natural charts. Thus, using the natural target triple supplied by Theorem 9.3, we obtain

$$(10.1) \quad d\psi \circ q_M = q_A \circ d\psi \quad (q = I, J, K).$$

The endpoint assertion of Theorem 5.10 applies to these precise one-framed spaces. In the covariant description of Definition 5.1, an allowed ambient gauge between image trajectories lifts uniquely to a source gauge with positive limit 1, negative limit in the simultaneous centralizer of dp_- , and convergence of every derivative at rates $O(e^{-2t})$ and $O(e^{2t})$ at the respective ends. Thus this identification includes the required gauge equivalences and endpoint conditions. The theorem also supplies injectivity of the map and its differential.

Step 2. In this step, we prove that the covariant pullbacks define a positive hyperkähler structure on M . Define g_M , $\omega_{M, q}$, Ω_M , and Φ_Z by the displayed formulas in the statement, taking the tensor pullbacks on M . For a nonzero real tangent vector v , immersion gives

$$g_M(v, v) = g_A(d\psi(v), d\psi(v)) > 0.$$

For real tangent vectors v, w , equation (10.1) gives

$$\omega_{M, q}(v, w) = g_A(q_A d\psi(v), d\psi(w)) = g_M(q_M v, w).$$

These forms are closed, since the ambient Kähler forms are closed. Each q_M is integrable, so g_M is Kähler for all three complex structures. Together with their quaternion relations, this proves that (g_M, I_M, J_M, K_M) is hyperkähler. In particular,

$$\Omega_M = \omega_{M, J} + i\omega_{M, K}$$

has type $(2, 0)$ for I_M . It is nondegenerate: for a nonzero real vector v ,

$$\Omega_M(v, J_M v) = g_M(v, v),$$

because $g_M(K_M v, J_M v) = 0$. A vector in the complex kernel would, by the $(2, 0)$ property, give a real vector in this kernel, which the displayed identity excludes.

Step 3. In this step, we use tangency of the raw Euler field to verify the pulled-back potential and gradient identities. Let $E_{A, \text{raw}}$ be the logarithmic positive-real generator on L_A . Differentiating the raw equivariance in Proposition 10.1 gives

$$(10.2) \quad d\psi(E_{\text{raw}}) = E_{A, \text{raw}} \circ \psi.$$

The map $\psi|_M$ is holomorphic for I_M and I_A , so pullback commutes with dd^c . The ambient potential identities of Theorem 9.3 therefore give

$$\text{dd}^c \Phi_Z = \psi^*(\text{dd}^c \Phi_{\text{nat}, A}) = \psi^* \omega_{A, I} = \omega_{M, I}.$$

For every real tangent vector v on M , the same theorem and equation (10.2) give

$$\begin{aligned} d\Phi_Z(v) &= d\Phi_{\text{nat}, A}(d\psi(v)) \\ &= g_A(E_{A, \text{raw}}, d\psi(v)) \\ &= g_M(E_{\text{raw}}, v). \end{aligned}$$

This proves $\nabla_{g_M} \Phi_Z = E_{\text{raw}}$. The ambient norm identity and the same tangency equation also give

$$\Phi_Z = \frac{1}{2} g_A(E_{A, \text{raw}}, E_{A, \text{raw}}) \circ \psi = \frac{1}{2} g_M(E_{\text{raw}}, E_{\text{raw}}).$$

Step 4. In this step, we verify the global continuity, properness, plurisubharmonicity, and homogeneity of Φ_Z . Theorem 9.3 supplies a continuous proper nonnegative $\Phi_{\text{nat}, A}$ on all of X_A ,

vanishing exactly at $e_A = \rho(e)$. Both this function and its logarithm, extended by $-\infty$ at e_A , are plurisubharmonic. Since ψ is holomorphic, their compositions are plurisubharmonic on Z . The function Φ_Z is continuous and nonnegative, and Proposition 10.1 gives

$$\Phi_Z^{-1}(0) = \psi^{-1}(e_A) = \{z_0\}.$$

It is smooth and positive on M because $\psi(M) \subset L_A$. For every real $R \geq 0$,

$$\{z \in Z : \Phi_Z(z) \leq R\} = \psi^{-1}(\{y \in X_A : \Phi_{\text{nat},A}(y) \leq R\})$$

is compact: the set on the right inside parentheses is compact by properness of the ambient potential, and ψ is proper. Thus Φ_Z is proper. Equivariance and the ambient weight give, for every $t \in \mathbb{C}^*$,

$$\Phi_Z(\kappa_t z) = \Phi_{\text{nat},A}(\kappa_t^A \psi(z)) = |t|^2 \Phi_Z(z).$$

Step 5. In this step, we verify the algebraic form, its reflexive extension and weight, and the volume normalization. The map $\psi|_M$ and the form $\Omega_{\text{nat},A}$ are algebraic, so Ω_M is algebraic. It is closed by pullback. The ambient weight and equivariance give

$$\kappa_t^* \Omega_M = \psi^*(\kappa_t^{A*} \Omega_{\text{nat},A}) = t^2 \Omega_M.$$

These facts, the nondegeneracy already proved, and the global potential properties verify every condition of Definition 2.4.

The pulled-back form also extends reflexively. If $j : M \hookrightarrow Z$ is the inclusion, normality and $\text{codim}_Z(Z \setminus M) \geq 2$ give

$$(\wedge^2 \Omega_Z^1)^{**} = j_*(\wedge^2 T_M^\vee).$$

Here Ω_Z^1 is the sheaf of Kähler differentials and $T_M^\vee$ is the cotangent bundle. Sections of a reflexive sheaf on a normal variety are determined by their regularity at codimension-one points, all of which belong to M . Hence the algebraic form Ω_M has a unique reflexive extension. Its raw weight remains two because both sides of the weight identity agree on the big open M and reflexive sheaves are torsion-free.

Writing $\dim_{\mathbb{C}} Z = 2n$, Lemma 4.7, applied to the metric and forms just constructed, yields the exact identity

$$\frac{\omega_{M,I}^{2n}}{(2n)!} = \frac{\Omega_M^n \wedge \overline{\Omega_M}^n}{4^n (n!)^2}.$$

Thus the algebraic form used in this identity has the normalization specified by the covariant pullback in the statement. $\square$

The raw action can now be effectivized without changing the potential or forms. If its kernel is μ_d , as in Proposition 3.4, then every $\zeta \in \mu_d$ satisfies

$$\Omega_M = \kappa_\zeta^* \Omega_M = \zeta^2 \Omega_M.$$

Since Ω_M is nonzero, $d \mid 2$. With effective parameter $s = t^d$, the potential and the algebraic form have weight $\ell = 2/d$. The gradient identity retains the raw generator: $\nabla_{g_M} \Phi_Z = E_{\text{raw}} = dE_{\text{eff}}$. On $F_Z = \text{Proj } \mathbb{C}[Z]$, the boundary for this package is therefore the boundary of Definition 2.2, computed from the actual generic stabilizers of the effective action on Z .

11. WEIGHTED PROJECTIVE DESCENT

We retain the grading and curvature conventions of Section 2. We pass from a homogeneous cone potential to a metric on its projective quotient. The homogeneous volume form determines the log-anticanonical class, including the divisorial stabilizer boundary. A determinant calculation then gives a Monge–Ampère equation. Bounded quotient weights allow us to recover its full mass and to prove klt singularities before applying the following two analytic results.

Theorem 11.1. *Let (F, B) be a log Fano pair of dimension N , set $L = -(K_F + B)$ and $V_L = L^N$, and fix a smooth positive reference weight ϕ_0 on L with adapted measure m_{ϕ_0} . A current $\varpi \in c_1(L)$ with full Monge–Ampère mass and bounded potential $\varphi = \phi - \phi_0$ is a weak log Kähler–Einstein current if*

$$\frac{\varpi^N}{V_L} = \frac{e^{-\varphi} m_{\phi_0}}{\int_F e^{-\varphi} m_{\phi_0}}.$$

Such a current has continuous potentials, is smooth and strictly Kähler on $F_{\text{reg}} \setminus \text{Supp } B$, and satisfies the log Kähler–Einstein equation there, with the corresponding Ricci current equation on F_{reg} .

The measure on each side has total mass one. We use the adapted measure, the definition of a weak solution, and its regularity in precisely this form from [BBE⁺19, Definition 3.1, equation (3.4), Definition 3.5, and Lemma 3.6].

Theorem 11.2. *A log Fano pair admitting a log Kähler–Einstein metric is log K -polystable: every normal relatively ample log test configuration has nonnegative log Donaldson–Futaki invariant, and equality occurs only for an equivariantly product test configuration.*

This is [Ber16, Section 4.3 and Theorem 4.8], with the normal total-space convention of that paper. Its equality statement allows a nontrivial one-parameter subgroup on the product, as in Definition 2.5.

Proposition 11.3. *Let $Z = \text{Spec } A$ be a normal effective positive affine cone of dimension $2n > 0$. Suppose an invariant smooth big open $M \subset Z \setminus \{z_0\}$ has a closed nondegenerate algebraic two-form Ω of weight $\ell > 0$, and a smooth positive potential Φ with*

$$\Phi(tz) = |t|^\ell \Phi(z), \quad \omega = \text{dd}^c \Phi > 0, \quad \omega^{2n} = c_0 \Omega^n \wedge \overline{\Omega}^n, \quad c_0 > 0.$$

For $F = \text{Proj } A$ and its actual boundary and polarization (B, H) , put $q = n\ell$. Then

$$-(K_F + B) \sim_{\mathbb{Q}} qH.$$

The local quotient weights $\psi_s = (2/\ell) \log(\Phi \circ s)$ define a positive metric on H over the smooth quotient charts. Let μ_ψ be its adapted measure for the canonical isomorphism induced by Ω^n : for an index r making $r(K_F + B)$ and $r q H$ Cartier, and a local generator σ of $r(K_F + B)$, it is

$$\mu_\psi = (i^{(2n-1)^2 r} \sigma \wedge \overline{\sigma})^{1/r} |\sigma^*|_{r q \psi}^{2/r}$$

on $F_{\text{reg}} \setminus \text{Supp } B$. There is one constant $C > 0$, independent of the divisorial stabilizer orders, such that

$$C_\psi^{2n-1} = C \mu_\psi, \quad T_\psi^{2n-1} = \frac{C}{(2\pi)^{2n-1}} \mu_\psi$$

on this smooth quotient open.

The boundary is determined by the effective action on the punctured cone. The calculation also fixes the same density normalization on every uniformizing chart.

Proof. We first descend the homogeneous canonical form and compute the coarse boundary. We then compute the quotient Hessian and compare its determinant with the cone volume form. Finally we check the normalization at each branch divisor. Write $U = Z \setminus \{z_0\}$, $m = 2n$, and $N = m - 1$.

Step 1. In this step, we descend the homogeneous canonical form to the quotient stack with weight $-q$. For every $u \in U$ there is a positive-degree homogeneous function f with $f(u) \neq 0$. If its degree is j , any element fixing u satisfies $t^j = 1$. All stabilizers on U are therefore finite. The generic stabilizer is trivial: at the generic point every nonzero homogeneous function is nonzero, and the gcd of their degrees is one by effectivity. The standard affine charts of F have degree-zero rings in normal homogeneous localizations of A . These rings are normal: an invariant fraction integral over such a ring belongs to the normal localization and still has degree

zero. The positive finitely generated grading also makes F projective. We use the quotient stack and its coarse map

$$p : U \longrightarrow \mathcal{F} = [U/\mathbb{G}_m], \quad \pi : \mathcal{F} \longrightarrow F.$$

The stack is normal and Deligne–Mumford. Its character-one line $\mathcal{O}_{\mathcal{F}}(1)$ is the quotient of $U \times \mathbb{C}$ by $t \cdot (u, v) = (tu, tv)$, and $\pi^*H = \mathcal{O}_{\mathcal{F}}(1)$ as rational line bundles.

On M the relative tangent line of p is generated by the holomorphic infinitesimal action ξ . Since $\mathbb{G}_m$ is abelian, this line has its trivial equivariant structure. The tangent sequence identifies $p^*K_{[M/\mathbb{G}_m]}$ with K_M , with their natural linearizations. Put $\Theta = \Omega^n$. It has no zeros on M and satisfies $t^*\Theta = t^q\Theta$. Consequently

$$(11.1) \quad K_{[M/\mathbb{G}_m]} \simeq \mathcal{O}_{\mathcal{F}}(-q)|_{[M/\mathbb{G}_m]}.$$

For the sign, a degree- j function represents a section of $\mathcal{O}_{\mathcal{F}}(j)$ in the character convention above; the dual of a canonical frame of weight q has the character of $\mathcal{O}_{\mathcal{F}}(q)$. All orbits in U have dimension one. Thus removing the invariant codimension-at-least-two subset $U \setminus M$ removes a subset of the same codimension in the quotient. Both sides of (11.1) extend as rank-one reflexive sheaves, so this isomorphism extends over $\mathcal{F}$.

Step 2. In this step, we compute the actual stabilizer boundary and the coarse log-anticanonical class. At the geometric generic point of a prime divisor $D \subset F$, choose a transverse uniformizing coordinate x . If the stabilizer has order $a = m_D$, its action is $x \mapsto \zeta x$, and the coarse coordinate is $y = x^a$. The transverse action is faithful because the generic stabilizer is trivial. Put $E = (x = 0)$ and denote this local coarse map by τ . The identities

$$\tau^*D = aE, \quad dy = ax^{a-1}dx$$

give, on this chart,

$$K_{\text{chart}} = \tau^*K_{\text{coarse}} + (a-1)E = \tau^*(K_{\text{coarse}} + (1-1/a)D).$$

This holds at every prime divisor. Normality and reflexive extension therefore identify the coarse log-canonical sheaf with the stack canonical sheaf after pullback, with

$$B = \sum_D (1 - 1/m_D)D.$$

Only finitely many stabilizer orders occur: if $f_1, \dots, f_k$ are positive-degree homogeneous algebra generators, a stabilizer order divides the degree of each f_j that is nonzero at the point. We may therefore take a common multiple that clears all inertia characters and makes a power of $\mathcal{O}_{\mathcal{F}}(1)$ descend to a line bundle on F . Combining this descent with (11.1) identifies the reflexive sheaf of a positive multiple of $-(K_F + B)$ with the corresponding power of H . The identification in codimension one extends over the normal space F . In particular, $K_F + B$ is rationally Cartier and

$$(11.2) \quad -(K_F + B) \sim_{\mathbb{Q}} qH.$$

This also specifies the isomorphism used to define the adapted measure.

Step 3. In this step, we verify the quotient metric's transition law and positivity by computing the Schur complement. On a smooth uniformizing chart of $[M/\mathbb{G}_m]$, choose a holomorphic section s and base coordinates $z = (z_1, \dots, z_N)$. A local logarithmic orbit coordinate w writes a cone point as $e^w s(z)$. Let $\alpha = \log \Phi$. Homogeneity gives

$$(11.3) \quad \alpha = \ell \operatorname{Re} w + \frac{\ell}{2} \psi_s(z), \quad dd^c \alpha = \frac{\ell}{2} C_{\psi}.$$

If $s' = gs$ for a nowhere-zero holomorphic function g , then

$$\psi_{s'} = \psi_s + 2 \log |g|, \quad e_{s'} = g^{-1} e_s,$$

where e_s is the corresponding frame of $\mathcal{O}_{\mathcal{F}}(1)$. These are exactly the transition laws for $|e_s|^2 = e^{-\psi_s}$. The equivariant transition laws on finite charts likewise give a metric on the rational line bundle H .

In the coordinates (w, z) , the complex Hessian of Φ , divided by Φ , is

$$\frac{1}{\Phi} \text{Hess}_{\mathbb{C}}(\Phi) = \begin{pmatrix} \ell^2/4 & (\ell^2/4)(\psi_s)_{\bar{j}} \\ (\ell^2/4)(\psi_s)_i & (\ell/2)(\psi_s)_{i\bar{j}} + (\ell^2/4)(\psi_s)_i(\psi_s)_{\bar{j}} \end{pmatrix}.$$

Its orbit entry is positive and its Schur complement is $(\ell/2)((\psi_s)_{i\bar{j}})$. The Hessian is positive definite by $\omega > 0$, so the Schur complement is positive definite. Hence $C_\psi = \text{dd}^c \psi_s$ is Kähler on the uniformizing chart.

Step 4. In this step, we derive the quotient determinant equation and its constant from the cone volume identity. Since $\Phi = e^\alpha$,

$$\omega = \Phi(\text{dd}^c \alpha + d\alpha \wedge d^c \alpha).$$

By (11.3), the first summand inside the parentheses has horizontal rank N , and the square of the second is zero. In its top exterior power only one factor of the second summand occurs. Its vertical part is $(i\ell^2/4)dw \wedge d\bar{w}$ with our fixed d^c convention. Thus

$$(11.4) \quad \omega^m = m \frac{\ell^2}{4} \left(\frac{\ell}{2}\right)^N \Phi^m i dw \wedge d\bar{w} \wedge C_\psi^N.$$

Define the nowhere-vanishing quotient canonical form $\beta = s^*(\iota_\xi \Theta)$. In these coordinates $\xi = \partial_w$, so homogeneity and contraction give

$$\Theta = e^{qw} dw \wedge \beta, \quad \Phi^m = |e^w|^{2q} e^{q\psi_s}.$$

Because m is even and $N = m - 1$ is odd, the complex orientations give

$$\Theta \wedge \bar{\Theta} = |e^w|^{2q} i dw \wedge d\bar{w} \wedge i^{N^2} \beta \wedge \bar{\beta}.$$

Substituting these identities into $\omega^m = c_0 \Theta \wedge \bar{\Theta}$ and cancelling the vertical and radial factors in (11.4) yields

$$(11.5) \quad C_\psi^N = C e^{-q\psi_s} i^{N^2} \beta \wedge \bar{\beta}, \quad C = \frac{c_0}{m(\ell^2/4)(\ell/2)^N} > 0.$$

In particular the constant depends only on n, ℓ, c_0 , and the curvature convention. For the hyperkähler packages of the preceding sections, Lemma 4.7 supplies $c_0 = (2n)!/(4^n(n!)^2)$ without changing that convention.

Under (11.1), β^* corresponds to e_s^q . Its squared norm is therefore $e^{-q\psi_s}$, and the density in (11.5) is the adapted measure on the stack. Under $s' = gs$ one has $\beta' = g^q \beta$: the derivative of g contributes an orbit vector, which is killed by contraction with ξ . Consequently $e^{-q\psi_s} i^{N^2} \beta \wedge \bar{\beta}$ is independent of s .

Step 5. In this step, we compare the coarse adapted measure across branch divisors and fix its integral first-Chern normalization. Return to the transverse chart $y = x^a$ at D , retaining unramified coordinates $z_2, \dots, z_N$ along D . The rational log-canonical generator

$$\eta_D = \frac{1}{a} \frac{dy \wedge dz_2 \wedge \dots \wedge dz_N}{y^{1-1/a}}$$

means the following genuine generator after taking a Cartier index r divisible by a :

$$\eta_D^{\otimes r} = a^{-r} \frac{(dy \wedge dz_2 \wedge \dots \wedge dz_N)^{\otimes r}}{y^{r(1-1/a)}}.$$

Its pullback is exactly $(dx \wedge dz_2 \wedge \dots \wedge dz_N)^{\otimes r}$. For $N = 1$ the factors in the z_j coordinates are absent. Take r also large enough to descend all relevant line-bundle frames. This calculation identifies the coarse generator with the stack generator without an a -dependent coefficient.

The root in the adapted-measure formula denotes the positive pluricanonical density: if $\sigma = h\theta^{\otimes r}$ for a local canonical form θ , its first factor is $|h|^{2/r} i^{N^2} \theta \wedge \bar{\theta}$. More explicitly, the adapted expression is unchanged when a generator σ is replaced by $h\sigma$. Its density factor is multiplied by $|h|^{2/r}$ and its dual-norm factor by $|h|^{-2/r}$. Thus the nowhere-zero local unit relating the pullback of $\eta_D^{\otimes r}$ to $\beta^{\otimes r}$ cancels as well. The coarse adapted measure pulls back to precisely the density in (11.5). This proves the determinant identity with the same C across every divisorial uniformizing chart. It is an identity under pullback on these charts; integrating a finite cover divides both sides by the same degree.

Finally, $C_\psi = 2\pi T_\psi$ by Notation 2.3. Hence

$$T_\psi^N = \frac{C}{(2\pi)^N} \mu_\psi,$$

which is the asserted identity in integral first-Chern normalization. $\square$

Theorem 11.4. *Under the hypotheses of Proposition 11.3, assume the quotient weights extend to a globally locally bounded plurisubharmonic metric on H . Then T_ψ has full Monge–Ampère mass H^{2n-1} , the pair (F, B) is klt, and, after adding a constant to ψ , the zero extension of the adapted measure satisfies*

$$T_\psi^{2n-1} = \mu_\psi$$

globally. The current qT_ψ is a weak log Kähler–Einstein current in $c_1(-(K_F + B))$, and (F, B) is log K -polystable with the all-normal product equality of Definition 2.5.

The boundedness hypothesis concerns the metric on the whole projective base. It supplies both the mass calculation and the integrability needed to establish the singularities of the pair.

Proof. We compute the mass on a smooth projective resolution, use the resulting finite adapted volume to prove klt, and then normalize the equation on the log-anticanonical class. The last step applies the two analytic theorems stated above. Put $N = 2n - 1$ and

$$F_0 = F_{\text{reg}} \setminus \text{Supp } B, \quad \gamma = \frac{C}{(2\pi)^N}.$$

The rational polarization H is ample: a sufficiently divisible Veronese grading gives an ample Cartier multiple on $\text{Proj } A$. Proposition 11.3 gives the canonical identification (11.2), with $q > 0$.

Step 1. In this step, we specify the open determinant equation and define the adapted measure before proving its finiteness. Choose a dense open $V \subset F_0$ over which the cone points lie in M and the effective action is free. Such an open exists because the action is generically free. It has algebraic complement: using a finite homogeneous generating set, the nonfree locus is a finite union of fixed loci of nontrivial roots of unity, and the images of these invariant closed sets are closed in the geometric quotient. The image of $U \setminus M$, where $U = Z \setminus \{z_0\}$, has codimension at least two, since quotienting lowers the dimensions of both invariant sets by one. Thus $A_F = F \setminus V$ is a proper analytic subset. On V ,

$$(11.6) \quad T_\psi^N = \gamma \mu_\psi.$$

Fix a Cartier index and the canonical isomorphism of Proposition 11.3. Its adapted-measure formula defines a positive measure on F_0 for the extended weight ψ . Independence of the generator follows from the cancellation of the two factors already checked in that proof, and replacing the index by a multiple gives the same measure. On relatively compact coordinate sets in F_0 this measure has a positive locally bounded density relative to smooth volume, because ψ is locally bounded. It therefore gives zero mass to the analytic set $F_0 \setminus V$. We write μ_ψ also for its zero extension to F ; finiteness of this extension will follow from the next calculation.

Step 2. In this step, we compute full mass on a smooth projective resolution and extend the determinant equation globally. Take a projective log resolution $f : \tilde{F} \rightarrow F$ of the pair and of

A_F , chosen to be an isomorphism over V . The pulled-back metric on f^*H has locally bounded plurisubharmonic weights. Its current is

$$\tilde{T} = \frac{1}{2\pi} dd^c(f^*\psi).$$

By the noncharging assertion of Theorem 4.5, applied on the smooth manifold $\tilde{F}$, its Bedford–Taylor measure satisfies

$$\tilde{T}^N(f^{-1}(A_F)) = 0.$$

Here $f^{-1}(A_F)$ is a proper analytic subset, hence locally pluripolar. We compute the top product on F by $f_*(\tilde{T}^N)$. It agrees with the Bedford–Taylor product on the open where f is an isomorphism. This description is independent of the resolution: on a common resolution the products agree off the exceptional analytic sets, and those sets have zero mass by the same noncharging assertion. In particular this is the non-pluripolar top product T_ψ^N , and

$$(11.7) \quad T_\psi^N(A_F) = 0.$$

The mass assertion of Theorem 4.5, again on $\tilde{F}$, gives

$$(11.8) \quad \int_F T_\psi^N = \int_{\tilde{F}} \tilde{T}^N = (f^*H)^N = H^N > 0.$$

For a rational line bundle this follows by taking a Cartier multiple and dividing its top mass by the corresponding N th power. All classes here use integral first Chern normalization. The two parts of the cited theorem are respectively [DS10, Appendix A, Corollary A.32] and [BGH⁺22, Lemma 2.24]; their manifold hypotheses are satisfied on $\tilde{F}$.

Equation (11.6), the zero mass in (11.7), and the zero extension of μ_ψ now imply

$$(11.9) \quad T_\psi^N = \gamma\mu_\psi \quad \text{on } F, \quad \mu_\psi(F) = \gamma^{-1}H^N < \infty.$$

The equality includes the branch and boundary divisors, the image of $U \setminus M$, and the singular locus: each lies in A_F and has zero mass. No integrability at these sets was assumed in defining the measure on F_0 .

Step 3. In this step, we prove klt singularities from integrability of the discrepancy density. On the same log resolution write

$$K_{\tilde{F}} = f^*(K_F + B) + \sum_j a_j E_j,$$

including both exceptional divisors and strict transforms of components of B . Enlarge the normal-crossing divisor to include $f^{-1}(A_F)$ if necessary, with coefficient zero on any added component. Near a point of $\tilde{F}$, choose coordinates in which its components are coordinate hyperplanes. For a local generator σ of $r(K_F + B)$, its pullback as a meromorphic pluricanonical form has the expression

$$f^*\sigma = u \prod_j z_j^{ra_j} (dz_1 \wedge \cdots \wedge dz_N)^{\otimes r},$$

where u is a holomorphic unit and r also clears the discrepancies. The pullback of the dual generator is a nonvanishing frame of $f^*(rqH)$. Since the pulled-back weight is locally bounded, the adapted density off the divisor is consequently

$$(11.10) \quad f^*\mu_\psi = b(z) \prod_j |z_j|^{2a_j} dV(z), \quad 0 < b_- \leq b(z) \leq b_+ < \infty$$

on a smaller coordinate neighborhood, for constants $b_-, b_+ > 0$ and smooth positive Euclidean volume dV .

Over V the map f is an isomorphism. The finite mass in (11.9) therefore bounds the integral of (11.10) off its normal-crossing divisor. At a general point of each E_j , the other divisor coordinates are nonzero, so finiteness requires

$$\int_0^\epsilon t^{2a_j+1} dt < \infty, \quad \text{equivalently} \quad a_j > -1.$$

The same criterion gives integrability at intersections of components. Thus all discrepancies on a log resolution exceed -1 , which is the klt criterion. Together with $-(K_F + B) \sim_{\mathbb{Q}} qH$ and ampleness, this makes (F, B) a log Fano pair.

Step 4. In this step, we choose an additive normalization of the weight so that $T_\psi^N = \mu_\psi$. For any real constant c , the adapted-measure formula gives

$$T_{\psi+c} = T_\psi, \quad \mu_{\psi+c} = e^{-qc} \mu_\psi.$$

Choose explicitly

$$c = -\frac{\log \gamma}{q} = \frac{N \log(2\pi) - \log C}{q}.$$

Then (11.9) becomes $T_{\psi+c}^N = \mu_{\psi+c}$. We henceforth denote the shifted weight by ψ . With this normalization,

$$(11.11) \quad T_\psi^N = \mu_\psi, \quad \mu_\psi(F) = H^N.$$

The factor $(2\pi)^{-N}$ has been absorbed into the adapted density by this explicit shift; the curvature and its cohomology class are unchanged.

Step 5. In this step, we prove the probability-normalized weak Kähler–Einstein equation for the anticanonical current. Use the fixed rational isomorphism to put

$$L = -(K_F + B) = qH, \quad \phi = q\psi, \quad \varpi = qT_\psi, \quad V_L = L^N = q^N H^N.$$

Choose a smooth positive reference weight ϕ_0 on L ; it can be obtained by restricting a Fubini–Study metric on a sufficiently high very ample multiple. The global difference $\varphi = \phi - \phi_0$ is bounded, since F is compact and ψ is locally bounded. Let m_{ϕ_0} be the adapted measure formed from this reference weight by the same Cartier-index formula. Replacing ϕ_0 by ϕ multiplies the squared norm of the dual rL -generator by $e^{-r\varphi}$. Taking its r th root proves the literal identity

$$(11.12) \quad \mu_\psi = e^{-\varphi} m_{\phi_0}.$$

Using (11.8) and (11.11), we have

$$\int_F \varpi^N = V_L, \quad \varpi^N = q^N \mu_\psi.$$

Together with (11.12), this gives

$$(11.13) \quad \frac{\varpi^N}{V_L} = \frac{\mu_\psi}{H^N} = \frac{e^{-\varphi} m_{\phi_0}}{\int_F e^{-\varphi} m_{\phi_0}}.$$

Equation (11.13) is the full-mass equation in Theorem 11.1. That theorem, with [BBE⁺19, Lemma 3.6], gives continuous potentials on F and a smooth strictly Kähler metric on F_0 . For clarity in our curvature convention, on F_{reg} its Ricci equation reads

$$-\frac{1}{2\pi} \text{dd}^c \log \vartheta = \varpi + [B],$$

where ϑ is the density of ϖ^N relative to the coordinate volume $i^{N^2} dv_1 \wedge \cdots \wedge dv_N \wedge d\bar{v}_1 \wedge \cdots \wedge d\bar{v}_N$, and $[B]$ is the integration current. Indeed, its local density is a positive constant times $e^{-\phi} \prod_j |h_j|^{-2\delta_j}$, where h_j locally defines a component of B of coefficient δ_j . Applying $-(2\pi)^{-1} \text{dd}^c \log$ gives the displayed equation, using $(2\pi)^{-1} \text{dd}^c \log |h_j|^2 = [h_j = 0]$.

Step 6. In this step, we deduce log K-polystability and the product equality case for every normal relatively ample log test configuration. We have proved that (F, B) is log Fano with

effective rational boundary and that ϖ has bounded potentials, is Kähler on F_0 , and satisfies the global weak equation. These are the log Kähler–Einstein hypotheses of Theorem 11.2, or equivalently [Ber16, Section 4.3 and Theorem 4.8]. For every normal relatively ample log test configuration the log Donaldson–Futaki invariant is therefore nonnegative. If it is zero, the configuration is equivariantly a product, with the allowed one-parameter automorphism action specified in Definition 2.5. This is the required log K-polystability conclusion. $\square$

12. SEMISIMPLE PRODUCTS AND THE STABILITY THEOREMS

We retain the grading and curvature conventions of Section 2. We first combine the potentials on affine products and verify the hypotheses of weighted projective descent. We then apply this construction to scalar orbit closures and to Kazhdan slices. Functions and differential forms from a factor are pulled back by the corresponding projection whenever they occur in a product formula.

Proposition 12.1. *Let Z_i be normal affine cones with homogeneous hyperkähler potential packages of the same weight a , with $\dim_{\mathbb{C}} Z_i = 2n_i > 0$. For arbitrary $c_i > 0$, their affine product has such a package on $\prod_i M_i$, given by*

$$g = \bigoplus_i c_i g_i, \quad \Phi = \sum_i c_i \Phi_i, \quad \omega_q = \sum_i c_i \omega_{i,q}, \quad \Omega = \sum_i c_i \Omega_i.$$

Its complement has codimension at least two, and, with $n = \sum_i n_i$,

$$\frac{\omega_I^{2n}}{(2n)!} = \frac{\Omega^n \wedge \overline{\Omega}^n}{4^n (n!)^2}.$$

If the common diagonal kernel has order d , then $d \mid a$ and the effective package has weight $\ell = a/d$. Its quotient weights are globally locally bounded and plurisubharmonic. Consequently its effective projective quotient with the actual boundary satisfies Theorem 11.4. Point factors can be inserted in the affine product and are omitted from the potential sum and kernel gcd.

The constants c_i allow independent positive normalizations on the factors. Each constant scales the metric, all three associated forms, and the potential together.

Proof. We construct the package on the affine product, identify its effective action and residual stabilizers, and then descend the potential.

Step 1. In this step, we construct the affine product and its hyperkähler tensors with the stated volume normalization. Write $A_i = \mathbb{C}[Z_i]$ with the given grading, and let $z_{i,0}$ be its vertex. The product $Z = \prod_{i=1}^r Z_i$ is irreducible and normal: over $\mathbb{C}$ the factors are geometrically integral and geometrically normal, and these properties are preserved by products. Its coordinate ring and diagonal action are

$$(12.1) \quad \begin{aligned} A &= \bigotimes_{i=1}^r A_i, & A_j &= \bigoplus_{j_1 + \dots + j_r = j} (A_1)_{j_1} \otimes \dots \otimes (A_r)_{j_r}, \\ t \cdot (z_1, \dots, z_r) &= (t \cdot z_1, \dots, t \cdot z_r). \end{aligned}$$

This is a finitely generated positive grading with $A_0 = \mathbb{C}$ and vertex $z_0 = (z_{1,0}, \dots, z_{r,0})$. In the orbit and slice applications, Proposition 3.4 identifies this affine product with the normalization of the corresponding product component.

The open set $M = \prod_i M_i$ is smooth and invariant. Its complement is

$$Z \setminus M = \bigcup_i Z_1 \times \dots \times (Z_i \setminus M_i) \times \dots \times Z_r.$$

Each member of this union has codimension at least two. On M put $I = \bigoplus_i I_i$, $J = \bigoplus_i J_i$, and $K = \bigoplus_i K_i$. Multiplication of g_i by the positive constant c_i preserves its Levi–Civita connection. The product connection therefore preserves I, J, K , and the quaternionic identities hold on each

tangent summand. The resulting metric is positive hyperkähler. Its associated forms are the stated sums, since for $q \in \{I, J, K\}$ and tangent vectors $v = (v_i)$, $w = (w_i)$,

$$g(qv, w) = \sum_i c_i g_i(q_i v_i, w_i).$$

Linearity of dd^c gives $\text{dd}^c \Phi = \omega_I$. The form $\Omega = \sum_i c_i \Omega_i$ is closed and algebraic. If $\Omega(v, w) = 0$ for every w , taking w supported in the i th summand gives $c_i \Omega_i(v_i, w_i) = 0$ for every w_i . Thus $v_i = 0$ for every i , proving nondegeneracy.

Lemma 4.7, applied to this product metric and its associated form Ω , gives the displayed volume identity. In particular the constant needed for descent is exactly

$$c_0 = \frac{(2n)!}{4^n (n!)^2}.$$

The same scaling occurs on both sides: the surviving term in ω_I^{2n} and that in $\Omega^n \wedge \bar{\Omega}^n$ each contain $\prod_i c_i^{2n_i}$. There is no additional factor depending on the choices of c_i .

Step 2. In this step, we prove the global properties of the product potential, including plurisubharmonicity of its logarithm. Extend Φ to Z by the same sum. It is continuous and nonnegative, and its zero set is $\{z_0\}$ because all c_i are positive. For $R \geq 0$,

$$\{\Phi \leq R\} \subseteq \prod_i \{\Phi_i \leq R/c_i\}.$$

The right side is compact by properness of the factor potentials; the left side is closed by continuity. Hence Φ is proper. It is smooth and positive on M , and is plurisubharmonic on all of Z as a positive sum of pullbacks of plurisubharmonic functions.

We check the logarithm also where one or more coordinates are vertices. On Z set

$$u_i = \log(c_i \Phi_i), \quad u_{i,k} = \max\{u_i, -k\} \quad (k \geq 1), \quad \Lambda(x_1, \dots, x_r) = \log\left(\sum_i e^{x_i}\right).$$

Here $u_i = -\infty$ on the locus where the i th coordinate is its vertex. Each $u_{i,k}$ is finite, locally bounded, and plurisubharmonic. The function Λ is convex and nondecreasing in each argument: its first derivatives are the positive numbers $p_i = e^{x_i} / \sum_j e^{x_j}$, and its Hessian has quadratic form

$$\sum_i p_i b_i^2 - \left(\sum_i p_i b_i\right)^2 \geq 0.$$

For any analytic disc, the submean inequalities for the $u_{i,k}$, followed by monotonicity and Jensen's inequality for Λ , give the submean inequality for $\Lambda(u_{1,k}, \dots, u_{r,k})$. These compositions are therefore plurisubharmonic. As k increases,

$$\Lambda(u_{1,k}, \dots, u_{r,k}) \searrow \log\left(\sum_i c_i \Phi_i\right) = \log \Phi.$$

The limit is finite off z_0 , and tends to $-\infty$ at z_0 by continuity of Φ . The decreasing-limit property proves that this extension of $\log \Phi$ is plurisubharmonic on Z . The common factor weight gives, for the diagonal action,

$$\Phi(t \cdot z) = |t|^a \Phi(z), \quad t^* \Omega = t^a \Omega.$$

This completes the package before effectivization.

Step 3. In this step, we compute the common diagonal kernel, the effective grading, and the actual residual boundary. Choose nonzero homogeneous algebra generators P_{ij} of A_i in positive degrees w_{ij} . Their pullbacks generate A . Let $d_i = \gcd_j w_{ij}$. An element acts trivially on Z_i exactly when its w_{ij} th powers are all one. The diagonal kernel is therefore

$$\bigcap_i \mu_{d_i} = \mu_d, \quad d = \gcd_i d_i = \gcd_{i,j} w_{ij}.$$

Equivalently, this is the gcd of all positive degrees in (12.1): each factor degree occurs in A by tensoring with units, and every total degree is a sum of factor degrees. If $\zeta \in \mu_d$, its action on M is the identity, whereas homogeneity gives $\zeta^* \Omega = \zeta^a \Omega$. Since Ω is nonzero, $\zeta^a = 1$ for every such ζ . Thus $d \mid a$.

Use the parameter $s = t^d$ to identify $\mathbb{G}_m/\mu_d$ with $\mathbb{G}_m$ and set $s \cdot_{\text{eff}} z = t \cdot z$. This is independent of the chosen root t and yields

$$A_j^{\text{eff}} = A_{dj}, \quad \ell = \frac{a}{d} \in \mathbb{Z}_{>0}, \quad \Phi(s \cdot_{\text{eff}} z) = |s|^\ell \Phi(z), \quad s^* \Omega = s^\ell \Omega.$$

Only now form $F = \text{Proj } A^{\text{eff}}$. Dividing all degrees by d leaves the degree-zero localization rings, and hence the coarse Proj, unchanged. The character-one line for the effective parameter pulls back to the character- d line for the raw parameter. Thus its polarization is $H_{\text{eff}} = dH_{\text{raw}}$. We write $H = H_{\text{eff}}$ below. A sufficiently divisible Veronese gives an ample line bundle on F representing a positive multiple of H ; in particular F is normal and projective and H is an ample rational polarization.

The same generators determine the residual stabilizers at each point. For $z \neq z_0$, at least one $P_{ij}(z)$ is nonzero, and the effective stabilizer is

$$(12.2) \quad \begin{aligned} \text{Stab}_{\text{eff}}(z) &= \{s \in \mathbb{C}^* : s^{w_{ij}/d} = 1 \text{ whenever } P_{ij}(z) \neq 0\} \\ &= \mu_{\text{gcd}\{w_{ij}/d : P_{ij}(z) \neq 0\}}. \end{aligned}$$

Indeed, fixing the values of all algebra generators is equivalent to fixing the point. The condition that a homogeneous generator vanish is constant along an orbit. At a geometric general point z above a prime divisor D of F , the right side of (12.2) therefore has order

$$m_D = \frac{1}{d} \text{gcd}\{w_{ij} : P_{ij}(z) \neq 0\}.$$

This computes the actual residual order after removal of the common kernel. At a general point of Z all the chosen generators are nonzero, so this order is one. The boundary for descent is consequently $B = \sum_D (1 - 1/m_D) D$, with these orders on the total quotient.

Step 4. In this step, we extend the quotient metric as continuous locally bounded plurisubharmonic weights on every coarse chart. Let σ be a local holomorphic section on a finite quotient chart of the effective punctured cone. Put

$$\psi_\sigma = \frac{2}{\ell} \log(\Phi \circ \sigma).$$

For another section $\sigma' = \gamma \cdot_{\text{eff}} \sigma$, where γ is nowhere zero and holomorphic, homogeneity gives

$$\psi_{\sigma'} = \psi_\sigma + 2 \log |\gamma|.$$

This is the transition law for the weight-one Seifert metric. Its weights are continuous and locally bounded because Φ is continuous and strictly positive on the punctured cone, and they are plurisubharmonic because $\log \Phi$ is plurisubharmonic there.

We make the extension over all coarse quotient charts explicit. Take a homogeneous $P \in A^{\text{eff}}$ of degree $b > 0$. The affine set $S_P = \{P = 1\} \subset Z$ avoids the vertex and maps finitely to $D_+(P) \subset F$ with quotient group μ_b . One obtains this description by choosing a b th root of $P(z)^{-1}$ on $\{P \neq 0\}$; the remaining choices act on S_P by μ_b . On S_P the function $(2/\ell) \log \Phi$ is invariant under this finite group. Its multiple $(2b/\ell) \log \Phi$ is the weight of the frame P of the character- b line. It descends to a continuous locally bounded weight on $D_+(P)$. The descended function is plurisubharmonic: after a finite branched cover an analytic disc in the quotient lifts to S_P , and the submean inequality descends. Clearing the finitely many character denominators makes these ordinary line-bundle weights. The section-change identity shows that they agree on overlaps as weights on H . Since the $D_+(P)$ cover F , this proves the required global extension, including over the image of $Z \setminus M$.

Step 5. In this step, we verify weighted descent for the product and account for point factors. We have a normal positive effective affine cone of dimension $2n$, an invariant smooth open M with complement of codimension at least two, and a closed nondegenerate algebraic form Ω of positive integral weight ℓ . The potential is smooth and positive on M , has the same modulus weight ℓ , and satisfies $\mathrm{dd}^c \Phi = \omega_I > 0$ there. The exact volume identity supplies $c_0 = (2n)!/[4^n(n!)^2]$. The preceding chart construction supplies the globally locally bounded plurisubharmonic quotient metric. All hypotheses of Proposition 11.3 and Theorem 11.4 are therefore satisfied with the actual boundary just computed.

For precision, write $N = 2n - 1$ and $T_\psi = (2\pi)^{-1} \mathrm{dd}^c \psi$. The canonical relation and the analytic conclusions of these results give

$$-(K_F + B) \sim_{\mathbb{Q}} n\ell H, \quad \int_F T_\psi^N = H^N, \quad T_\psi^N = \mu_\psi$$

after an additive normalization of ψ . The pair is klt, and the canonical relation with $n\ell > 0$ and H ample makes it log Fano. If $L = -(K_F + B)$ and $\varpi = n\ell T_\psi$, the full-mass normalization reads

$$\frac{\varpi^N}{L^N} = \frac{\mu_\psi}{H^N} = \frac{\mu_\psi}{\int_F \mu_\psi}.$$

Thus it is the probability normalization of the weak log Kähler–Einstein current supplied by Theorem 11.4. Its stability conclusion applies to every normal relatively ample log test configuration, with equality only for an equivariantly product configuration.

Finally, inserting a point factor tensors A with $\mathbb{C}$ and adds neither a tangent summand nor a nonzero term to Φ . Its action has the whole group as kernel and supplies no positive degrees. It is therefore omitted from both the potential sum and the gcd calculation. $\square$

Theorem 12.2. *Let $\mathfrak{g}$ be any complex semisimple Lie algebra, let $\mathcal{O} \subset \mathfrak{g}$ be a nonzero nilpotent adjoint orbit, and let Y be the normalization of $\overline{\mathcal{O}}$. Equip Y with the lifted scalar action, and write $\dim_{\mathbb{C}} Y = 2n$. The action on the punctured cone is free. Its projective base $F = \mathrm{Proj} \mathbb{C}[Y]$, with rational weight-one Seifert polarization H , is a klt Fano variety and satisfies*

$$-K_F \sim_{\mathbb{Q}} nH.$$

It admits a weak Kähler–Einstein current and is K-polystable: every normal relatively ample test configuration for $-K_F$ has nonnegative Donaldson–Futaki invariant, with equality only for an equivariantly product configuration.

Here the scalar action itself is already effective. Its free punctured action fixes both the weight-one polarization and the zero boundary in the conclusion.

Proof. We assemble the simple orbit potentials with the scalar grading and then apply weighted descent with weight one.

Step 1. In this step, we identify the normalized scalar product cone and its zero stabilizer boundary. Decompose $\mathfrak{g} = \bigoplus_{i=1}^r \mathfrak{g}_i$ into simple ideals and write $\mathcal{O} = \prod_i \mathcal{O}_i$. Proposition 3.4 gives

$$\overline{\mathcal{O}} = \prod_i \overline{\mathcal{O}_i}, \quad Y = \prod_i Y_i, \quad Y_i = \mathrm{Norm}(\overline{\mathcal{O}_i}),$$

with the diagonal scalar action and the total tensor grading (12.1). Denote the vertex of Y by y_0 . Let $\mathcal{I} = \{i : \mathcal{O}_i \neq \{0\}\}$, a nonempty set. For $i \notin \mathcal{I}$ the factor Y_i is a point; all subsequent sums run over $\mathcal{I}$.

The punctured scalar action is free by Proposition 3.4(i). Thus its kernel has order one, every divisorial stabilizer has order one, and the boundary is $B = 0$.

Step 2. In this step, we combine the simple-factor weight-one potentials and deduce the Fano and K-polystability conclusions. For every $i \in \mathcal{I}$, Theorem 6.3 gives a weight-one package on Y_i with smooth big open $\mathcal{O}_i$. Its metric, symplectic form, and potential have the common

normalization specified there. Write $\dim_{\mathbb{C}} Y_i = 2n_i$; the affine product gives $n = \sum_{i \in \mathcal{I}} n_i$. Choose $c_i > 0$ and apply Proposition 12.1. On the smooth big open $M = \prod_{i \in \mathcal{I}} \mathcal{O}_i$, with point factors inserted, we obtain

$$\Phi = \sum_{i \in \mathcal{I}} c_i \Phi_i, \quad \Omega = \sum_{i \in \mathcal{I}} c_i \Omega_i, \quad g = \bigoplus_{i \in \mathcal{I}} c_i g_i.$$

The product potential is continuous, proper, and nonnegative on Y , vanishes only at y_0 , and both Φ and $\log \Phi$ are plurisubharmonic. On M it satisfies $\mathrm{dd}^c \Phi = \omega_I > 0$ and

$$\begin{aligned} \Phi(t \cdot y) &= |t| \Phi(y), & t^* \Omega &= t \Omega, \\ \omega_I^{2n} &= \frac{(2n)!}{4^n (n!)^2} \Omega^n \wedge \bar{\Omega}^n. \end{aligned}$$

The form is closed, algebraic, and nondegenerate, and $Y \setminus M$ has codimension at least two. The quotient weights are $\psi_\sigma = 2 \log(\Phi \circ \sigma)$ and extend to a continuous locally bounded plurisubharmonic metric on H .

These are exactly the hypotheses of Theorem 11.4 with $\ell = 1$, $B = 0$, and $c_0 = (2n)!/[4^n (n!)^2]$. It gives $-K_F \sim_{\mathbb{Q}} nH$ and klt singularities. Since H is ample, F is Fano. It also gives full mass $\int_F T_\psi^{2n-1} = H^{2n-1}$ and the weak Kähler–Einstein current nT_ψ in $c_1(-K_F)$. Its all-normal equality assertion gives precisely the stated K-polystability and equivariantly product equality case. $\square$

Theorem 12.3. *Let $\mathfrak{g}$ be any complex semisimple Lie algebra, $\mathcal{O} \subset \mathfrak{g}$ a nilpotent adjoint orbit, and (e, h, f) an $\mathfrak{sl}_2$ -triple with $e \in \mathcal{O}$. Let Z normalize an irreducible component of*

$$(\overline{\mathcal{O}} \cap (e + \mathfrak{g}^f))_{\mathrm{red}}.$$

If $\dim Z = 0$, Z is the point above e and has no projective base. If $\dim_{\mathbb{C}} Z = 2n > 0$, first give Z its lifted raw Kazhdan action $\kappa_t(x) = t^2 \mathrm{Ad}(t^{-h})x$, and then remove its common finite kernel μ_d . For the resulting effective total grading, put

$$F = \mathrm{Proj} \mathbb{C}[Z], \quad \ell = 2/d, \quad B = \sum_D \left(1 - \frac{1}{m_D}\right) D,$$

where m_D is the actual generic stabilizer order above D for the effective action. Then ℓ is a positive integer and, for the effective weight-one Seifert polarization H ,

$$-(K_F + B) \sim_{\mathbb{Q}} n\ell H.$$

The pair (F, B) is log Fano, admits a weak log Kähler–Einstein current, and is log K-polystable: every normal relatively ample log test configuration has nonnegative log Donaldson–Futaki invariant, with equality only for an equivariantly product configuration. In a decomposition into simple ideals, the affine factors are combined before taking this effective grading, projective base, and boundary.

The raw symplectic weight is two. The effective weight and boundary are determined by the diagonal action on the chosen normalized component.

Proof. We separate the point case, construct the raw product package in positive dimension, and then compute the effective data for descent.

Step 1. In this step, we factor the normalized slice component and settle the zero-dimensional case. Write $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$, $\mathcal{O} = \prod_i \mathcal{O}_i$, and $(e, h, f) = ((e_i), (h_i), (f_i))$. If X denotes the reduced irreducible component normalized by Z , Proposition 3.4 gives

$$\begin{aligned} X &= \prod_i X_i, & Z &= \prod_i Z_i, & Z_i &= \mathrm{Norm}(X_i), \\ X_i &\subseteq (\overline{\mathcal{O}_i} \cap (e_i + \mathfrak{g}_i^{f_i}))_{\mathrm{red}}. \end{aligned}$$

Each X_i is an irreducible component of the displayed reduced intersection. The raw action on Z is the lift of

$$(x_i)_i \longmapsto (t^2 \operatorname{Ad}(t^{-h_i})x_i)_i,$$

so its ring has the total raw grading (12.1). The same proposition gives a unique vertex over e . If $\dim Z = 0$, then Z is that point, its punctured cone is empty, and $\operatorname{Proj} \mathbb{C} = \emptyset$. This proves the zero-dimensional assertion.

Assume now $\dim Z > 0$ and put $\mathcal{I} = \{i : \dim Z_i > 0\}$. This set is nonempty. The other factors are their vertices and are omitted from all metric sums and all kernel gcds below.

Step 2. In this step, we construct the raw weight-two product package and verify its Euler identities. For each $i \in \mathcal{I}$, Theorem 10.2 supplies the natural hyperkähler package on the smooth big open $M_i \subset Z_i$. Thus $\dim_{\mathbb{C}} Z_i = 2n_i$ is even and $n = \sum_{i \in \mathcal{I}} n_i$. In particular Ω_i is its algebraic holomorphic symplectic form, $\operatorname{dd}^c \Phi_i = \omega_{i,I}$, and

$$\Phi_i(\kappa_t z_i) = |t|^2 \Phi_i(z_i), \quad \kappa_t^* \Omega_i = t^2 \Omega_i.$$

The potential extends continuously and properly over Z_i , vanishes only at its vertex, and is plurisubharmonic with plurisubharmonic logarithm. These are the full factor hypotheses of Proposition 12.1 with $a = 2$. Choose $c_i > 0$ and form the product on $M = \prod_{i \in \mathcal{I}} M_i$:

$$g = \bigoplus_{i \in \mathcal{I}} c_i g_i, \quad \Phi = \sum_{i \in \mathcal{I}} c_i \Phi_i, \quad \Omega = \sum_{i \in \mathcal{I}} c_i \Omega_i.$$

Point factors are inserted in M and Z . The proposition gives the global potential, the smooth big open, the algebraic symplectic form, and the exact volume coefficient $(2n)!/[4^n(n!)^2]$.

The gradient identities also retain their raw parameter. Indeed, if E_i is the raw positive-real generator on M_i and $E_{\text{raw}} = (E_i)_i$, then for $v = (v_i)$ the simple-factor identities give

$$\operatorname{d}\Phi(v) = \sum_i c_i g_i(E_i, v_i) = g(E_{\text{raw}}, v), \quad \Phi = \frac{1}{2} g(E_{\text{raw}}, E_{\text{raw}}).$$

Thus $\nabla_g \Phi = E_{\text{raw}}$ with this product metric.

Step 3. In this step, we compute the effective total grading, polarization, and divisorial stabilizer orders. Let μ_{d_i} be the raw kernel on Z_i for $i \in \mathcal{I}$. Every $\zeta \in \mu_{d_i}$ fixes the nonzero form Ω_i while homogeneity multiplies it by ζ^2 . Hence $d_i \mid 2$. The kernel on the affine product is $\mu_d = \bigcap_{i \in \mathcal{I}} \mu_{d_i}$, with

$$d = \gcd_{i \in \mathcal{I}} d_i, \quad d \mid 2.$$

Dividing the total raw degrees by this single d gives the effective grading and parameter $s = t^d$. Accordingly

$$\ell = \frac{2}{d} \in \mathbb{Z}_{>0}, \quad E_{\text{raw}} = d E_{\text{eff}},$$

$$\Phi(s \cdot_{\text{eff}} z) = |s|^\ell \Phi(z), \quad s^* \Omega = s^\ell \Omega.$$

The generator identity records why the preceding gradient formula keeps its raw generator.

Form $F = \operatorname{Proj} \mathbb{C}[Z]$ with this effective total grading. To compute its boundary, choose homogeneous generators P_{ij} of the positive-dimensional factor rings with raw weights w_{ij} . For a geometric general point z above a prime divisor $D \subset F$, formula (12.2) gives

$$m_D = \gcd\{w_{ij}/d : P_{ij}(z_i) \neq 0\}.$$

This uses exactly the coordinates nonzero above D after effectivization. Taking $B = \sum_D (1 - 1/m_D) D$ therefore gives the actual boundary of this total quotient. The effective character-one polarization is $H = d H_{\text{raw}}$.

Step 4. In this step, we apply weighted descent to obtain the log Fano, weak Kähler–Einstein, and log K-polystability conclusions. The effective cone is normal, affine, and positive. Its invariant smooth open M has complement of codimension at least two. The form Ω is closed,

nondegenerate, algebraic, and of weight ℓ ; the smooth positive potential has the same modulus weight and satisfies

$$\mathrm{dd}^c \Phi = \omega_I > 0, \quad \omega_I^{2n} = \frac{(2n)!}{4^n (n!)^2} \Omega^n \wedge \bar{\Omega}^n.$$

The quotient weights $(2/\ell) \log(\Phi \circ \sigma)$ extend globally as continuous locally bounded plurisubharmonic weights by Proposition 12.1. Thus Theorem 11.4 applies with the effective weight $\ell = 2/d$ and the boundary just computed.

It gives klt singularities and $-(K_F + B) \sim_{\mathbb{Q}} n\ell H$. Since H is ample and $n\ell > 0$, the pair is log Fano. With $T_\psi = (2\pi)^{-1} \mathrm{dd}^c \psi$, it gives full mass H^{2n-1} and, after the stated additive normalization, a weak log Kähler–Einstein current $n\ell T_\psi$ in $c_1(-(K_F + B))$. Its conclusion for every normal relatively ample log test configuration gives nonnegative log Donaldson–Futaki invariant and the equivariantly product equality case. Every positive-dimensional component enters this construction; the point case was settled before any effective weight or quotient was assigned. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. We distinguish the earlier proof-development project from the subsequent manuscript-writing run. The latter used a frozen copy of the repaired proof records and did not restart mathematical research.

Technical service and project identities in this appendix have been anonymized for privacy. Consistent neutral labels preserve the links between the records. The mathematical content, model names, reasoning configurations and remaining narrative are retained.

A.1. How the system was run. The retained metadata for the earlier proof-development project records the base model `gpt-5.6-sol`, with four research workers: two at reasoning effort `high` and two at `xhigh`. It identifies a repair stage dated 11 August 2026. Those metadata do not supply the original start time, the original main-agent and verifier effort settings, or the elapsed time to target verification. We therefore report no historical discovery duration or comparison-run result.

The new writing view was created on 20 September 2026 at 05:36:40 UTC. Its main agent used `gpt-6-astra` at effort `ultra`; its native section writers, independent verifiers and reading helpers used the same model at effort `max`, through the configured private endpoint. Ten technical sections were assigned to native writing homes, each with an independent section verifier. Three helper homes supported source reading, bibliographic checks and exposition review. The main agent supplied the notation section, introduction and front matter. No research worker or mathematical computation was launched in this writing run.

The frozen source contained 235 active facts and 36 revoked records. The declared and literal support inspected for the manuscript comprised 109 active facts. Revoked records were retained for provenance and were not positive premises. Every technical section passed its independent native review. A subsequent whole-paper review passed before the readability and front-matter pass. These automated reviews examined the written arguments and the hypotheses of their interfaces; acceptance markers alone were not treated as proofs. They do not constitute a claim of independent human verification. The writing-view creation time and manuscript-review times are not dates of mathematical discovery.

A.2. The problem as posed. The historical mathematical target was to establish both parts of Xu–Zhuang Problem 23 in the semisimple scope, with the repaired hyperkähler comparison and descent arguments. The following is the current writing assignment, quoted from the fixed problem file with technical identifiers anonymized. Its identifiers and filenames are archival references in that assignment, not mathematical premises or citations in the proofs.

Fixed writing-only assignment

Read WRITING_REQUEST.md completely, then enter the installed native paper
 ↪ workflow now. Prove both parts of Xu-Zhuang Problem 23 for all complex
 ↪ semisimple Lie algebras, with the exact normalization, effectivization,
 ↪ actual divisorial-boundary, and zero-dimensional no-base conventions in
 ↪ that request.

Target roots are e5a7c7de4deed881, 6aef1cedc3a0eece, d5cb641bf6fae98b in this
 ↪ independently frozen clean writing-project graph. Read all support and
 ↪ SOURCE_SUPPORT_GATE review notes before architecture. All235 active facts
 ↪ and36 revoked records are retained; no archived mixed earlier-project
 ↪ material is positive authority.

This is a new writing-only view. SOURCE_PROBLEM.md and
 ↪ source_provenance/SOURCE_PROJECT.json retain prior research history;
 ↪ historical pause/research instructions do not replace the current writing
 ↪ authorization. Do not launch research, mutate source, or modify global
 ↪ configs. Empty workers is scope scaffolding only. Main MCP stays reader.
 ↪ Use native v3.1 style and complete sectional write/verify, global review,
 ↪ polish and checked build. Read NATIVE_COMMANDS.md before any native CLI
 ↪ command. All model calls use a private inference service. No external
 ↪ publication.

Read NATIVE_COMMANDS.md before any native CLI command.

A.3. Guidance given during the run. The current instruction record contains the following automatically generated startup message, dated 20 September 2026:

You are the main agent for Danus project writing-project (this directory). Read AGENTS.md and begin.

This initiated the fixed writing assignment. A separate delegated writing directive, also dated 20 September 2026, required the native section-writing and verification workflow, the default style, an explicit review of the repaired comparison and descent, preservation of the full semisimple scope, accurate prior-work coverage, and final compilation with delivery receipts. It required the specified model and endpoint, prohibited source and global-configuration changes, and authorized no external publication. The source-reading checkpoint, section reviews and whole-paper review were performed under that directive. Generated startup instructions are not recorded as human utterances. Historical repair and pause directions belong to the earlier project and were not interpreted as instructions to restart research.

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DEPARTMENT OF MATHEMATICS, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

BEIJING INTERNATIONAL CENTER FOR MATHEMATICAL RESEARCH, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

Email address: liujiahao@math.pku.edu.cn